

8		7		6		5		4		3		2		1	
1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%. 2. ALL CAPACITANCE VALUES ARE IN MICROFARADS. 3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.												REV	ECN	DESCRIPTION OF REVISION	CK APPD DATE
												10	0008927012	ENGINEERING RELEASED	2017-06-06
D21 MLB - DVT (OK2FAB)															
LAST_MODIFICATION= Tue Jun 6 18:57:04 2017															
PAGE CSA CONTENTS				SYNC		DATE		PAGE CSA CONTENTS				SYNC		DATE	
1	1	TABLE OF CONTENTS		sync		06/19/2016		46	62	I/O: USB PD		sync		04/14/2017	
2	2	SYSTEM: BOM TABLES (1/2)		sync		02/24/2017		47	63	I/O: Hydra		sync		04/14/2017	
3	3	SYSTEM: BOM TABLES (2/2)		test_mlb		03/17/2017		48	64	I/O: B2B DOCK		sync		10/10/2016	
4	4	SYSTEM:MECH COMPONENTS		sync		04/14/2017		49	65	SYSTEM: I2C MAP		sync		04/14/2017	
5	5	SYSTEM: Testpoints		sync		04/14/2017		50	80	RADIOS		D21_MLB_DOE		07/18/2016	
6	6	BOOTSTRAPPING		sync		09/19/2016		51	1	page1					
7	10	SOC: JTAG, USB, XTAL		sync		04/14/2017		52	75	NFC					
8	11	SOC: PCIE		sync		04/14/2017		53	1	FRONT PAGE					
9	12	SOC: MIPI + ISP		sync		04/14/2017		54	2	HIERARCHY					
10	13	SOC: LPDP		sync		04/14/2017		55	3	PMIC: BUCKS & LDOS					
11	14	SOC: SERIAL + SMC		sync		04/14/2017		56	4	PMIC: CLOCKS & CONTROL					
12	15	SOC: GPIO + UART		sync		04/14/2017		57	5	BB: POWER					
13	16	SOC: AOP		sync		04/14/2017		58	6	BB: CONTROL & HS PERIPHERALS [6]					
14	17	SOC: Power (1/3)		sync		04/14/2017		59	7	BB: GPIOs & QLINK					
15	18	SOC: Power (2/3)		sync		04/14/2017		60	8	XCVR					
16	19	SOC: Power (3/3)		sync		04/14/2017		61	9	XCVR					
17	26	NAND		sync		04/14/2017		62	10	QPOET MODULE					
18	27	SYSTEM POWER: PMU Bucks (1/4)		sync		04/14/2017		63	11	LB PAD					
19	28	SYSTEM POWER: PMU Bucks (2/4)		sync		04/14/2017		64	12	HB PAD					
20	29	SYSTEM POWER: PMU LDos (3/4)		sync		04/14/2017		65	13	UHB PAD					
21	30	SYSTEM POWER: PMU (4/4)		sync		04/14/2017		66	14	2G PA					
22	31	SYSTEM POWER: Boost		sync		04/14/2017		67	15	LB DSM					
23	32	SYSTEM POWER: B2B Battery		sync		04/14/2017		68	16	HB DSM					
24	33	SYSTEM POWER: Charger		sync		04/14/2017		69	17	COUPLER2					
25	34	SYSTEM POWER: Iktara		sync		04/14/2017		70	18	LOWER ANTENNA					
26	35	SYSTEM POWER: B2B Cyclone		sync		04/14/2017		71	19	UPPER ANTENNA					
27	36	SENSORS		sync		04/14/2017		72	20	GNSS					
28	37	CAMERA: PMU (1/2)		sync		04/14/2017		73	21	SIM, DEBUG CONN					
29	38	CAMERA: PMU (2/2)		sync		04/14/2017		74	22	TEST POINTS		PROBE POINTS		07/18/2016	
30	39	CAMERA: B2B Wide (KY)		sync		04/14/2017		75	1	FRONT PAGE					
31	40	CAMERA: B2B Tele (TN)		sync		04/14/2017		76	2	METROCIRC					
32	41	CAMERA: Strobe Drivers		sync		04/14/2017		77	3	UAT MATCH AND TUNER					
33	42	CAMERA: FCAM + FOREHEAD		sync		04/14/2017		78	4	RFFE TO GPO TRANSLATOR					
34	43	CAMERA: B2B Strobe + Buttons		sync		04/14/2017		79	1	SymbolPorts					
35	47	AUDIO: CODEC (1/2)		sync		04/14/2017		80	4	Guinness		WIFI		11/28/2016	
36	48	AUDIO: CODEC (2/2)		sync		04/14/2017		81	5	WiFIANTFeeds				11/28/2016	
37	49	AUDIO: Speaker Amp Bottom		sync		04/14/2017									
38	50	AUDIO: Speaker Amp Top		sync		04/14/2017									
39	51	ARC: Driver		sync		04/14/2017									
40	55	CG: Niki		sync		04/14/2017									
41	56	CG: PMUs		sync		04/14/2017									
42	57	CG: B2B DISPLAY + TOUCH		sync		04/14/2017									
43	58	CG: B2B ORB + MESA		sync		04/14/2017									
44	59	I/O: Overvoltage Cut-Off Circuit		sync		04/14/2017									
45	61	I/O: Accessory Buck		sync		04/14/2017									
Sub Designs															
PART#		QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION		SYNC_MASTER=sync SYNC_DATE=06/19/2016								
051-02159		1	SCH, MLB, D21	SCH	?		DRAWING TITLE SCH, MLB, D21								
020-00847		1	PCBF, MLB, D21	PCB	?		SOURCE PROJECT		SUB-DESIGN NAME		VERSION	HARD/ SOFT	SYNC_DATE/TIME		
							D21	RADIO_MLB		0.122.0	S	2017_06_06_17:14:31			
							D20	NFC_MLB		0.50.0	S	2017_05_26_07:54:42			
							D21	WIFI_MLB		0.20.0	S	2017_05_16_02:56:32			
							D21	RADIO_MLB_FF		0.35.0	S	2017_06_02_14:56:02			
												DRAWING NUMBER 051-02159		SIZE D	
												REVISION 10.0.0			
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												PAGE 1 OF 80			
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2

1

EEEE Codes

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
825-7691	1	EEEE FOR (MLB, 639-94441, JP, ULTRA)	EEEE_J08R	CRITICAL	EEEE_ULTIMATE
825-7691	1	EEEE FOR (MLB, 639-93377, JP, EXTREME)	EEEE_HNV8	CRITICAL	EEEE_EXTREME
825-7691	1	EEEE FOR (MLB, 639-94442, ROW, ULTRA)	EEEE_J08X	CRITICAL	EEEE_ULTIMATE_ROW
825-7691	1	EEEE FOR (MLB, 639-93389, ROW, EXTREME)	EEEE_HNVF	CRITICAL	EEEE_EXTREME_ROW

D21x Specific

D21x Specific

D21x Specific

D21x Specific

SOC

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
339S00416	1	SoC, H, 3GB 23nm DDR, B1	U1000	CRITICAL	COMMON

D21x Specific

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
339S00415	339S00416	BOM_TABLE_ALTS	U1000	SoC, M, 3GB 20nm DDR, B1
339S00417	339S00416	BOM_TABLE_ALTS	U1000	SoC, S, 3GB 20nm DDR, B1
339S00418	339S00416	BOM_TABLE_ALTS	U1000	SoC, S, 3GB 1xnm DDR, B1
339S00429	339S00416	BOM_TABLE_ALTS	U1000	SoC, H, 3GB 20nm DDR, B1
339S00430	339S00416	BOM_TABLE_ALTS	U1000	SoC, M, 3GB 20nm DDR, B1
339S00431	339S00416	BOM_TABLE_ALTS	U1000	SoC, S, 3GB 20nm DDR, B1
339S00432	339S00416	BOM_TABLE_ALTS	U1000	SoC, S, 3GB 18nm DDR, B1

SIPs

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
339M00011	339M00010	BOM_TABLE_ALTS	M5500	NIKI, STATS SIP
339M00013	339M00012	BOM_TABLE_ALTS	M4100	HUNT, STATS SIP

D21x Specific

SOFT-TERM CAP SUB BOMS

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
685-00152	685-00151	BOM_TABLE_ALTS	SUBBOM_CAP	SUBBOM, MLB, CAP, SOFT, D21
138S00049	138S0031	TYPICAL_CAP	ALL	CAP, D20S1, 2.2UF, 6.3V, KYOCERA

To enable alt in SUBBOM

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
685-00151	1	SUBBOM, MLB, SOFT-TERM, D21	SUBBOM_CAP		
138S00159	9	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C2000, C2001, C2003, C2005, C2006, C2007, C2008, C2013		SOFT_CAP
138S0031	9	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C2000, C2001, C2003, C2005, C2006, C2007, C2008, C2013		TYPICAL_CAP
138S00159	15	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C2051, C2052, C2054, C2055, C2062, C2066, C2068		SOFT_CAP
138S0031	15	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C2051, C2052, C2054, C2065, C2066, C2067, C2068, C2069		TYPICAL_CAP
138S00159	10	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C2070, C2073, C2074, C2076, C2078, C2071, C2080, C2081		SOFT_CAP
138S0031	10	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C2070, C2073, C2074, C2076, C2078, C2071, C2080, C2081		TYPICAL_CAP
138S00159	2	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C3351, C3352		SOFT_CAP
138S0031	2	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C3351, C3352		TYPICAL_CAP
138S00159	3	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C3602, C3611, C3622		SOFT_CAP
138S0031	3	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C3602, C3611, C3622		TYPICAL_CAP
138S00159	5	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C3791, C3795, C3798		SOFT_CAP
138S0031	5	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C3791, C3795, C3798		TYPICAL_CAP
138S00159	3	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C3909, C3925, C3926		SOFT_CAP
138S0031	3	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C3909, C3925, C3926		TYPICAL_CAP
138S00159	2	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C4250, C4261		SOFT_CAP
138S0031	2	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C4250, C4261		TYPICAL_CAP
138S00159	1	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C4303		SOFT_CAP
138S0031	1	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C4303		TYPICAL_CAP
138S00159	5	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C4805, C4809, C4914, C4926, C4929		SOFT_CAP
138S0031	5	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C4805, C4809, C4914, C4926, C4929		TYPICAL_CAP
138S00159	5	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C5016, C5018, C5134, C5029, C5136		SOFT_CAP
138S0031	5	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C5016, C5018, C5134, C5029, C5136		TYPICAL_CAP
138S00159	1	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C5611		SOFT_CAP
138S0031	1	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C5611		TYPICAL_CAP
138S00159	4	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C5700, C5705, C5716, C5721		SOFT_CAP
138S0031	4	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C5700, C5705, C5716, C5721		TYPICAL_CAP
138S00159	5	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C5802, C5804, C5811, C5808		SOFT_CAP
138S0031	5	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C5802, C5804, C5811, C5808		TYPICAL_CAP
138S00159	1	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C5890		SOFT_CAP
138S0031	1	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C5890		TYPICAL_CAP
138S00159	5	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C6320, C6321, C6322, C6323, C6324		SOFT_CAP
138S0031	5	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C6320, C6321, C6322, C6323, C6324		TYPICAL_CAP
138S00160	4	CAP, SOFT-TERM, 10UF, 10V, 0402, MURATA	C4907, C4903, C4904, C4931		SOFT_CAP
138S0979	4	CAP, TYPICAL, 10UF, 10V, 0402, MUR/KYO	C4907, C4903, C4904, C4931		TYPICAL_CAP
138S00160	5	CAP, SOFT-TERM, 10UF, 10V, 0402, MURATA	C5600, C5602, C5603, C5702, C5703		SOFT_CAP
138S0979	5	CAP, TYPICAL, 10UF, 10V, 0402, MUR/KYO	C5600, C5602, C5603, C5702, C5703		TYPICAL_CAP
138S00159	2	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201, KYOCERA	C4025, C4026		SOFT_CAP
138S0031	2	CAP, TYPICAL, 2.2UF, 6.3V, 0201, MURATA	C4025, C4026		TYPICAL_CAP

Default is typical

D21x Specific

D21x Specific

IKTARA DIODES SUB BOMS

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
685-00172	685-00171	BOM_TABLE_ALTS	SUBBOM_DS	SUBBOM, MLB, SCHOTTKY DIODES, D21

#31526972: Update Iktara Diodes APNs

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
685-00171	1	SUBBOM, MLB, SCHOTTKY DIODES, D100ES, D21	SUBBOM_DS		
371S00133	4	DIODES, SHOTTKY DIODE, 30V, 2A, 9603	D3400, D3401, D3402, D3403		DIODES_DS
371S00132	4	ONSEMI, SHOTTKY DIODE, 30V, 2A, 9603	D3400, D3401, D3402, D3403		ONSEMI_DS

Footprint is ONSMT

Primary is D100ES

PAGE TITLE		
SYSTEM: BOM TABLES (1/2)		
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2

1

Current as of D21 MC0 056-03352 REV 42

D

C

B

A

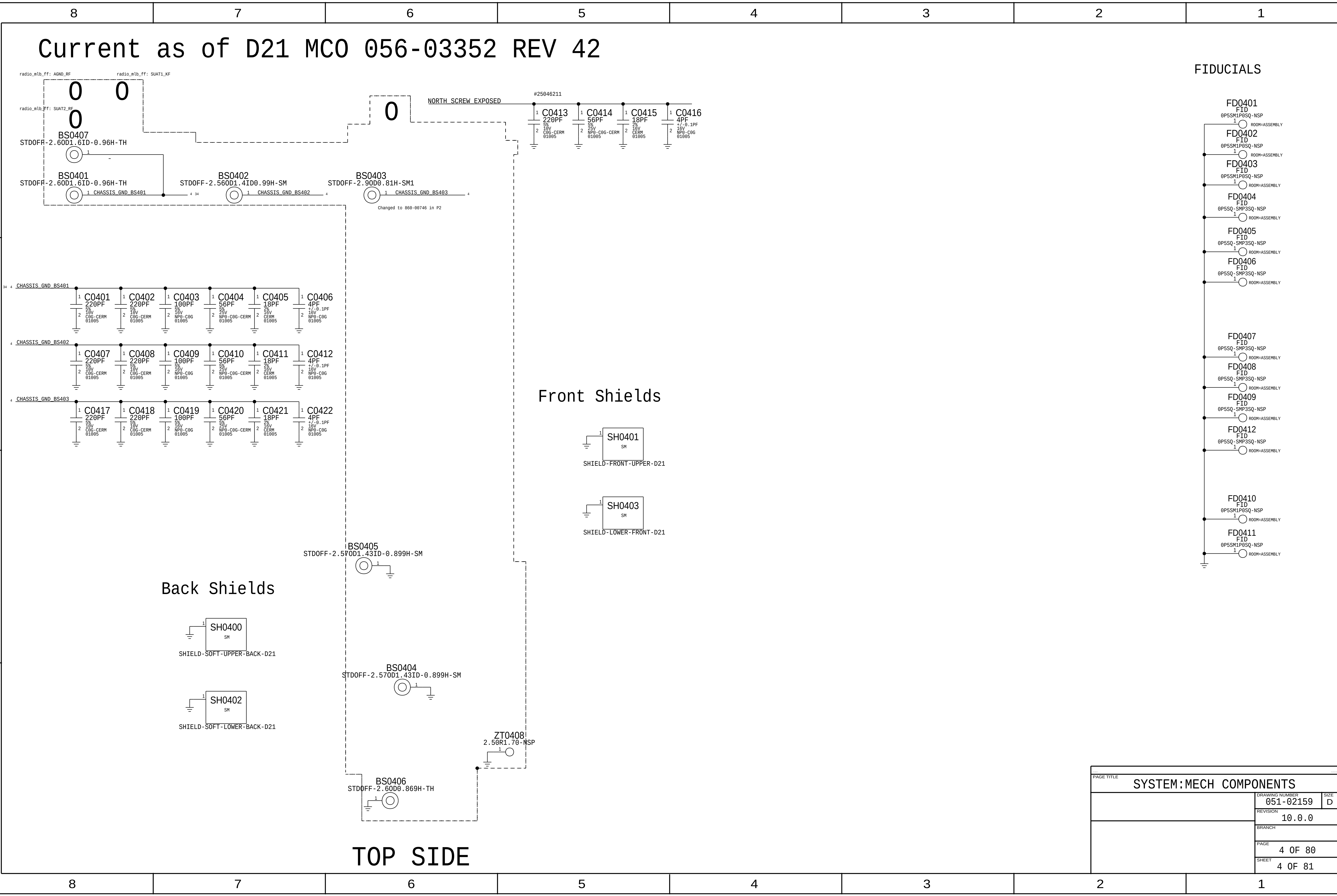
FIDUCIALS

D

C

B

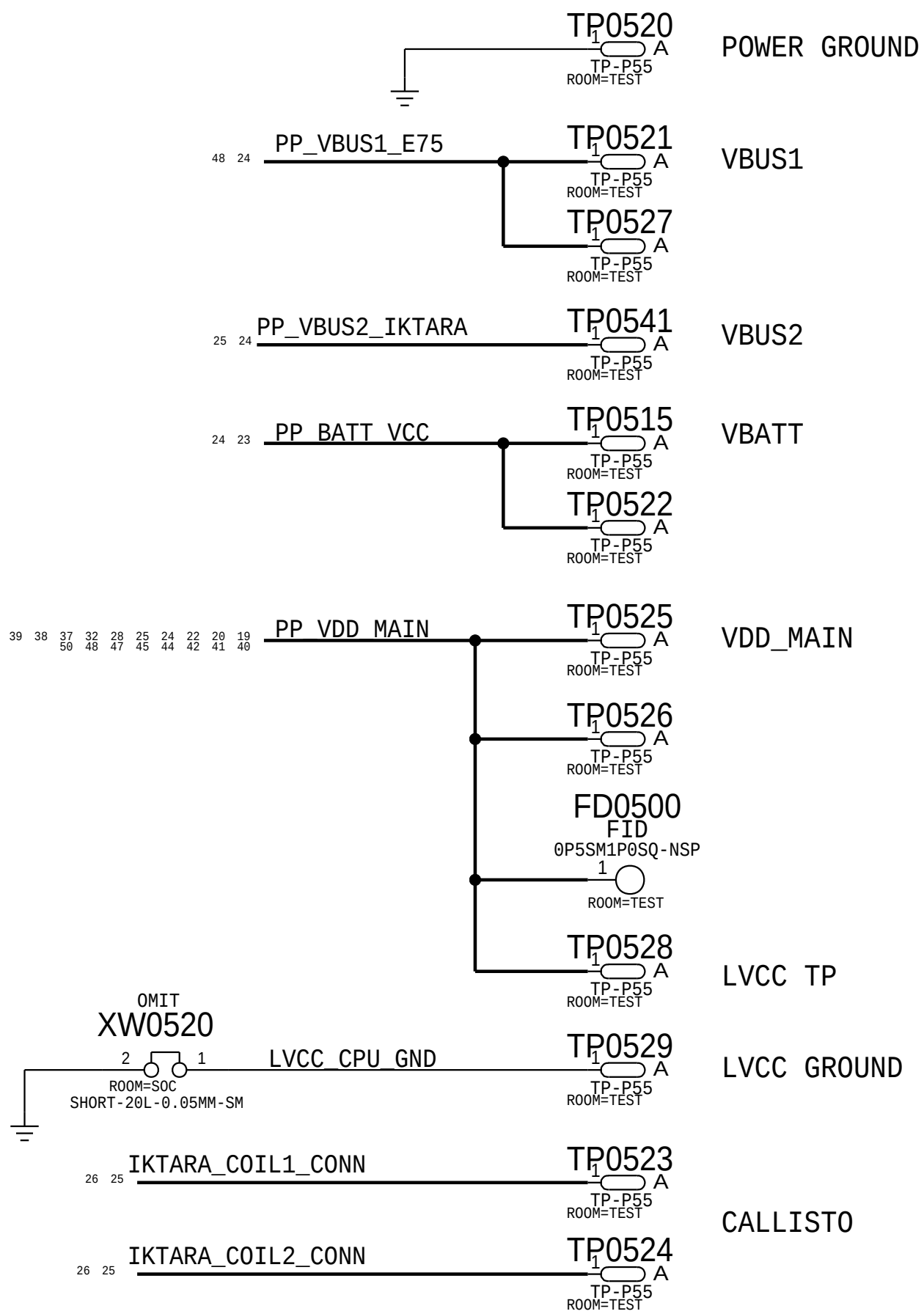
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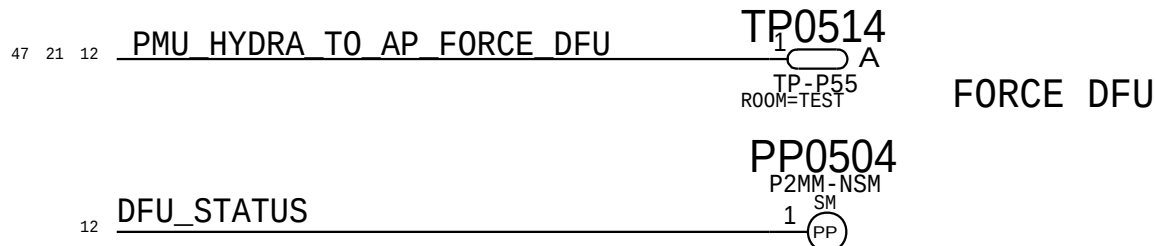
TOP SIDE

Test Points

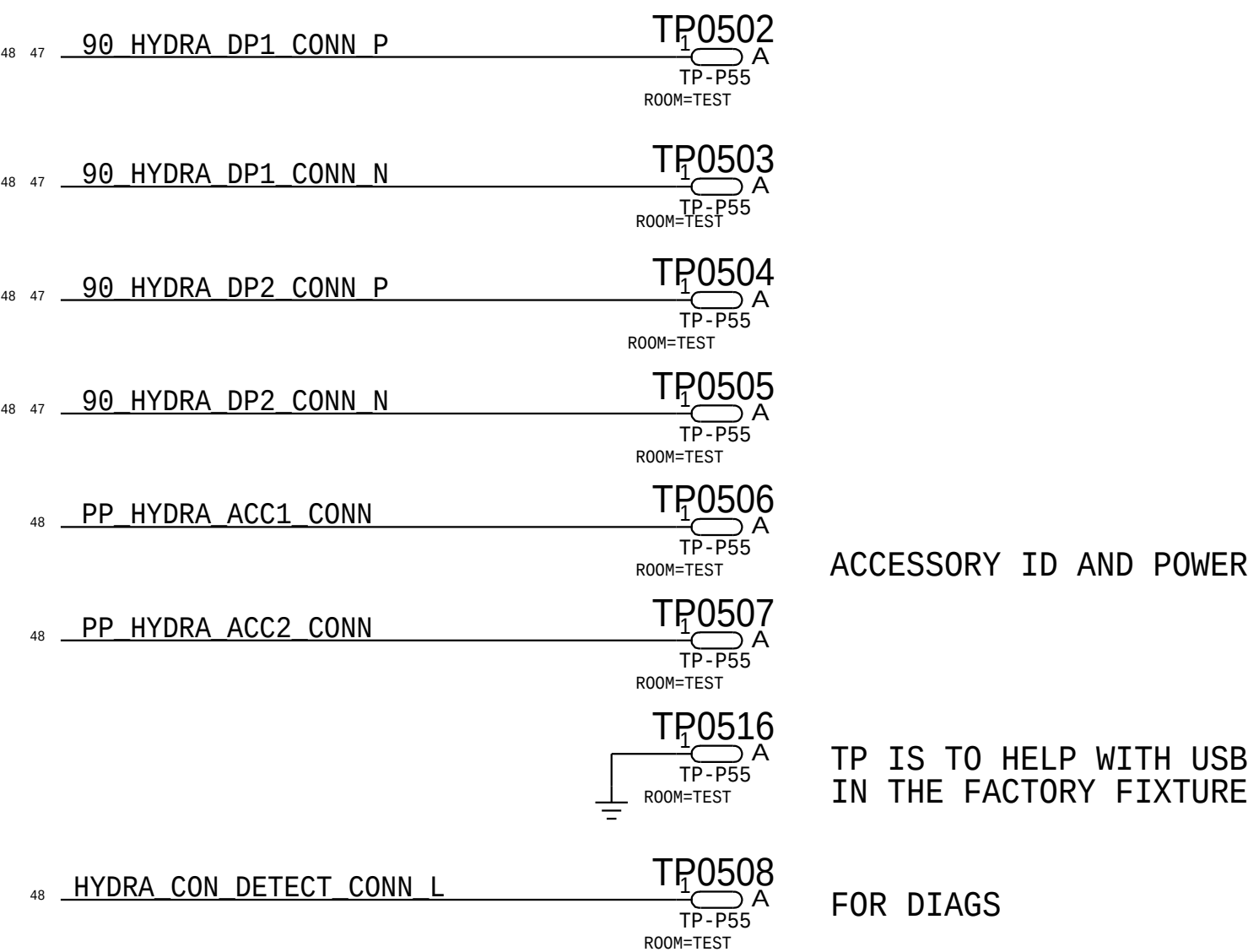
POWER



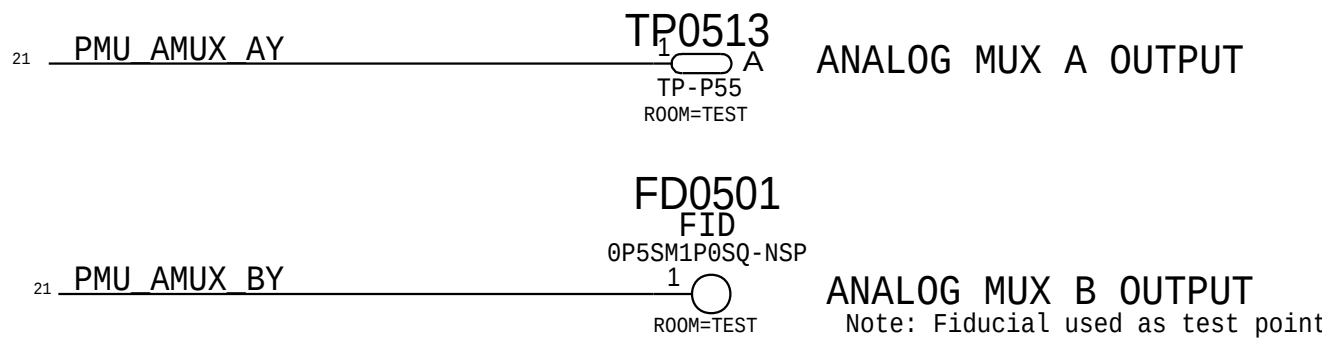
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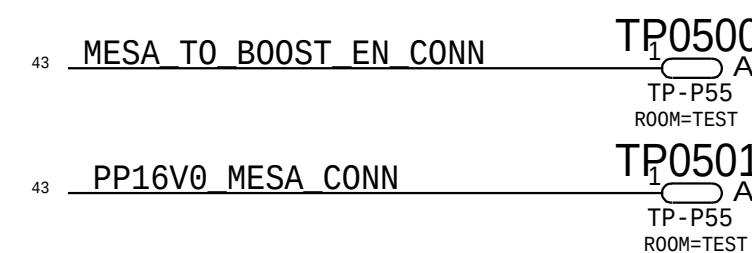
E75



AMUX



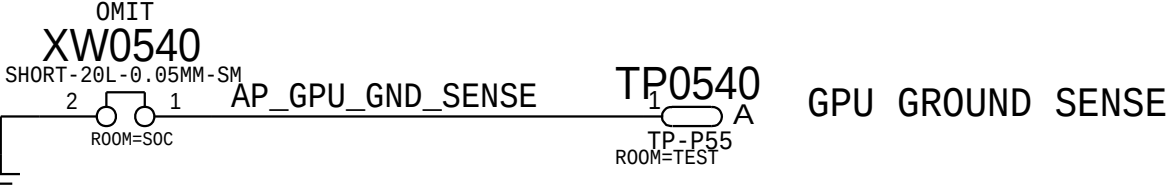
MOJAVE



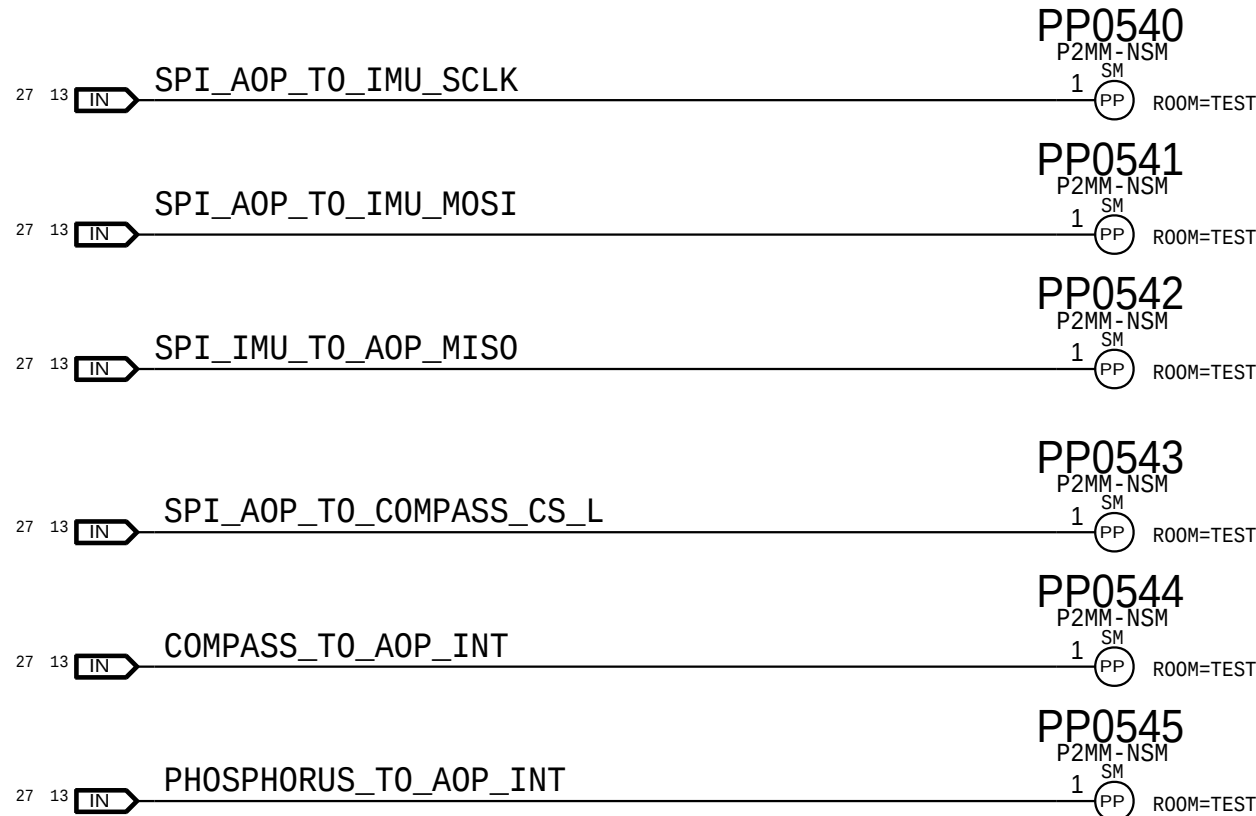
LCM



SENSE TP

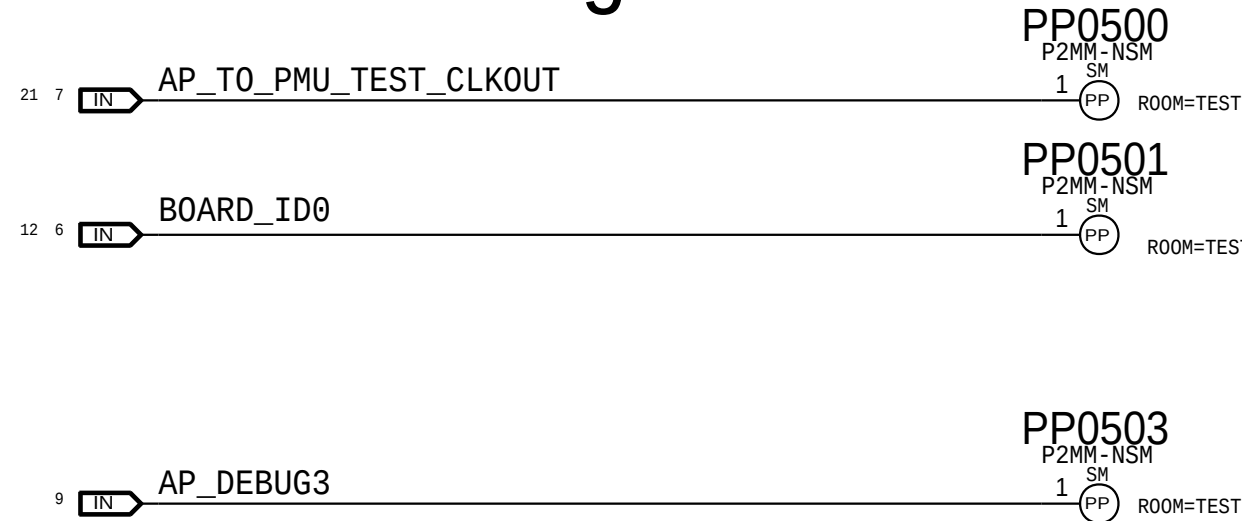


Sensor SPI

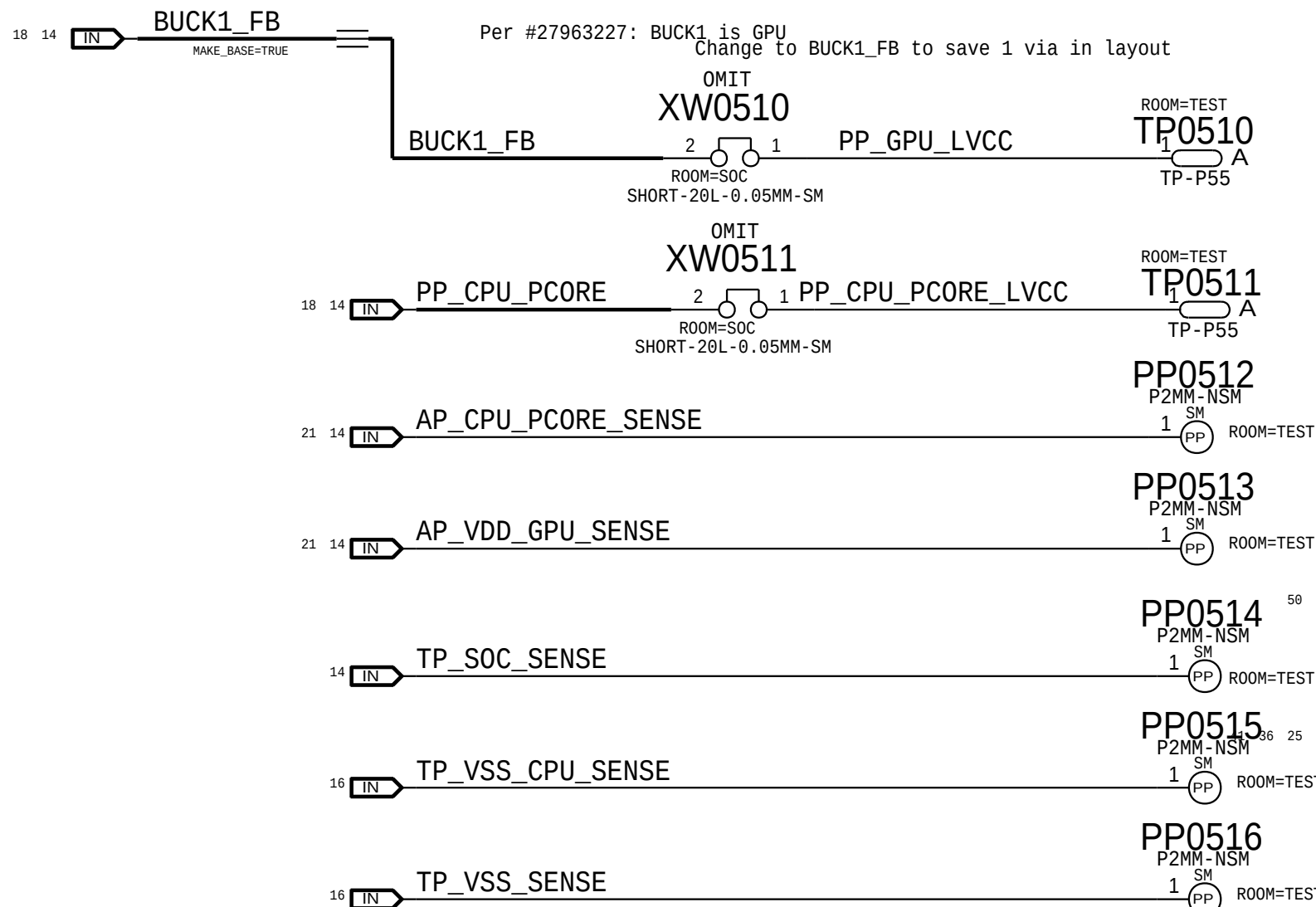


Probe Points

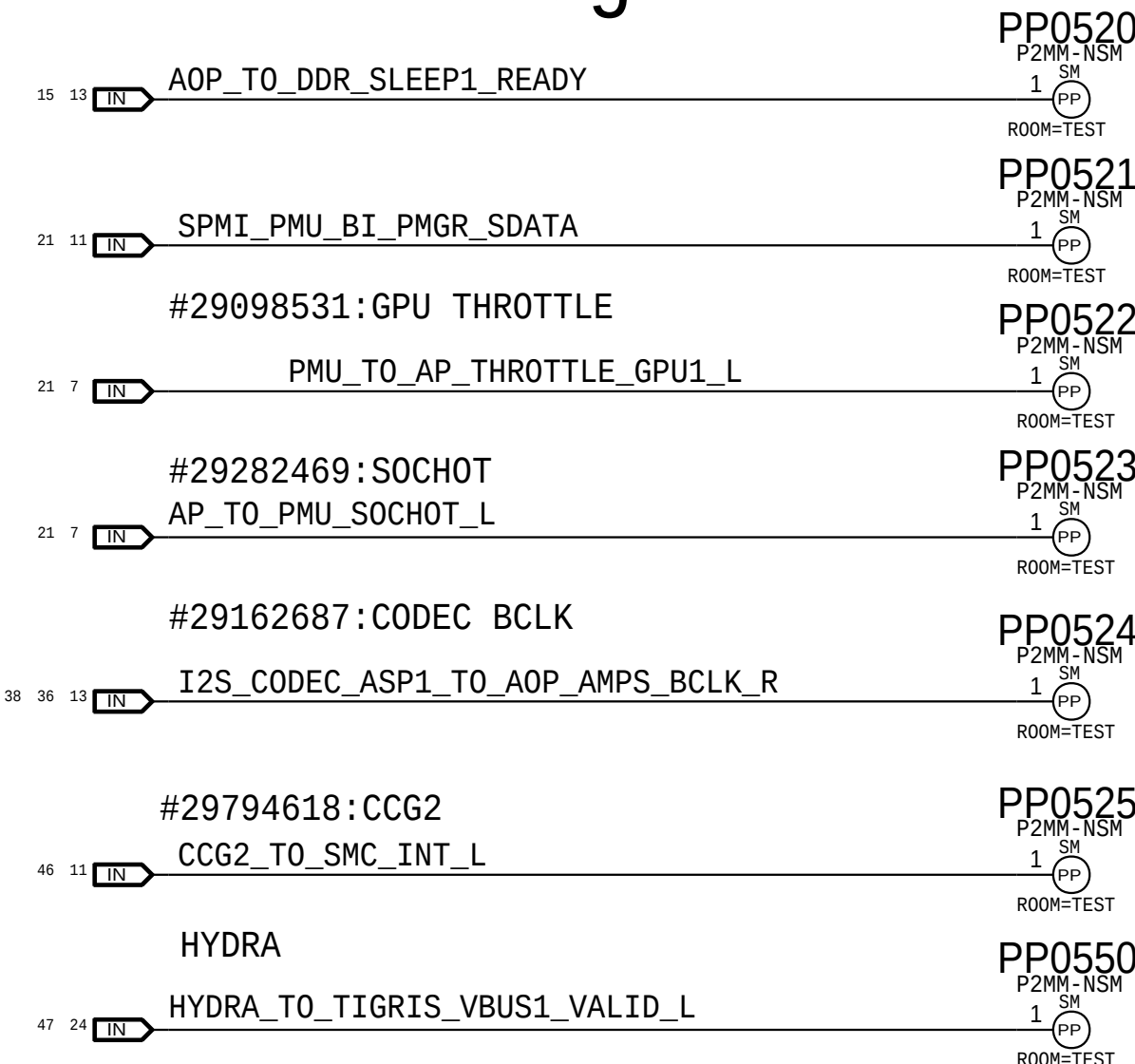
SOC Debug



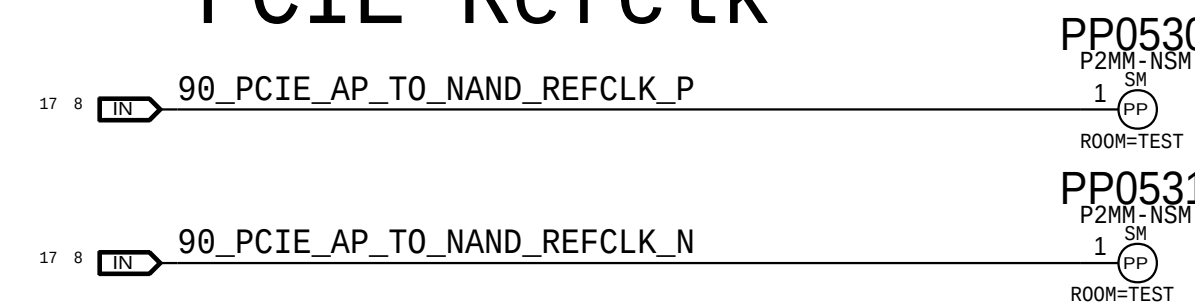
SOC CPU/GPU



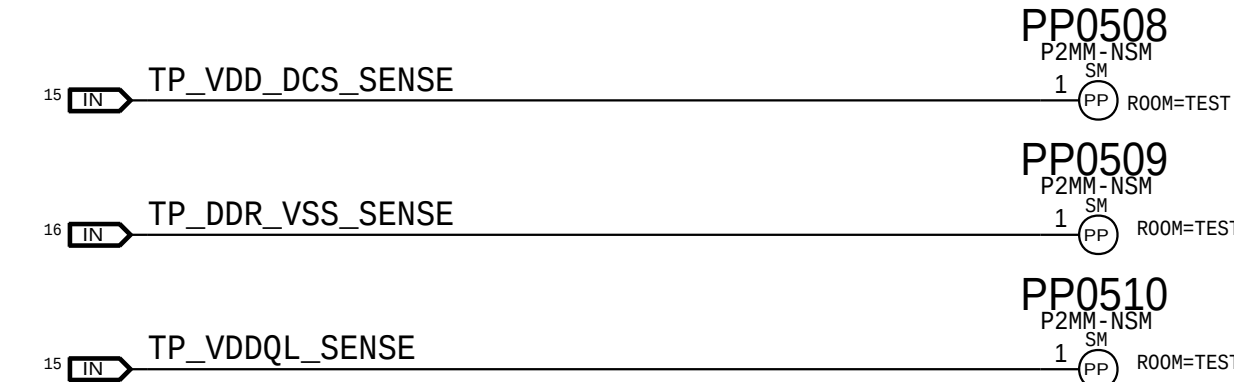
PMU Debug



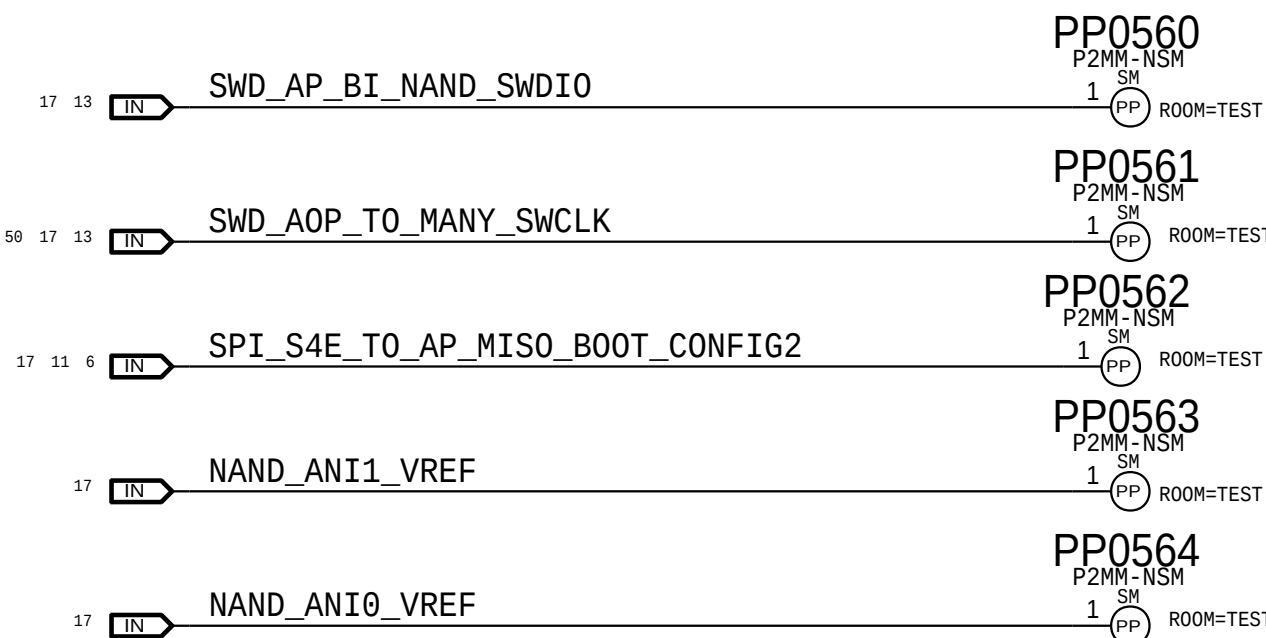
PCIE Refclk



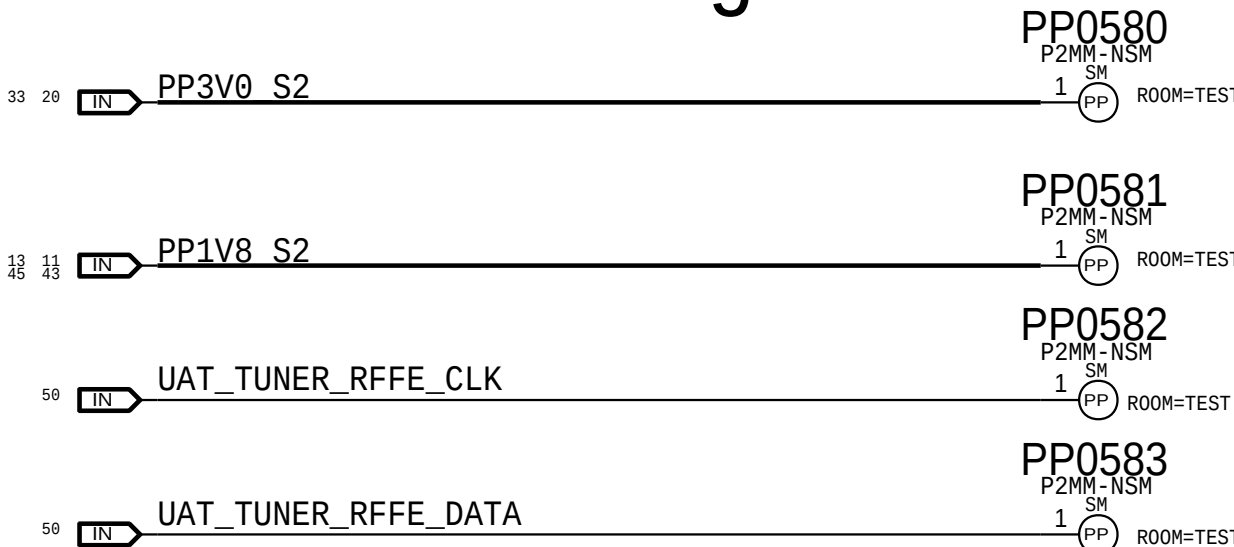
SOC DDR



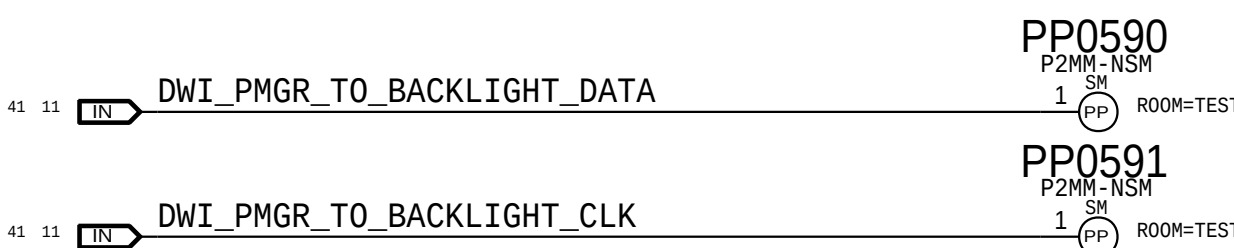
NAND Debug



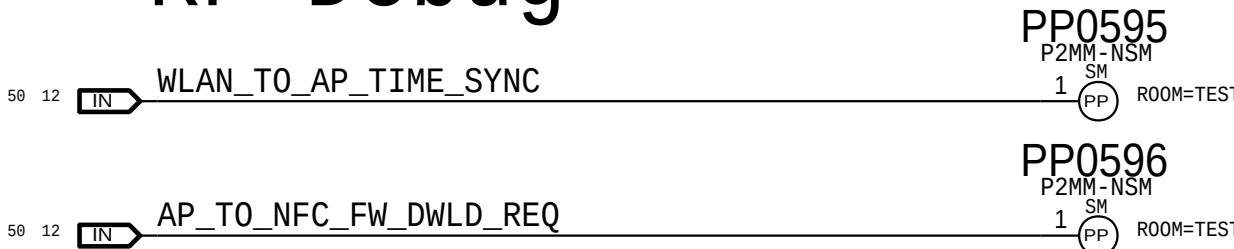
BB UAT Debug



BACKLIGHT Debug

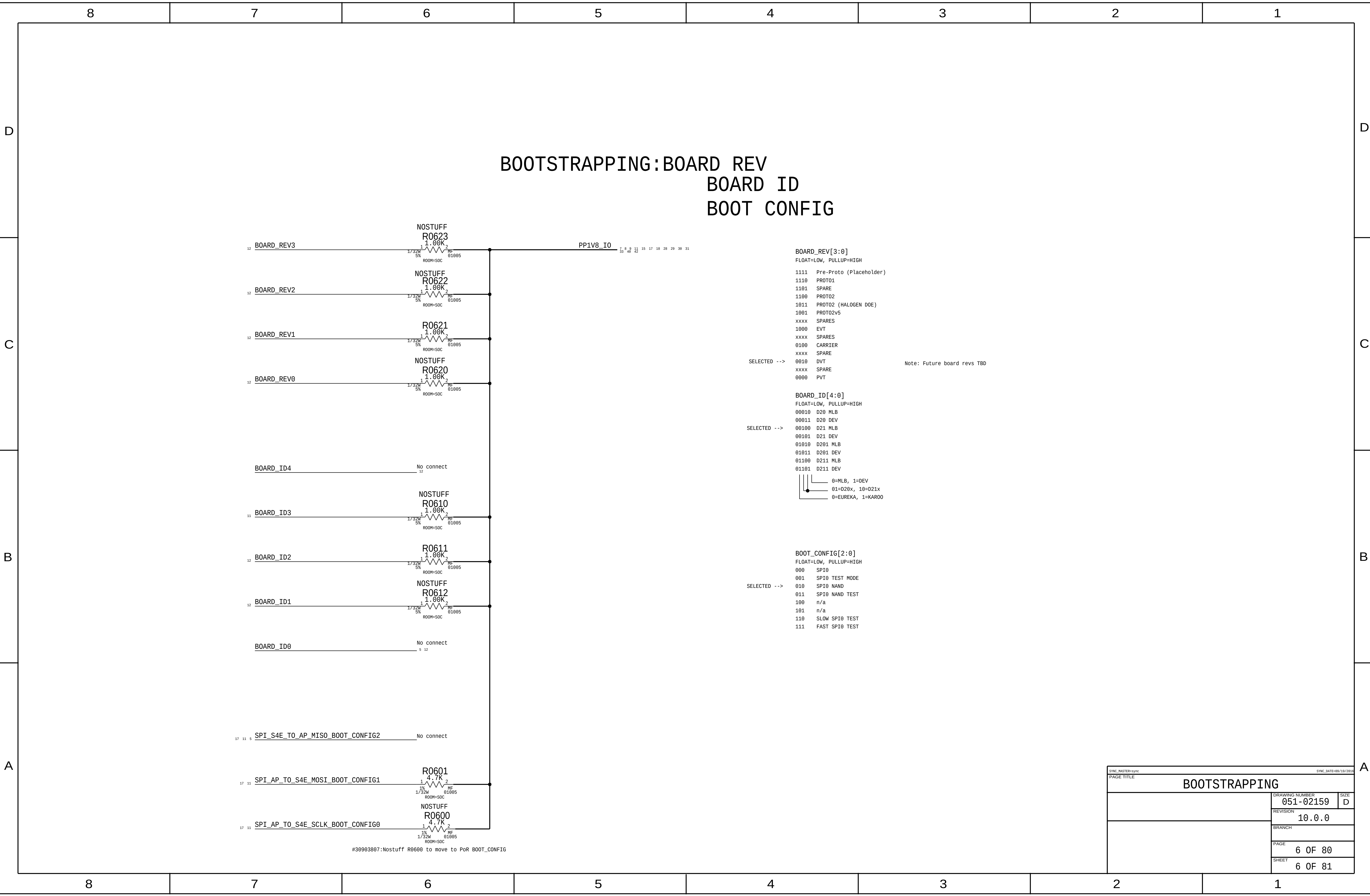


RF Debug



SYSTEM: Testpoints

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BOARD_ID4

12

No connect

BOARD_ID3

11

1

1.00K

2

1/32W

5%

R0610

NOSTUFF

R00M=SOC

01005

BOARD_ID2

12

1

1.00K

2

1/32W

5%

R0611

NOSTUFF

R00M=SOC

01005

BOARD_ID1

12

1

1.00K

2

1/32W

5%

R0612

NOSTUFF

R00M=SOC

01005

BOARD_ID0

5 12

No connect

SPI_S4E_TO_AP_MISO_BOOT_CONFIG2

17 11 5

No connect

SPI_AP_TO_S4E_MOSI_BOOT_CONFIG1

17 11

1

4.7K

2

1/32W

1%

R0601

NOSTUFF

R00M=SOC

01005

SPI_AP_TO_S4E_SCLK_BOOT_CONFIG0

17 11

1

4.7K

2

1/32W

1%

R0600

NOSTUFF

R00M=SOC

01005

PP1V8_I0

1 8 9 11 15 17 18 28 29 30 31

13 40 12

BOARD_REV[3:0]

FLOAT=LOW, PULLUP=HIGH

1111 Pre-Proto (Placeholder)

1110 PROT01

1101 SPARE

1100 PROT02

1011 PROT02 (HALOGEN D0E)

1001 PROT02v5

xxxx SPARES

1000 EVT

xxxx SPARES

0100 CARRIER

xxxx SPARE

0010 DVT

xxxx SPARE

0000 PVT

SELECTED -->

Note: Future board revs TBD

BOARD_ID[4:0]

FLOAT=LOW, PULLUP=HIGH

00010 D20 MLB

00011 D20 DEV

00100 D21 MLB

00101 D21 DEV

01010 D201 MLB

01011 D201 DEV

01100 D211 MLB

01101 D211 DEV

SELECTED -->

0=MLB, 1=DEV

01=D20x, 10=D21x

0=EUREKA, 1=KAR00

BOOT_CONFIG[2:0]

FLOAT=LOW, PULLUP=HIGH

000 SPI0

001 SPI0 TEST MODE

010 SPI0 NAND

011 SPI0 NAND TEST

100 n/a

101 n/a

110 SLOW SPI0 TEST

111 FAST SPI0 TEST

SELECTED -->

#30903807:Nostuff R0600 to move to PoR BOOT_CONFIG

SYNC_MASTER=sync

SYNC_DATE=09/19/2016

PAGE TITLE

BOOTSTRAPPING

DRAWING NUMBER

051-02159

SIZE

D

REVISION

10.0.0

BRANCH

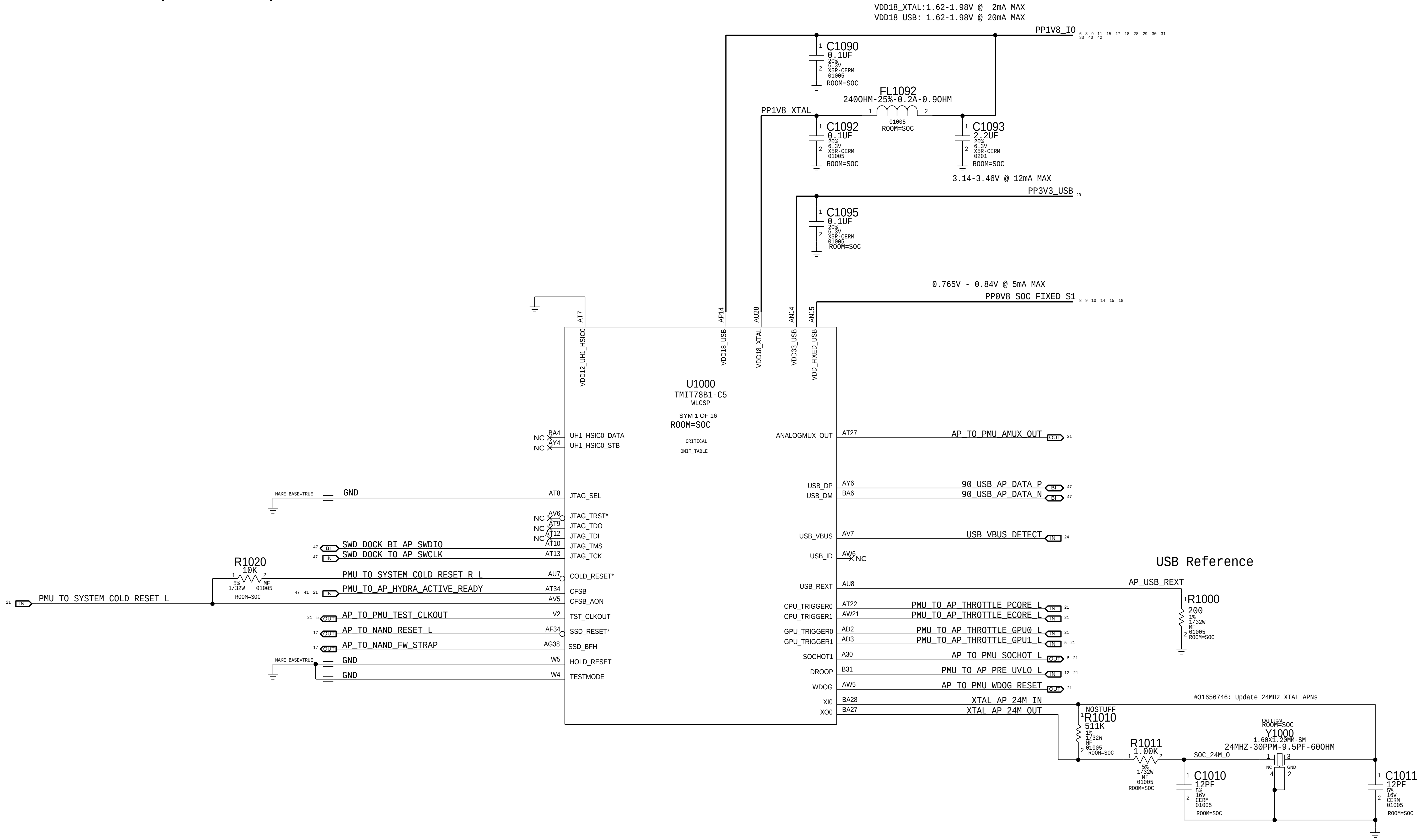
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SOC - USB, JTAG, XTAL

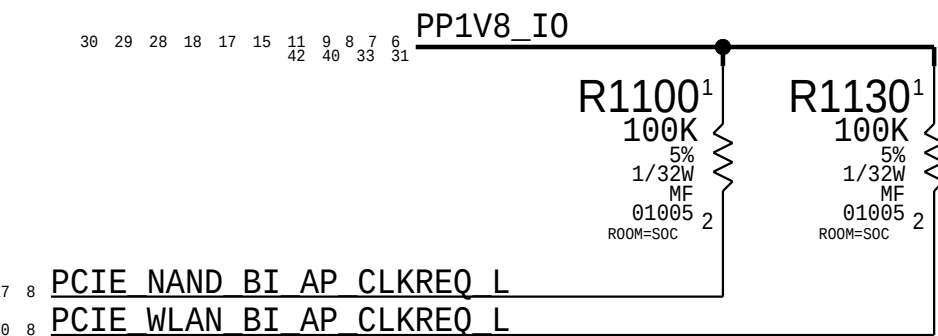


PAGE TITLE		
SOC: JTAG, USB, XTAL		
DRAWING NUMBER	051-02159	SIZE D
REVISION	10.0.0	
BRANCH		
PAGE	10 OF 80	
SHEET	7 OF 81	

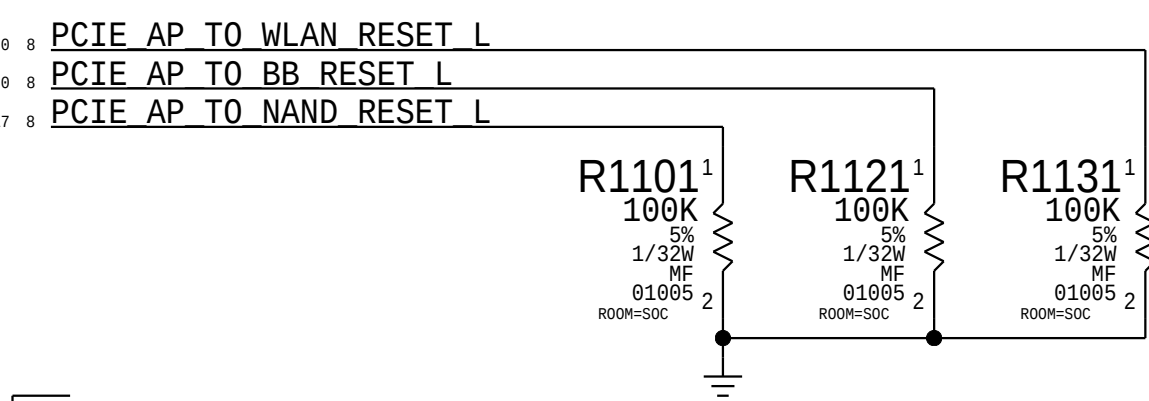
SOC - PCIE INTERFACES

D

PCie Clock Request Pull-Ups



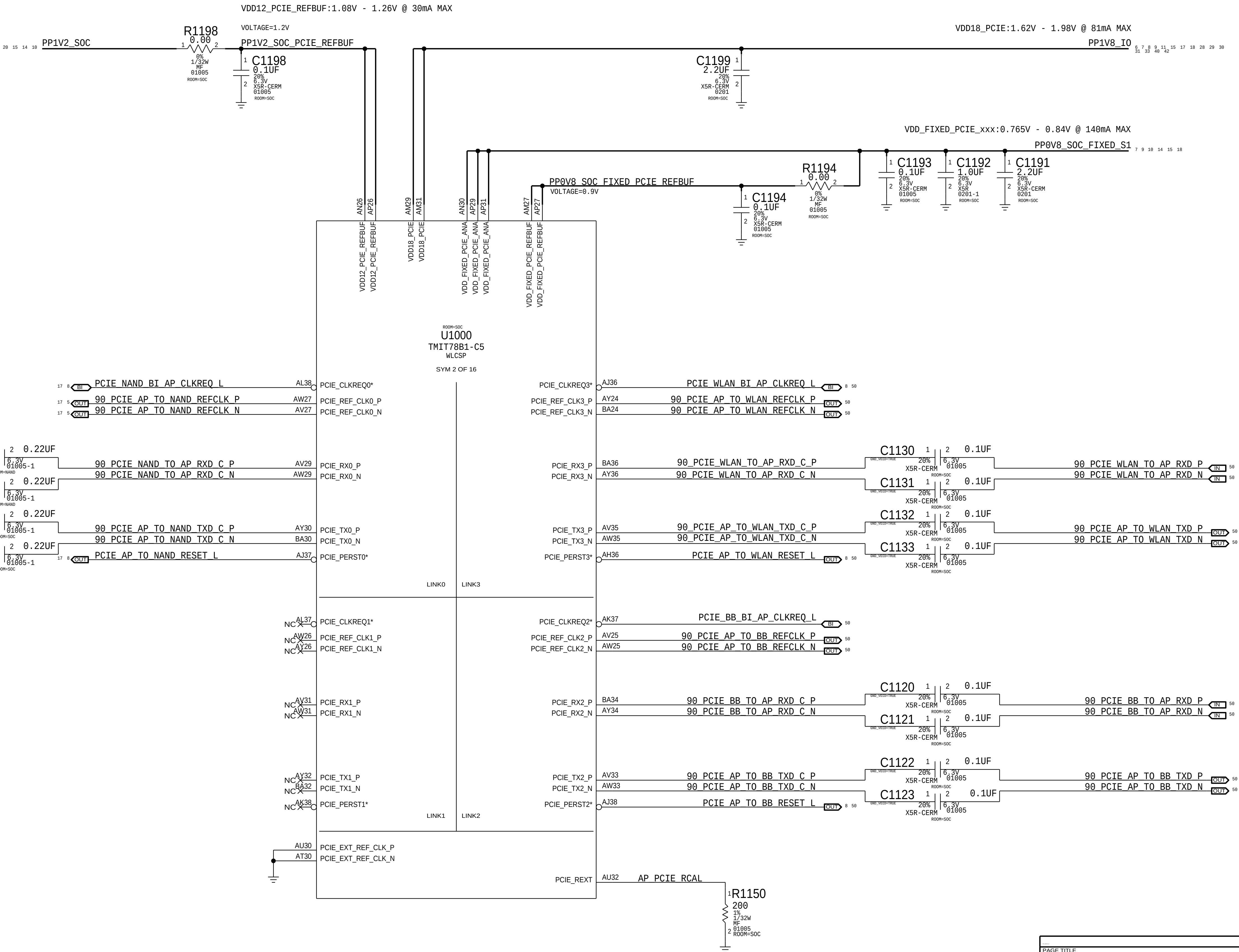
PCie Reset Pull-Downs



PCIE LINK 0

B

A



D

C

B

A

PAGE TITLE		
SOC: PCIE		
	DRAWING NUMBER	051-02159
	REVISION	10.0.0
	BRANCH	
	PAGE	11 OF 80
	SHEET	8 OF 81

SOC - MIPI & ISP INTERFACES

D

C

B

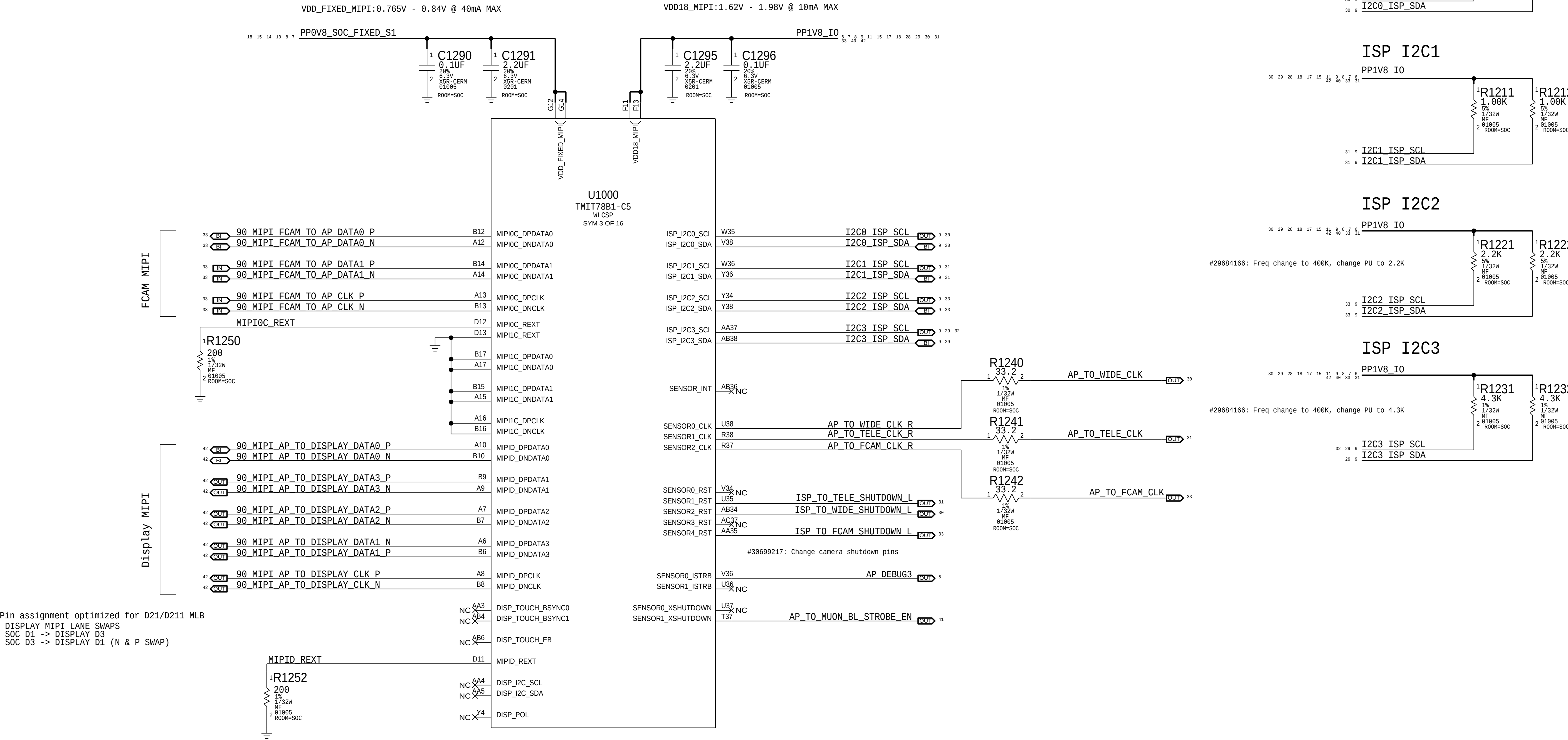
A

D

C

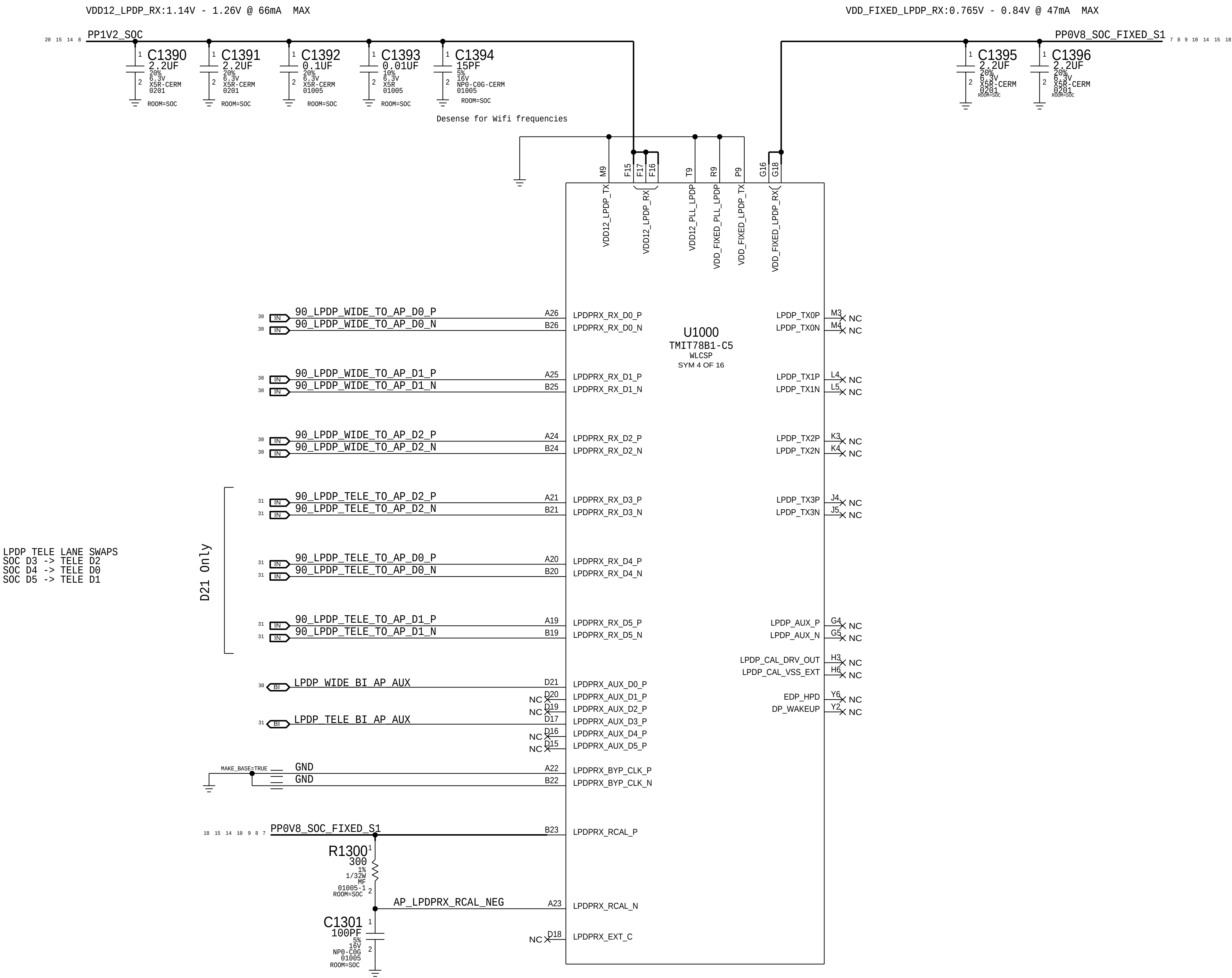
B

A



PAGE TITLE		
SOC: MIPI + ISP		
	DRAWING NUMBER	051-02159
	REVISION	10.0.0
	BRANCH	
	PAGE	12 OF 80
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SOC - LPDP INTERFACES



PAGE TITLE		
SOC: LPDP		
	DRAWING NUMBER	051-02159
	REVISION	10.0.0
	BRANCH	
	PAGE	13 OF 80
	SHEET	10 OF 81

SOC - SERIAL INTERFACES

D

C

B

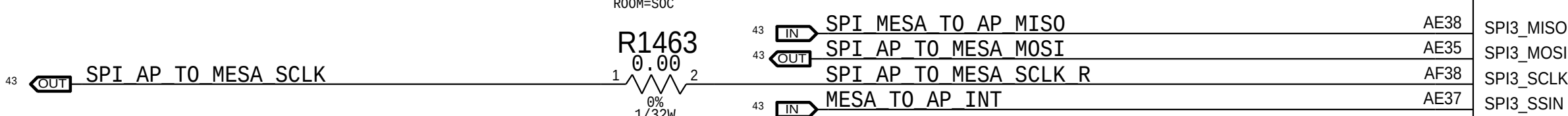
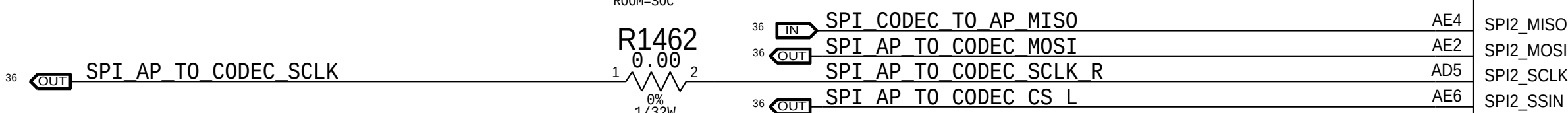
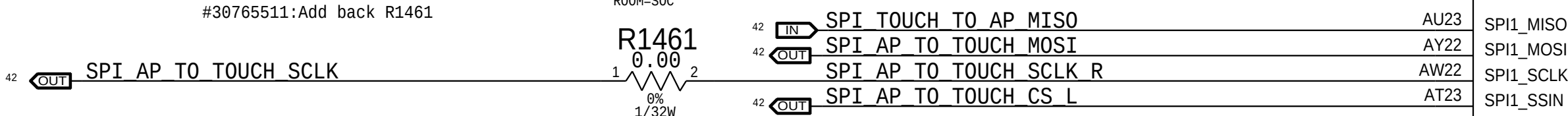
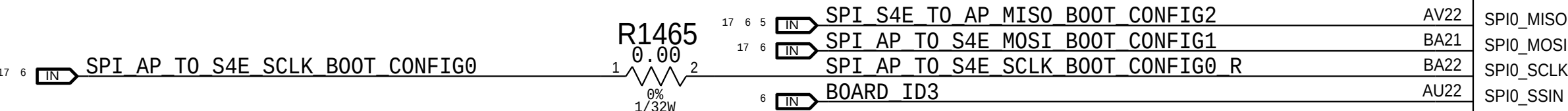
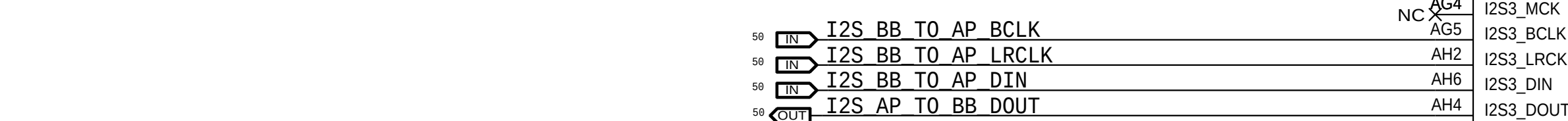
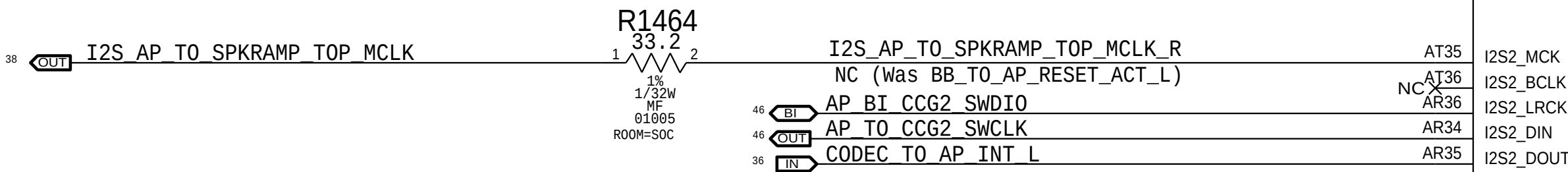
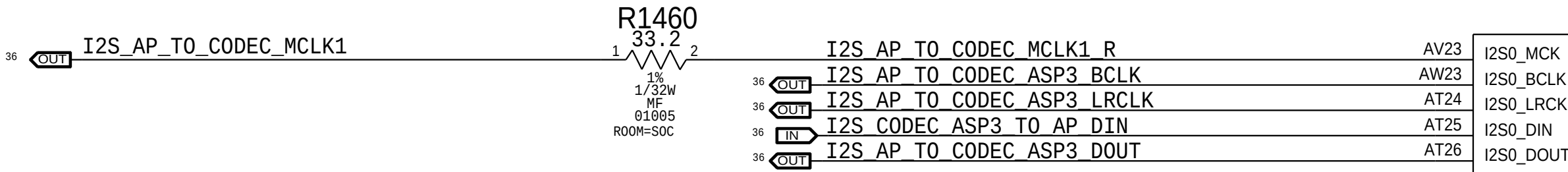
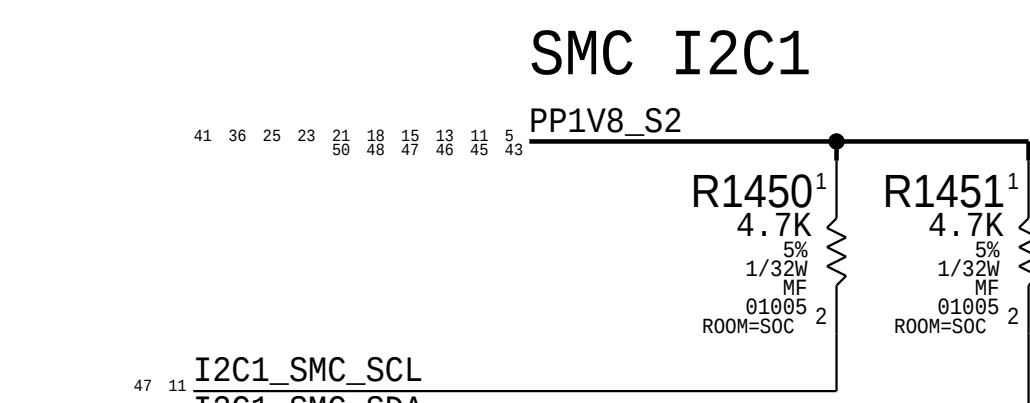
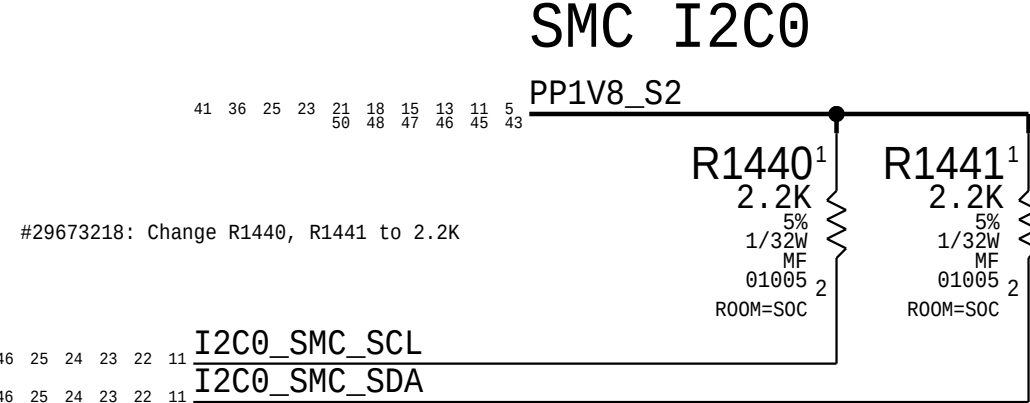
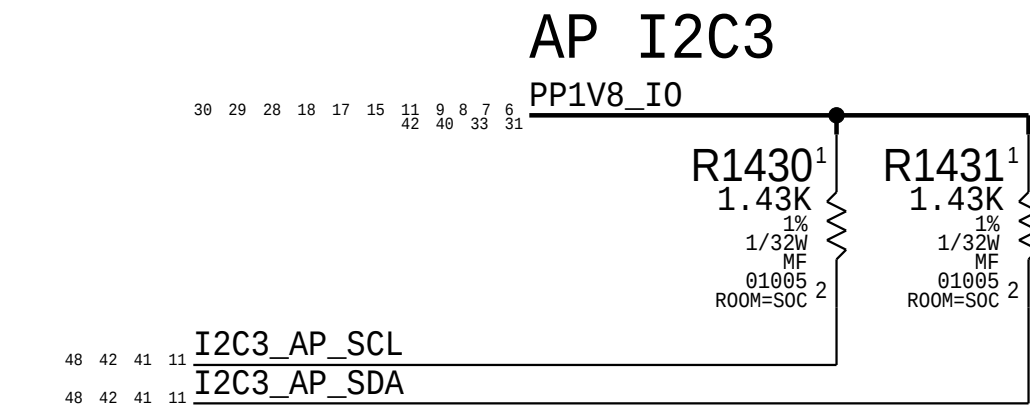
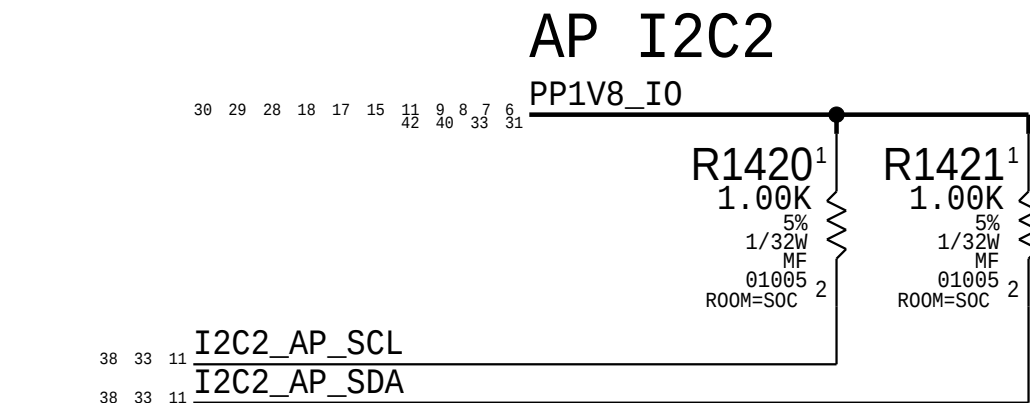
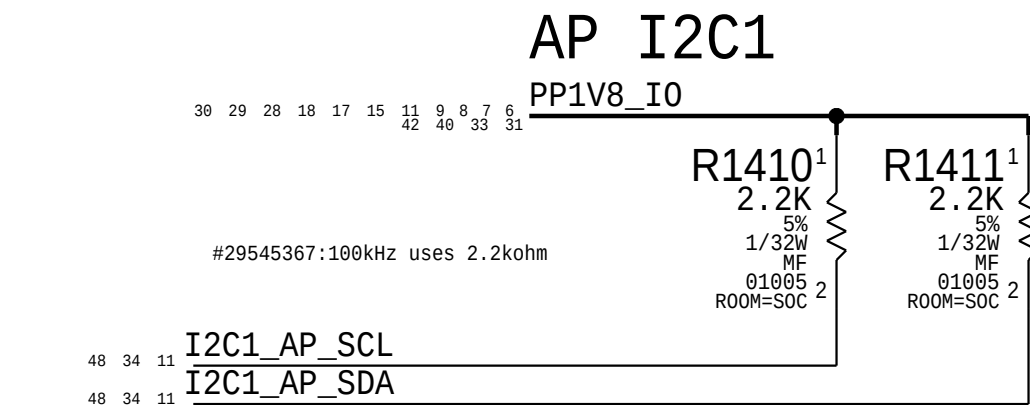
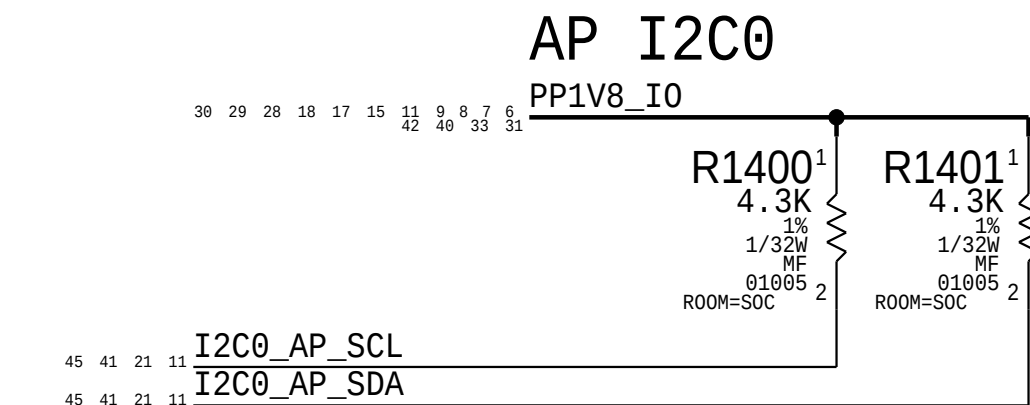
A

D

C

B

A

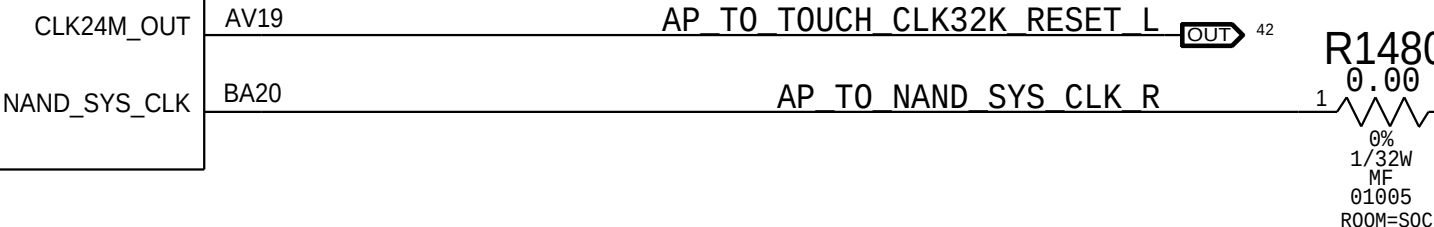
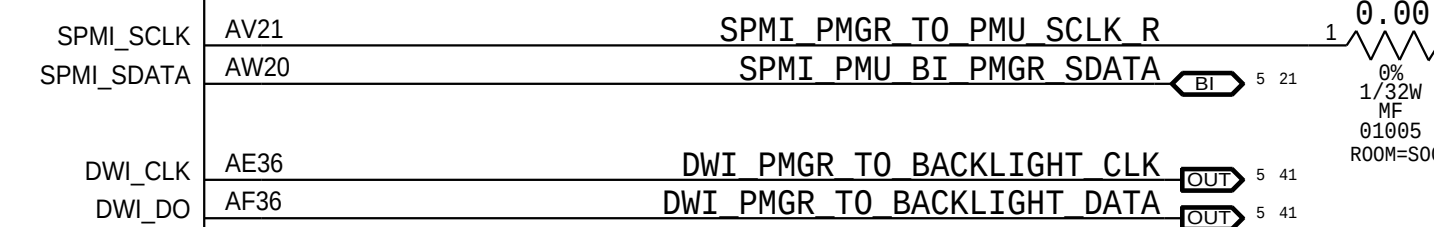
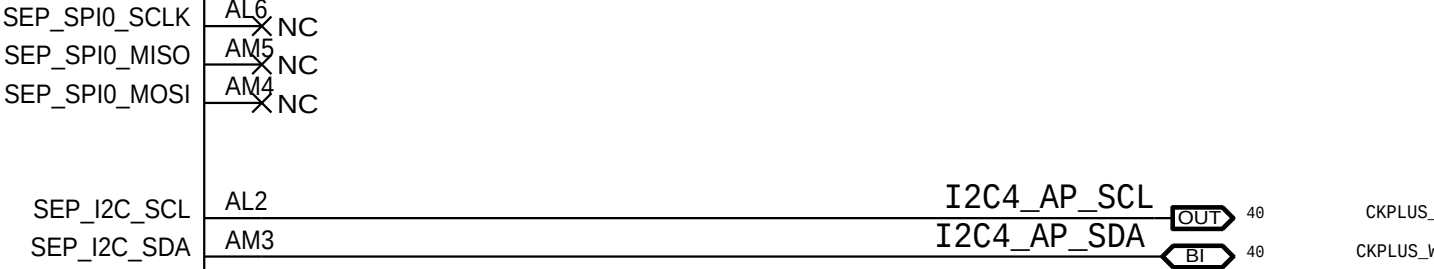
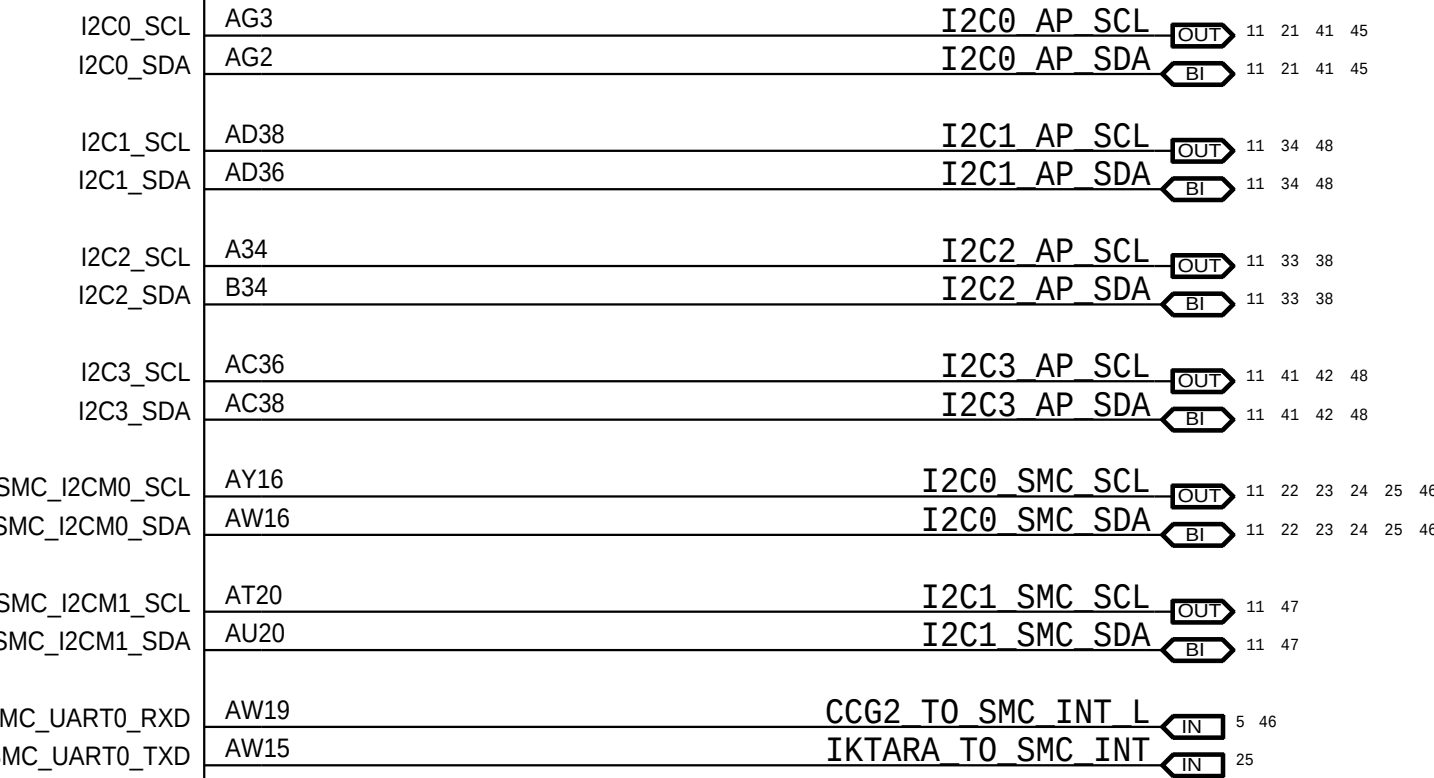


U1000

TMIT78B1-C5

WLCSP

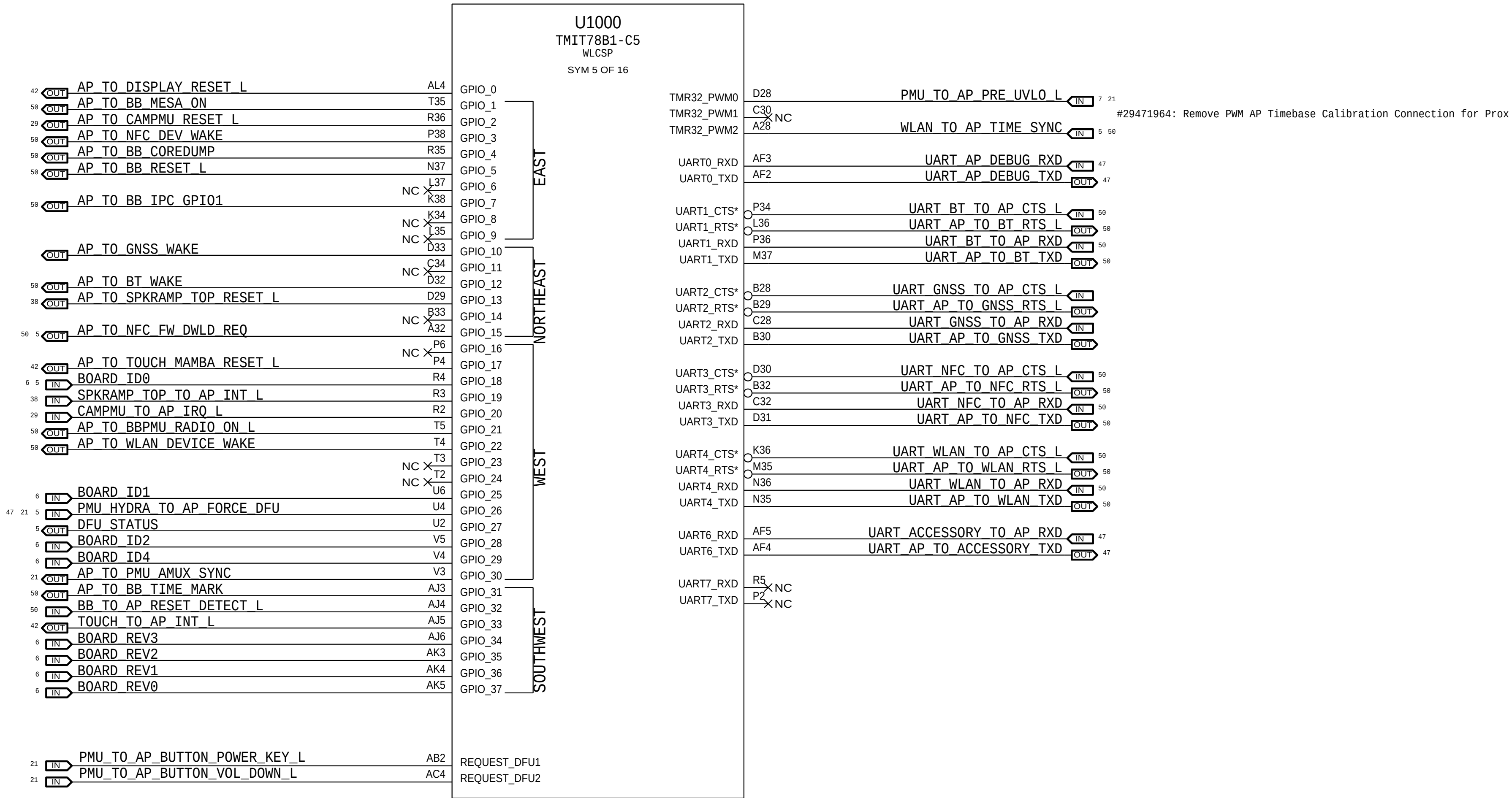
SYM 6 OF 16



SPI: Route as Daisy-Chain (No T's Allowed)

PAGE TITLE		
SOC: SERIAL + SMC		
	DRAWING NUMBER	051-02159
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	BRANCH	
	PAGE	14 OF 80
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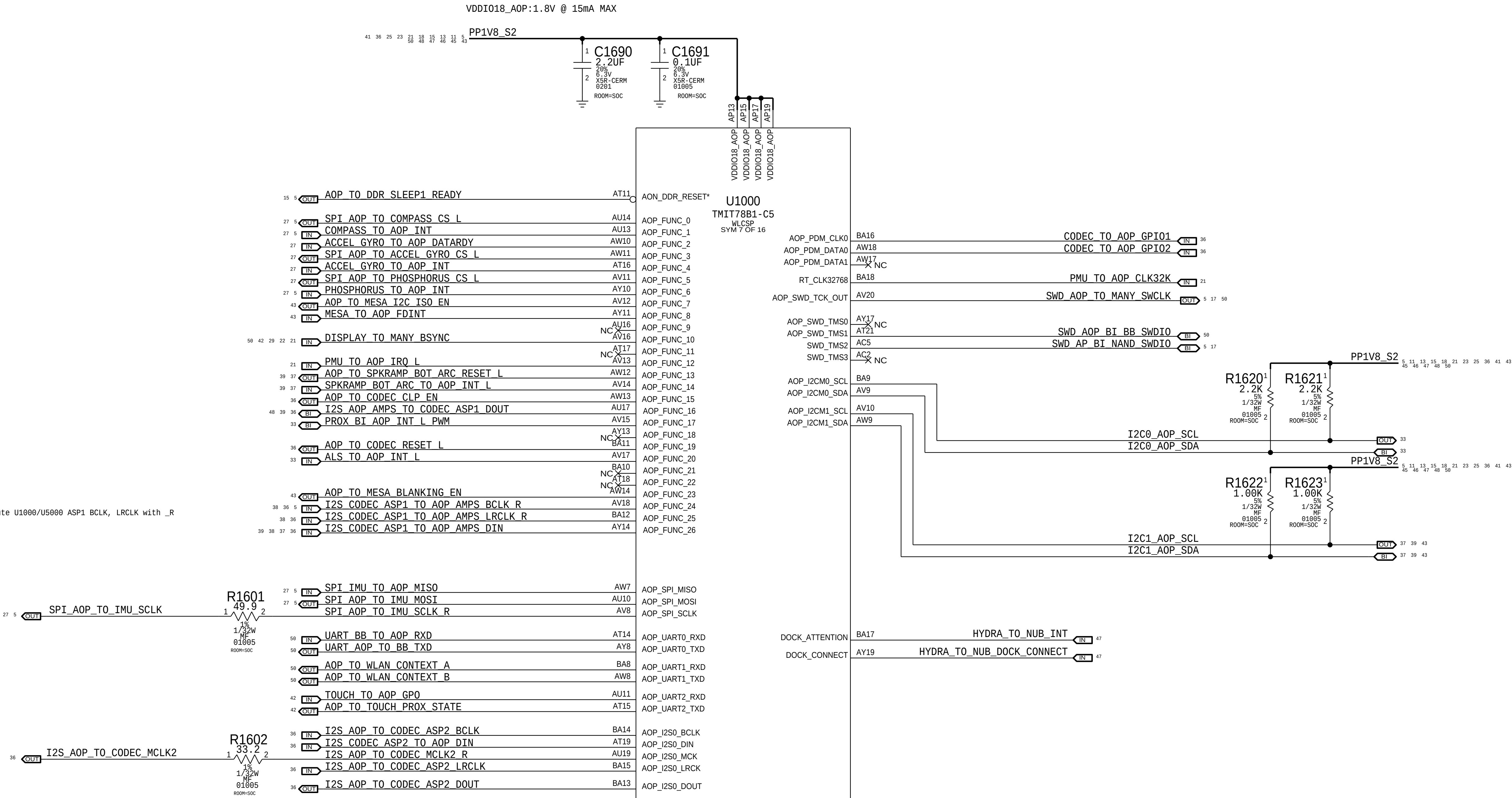
SOC - GPIO INTERFACES



PAGE TITLE		
SOC: GPIO + UART		
	DRAWING NUMBER	051-02159
	REVISION	10.0.0
	BRANCH	
	PAGE	15 OF 80
	SHEET	12 OF 81

SOC - AOP

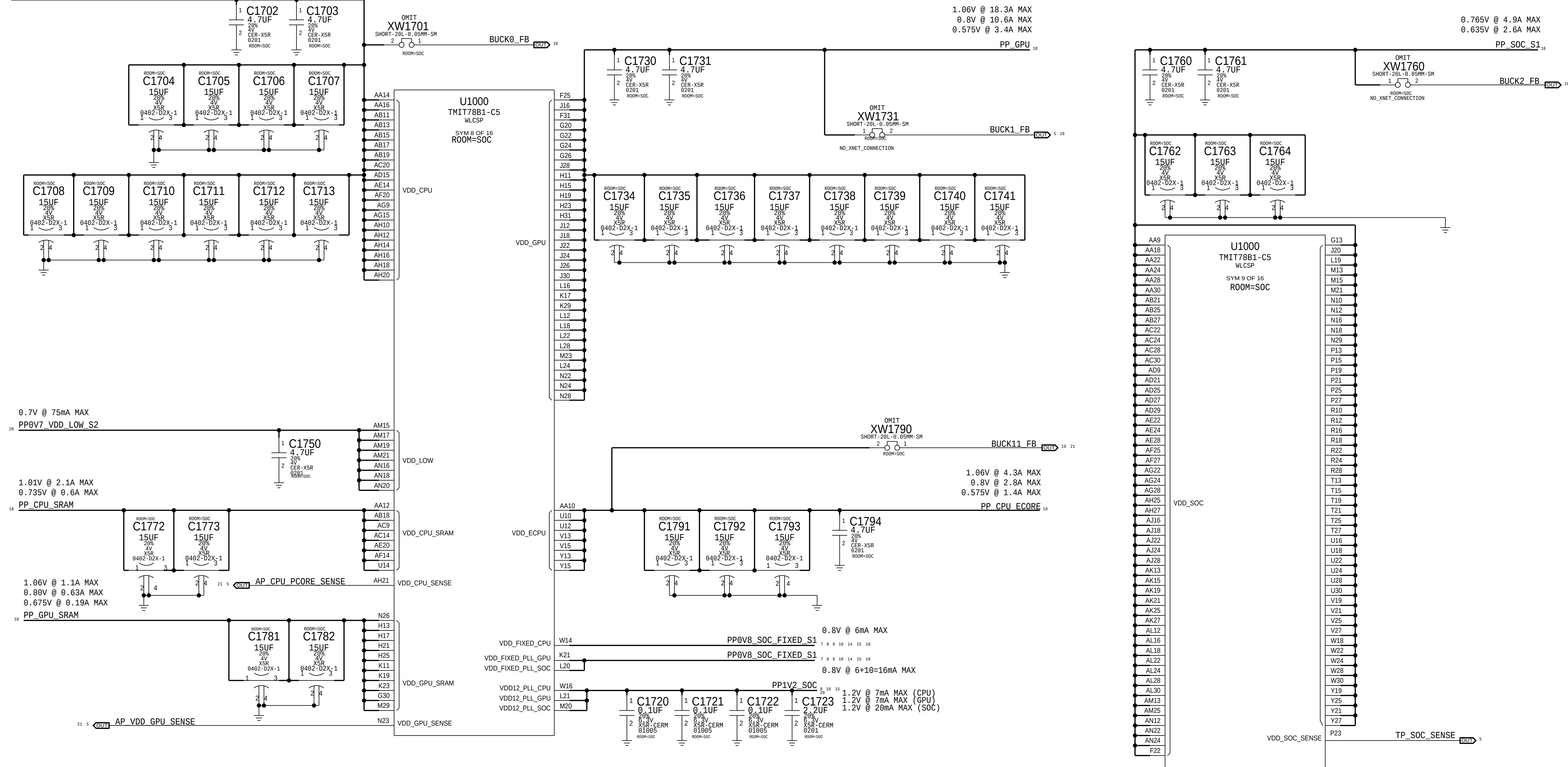
#38638490: Route U1000/U5000 ASP1 BCLK, LRCLK with _R



PAGE TITLE		
SOC: AOP		
	DRAWING NUMBER	051-02159
	REVISION	10.0.0
	BRANCH	
	PAGE	16 OF 80
	SHEET	13 OF 81
	SIZE	D

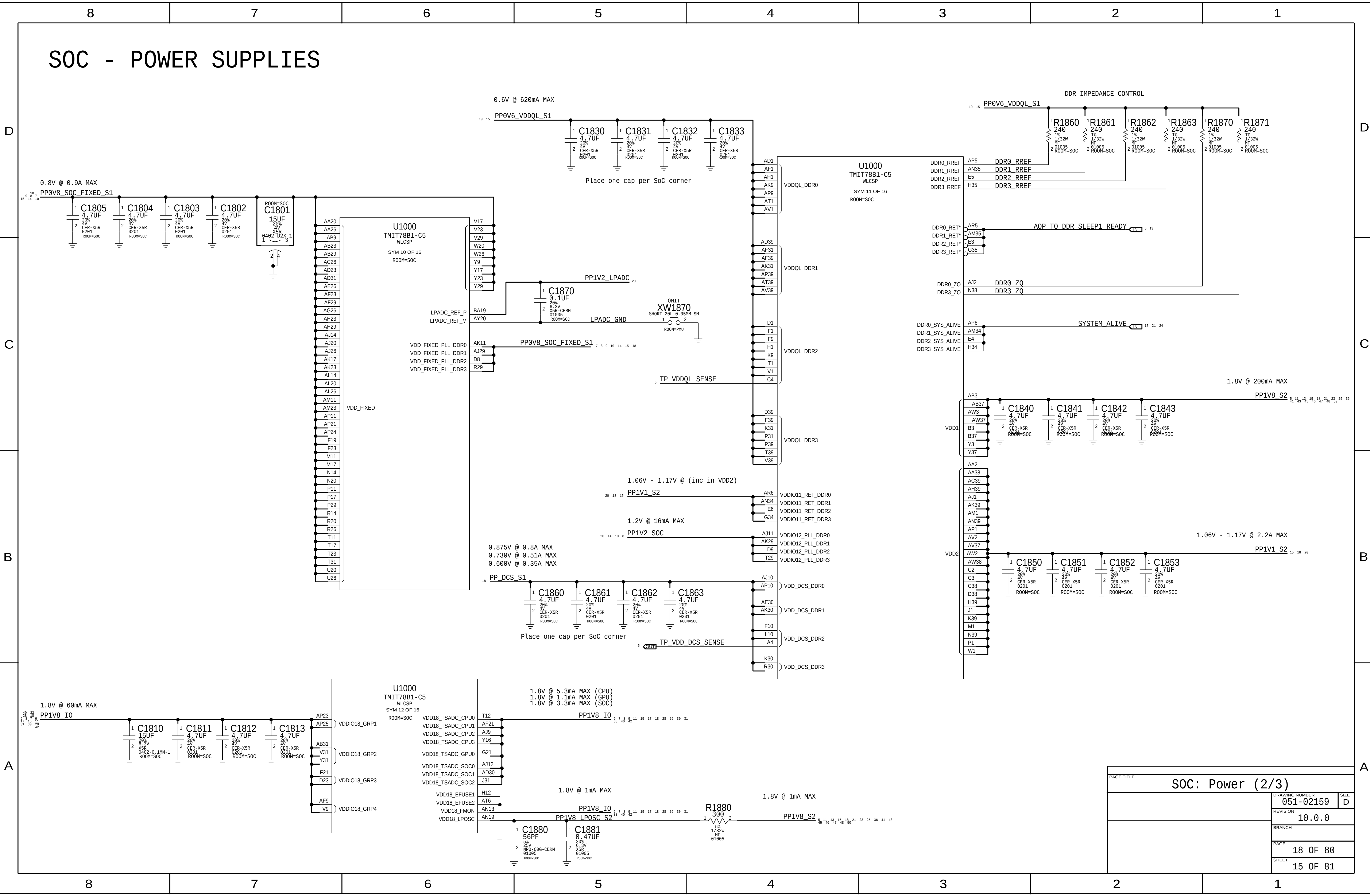
SOC - CPU, GPU & SOC RAILS

1.06V @ 11.0A MAX
0.8V @ 6A MAX
0.575V @ 2.7A MAX

18 5 PP_CPU_PCORE

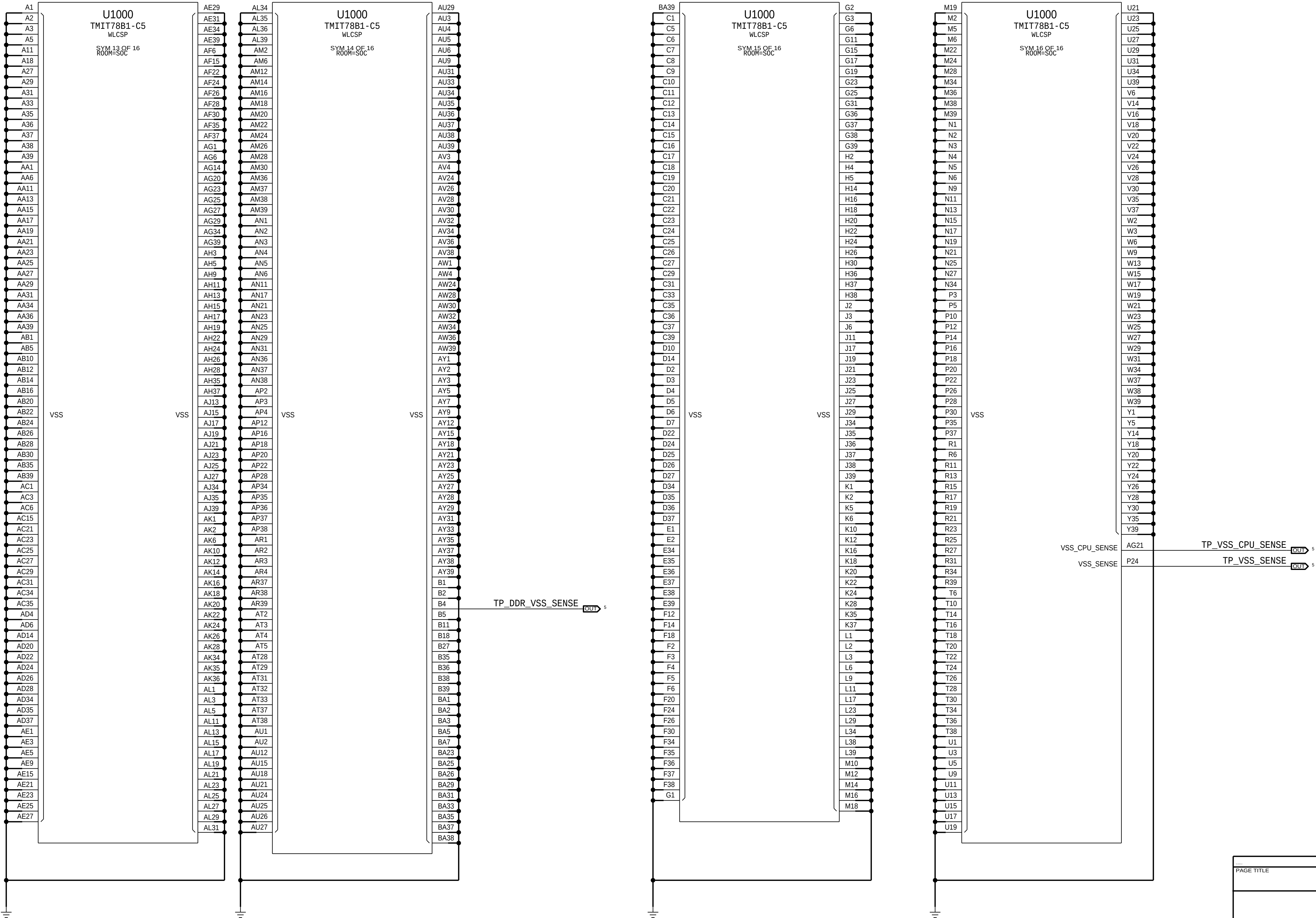
PAGE TITLE		SOC: Power (1/3)	
	DRAWING NUMBER	051-02159	SIZE
	REVISION	10.0.0	
	BRANCH		
	PAGE	17 OF 80	
		SHEET	14 OF 81

SOC - POWER SUPPLIES

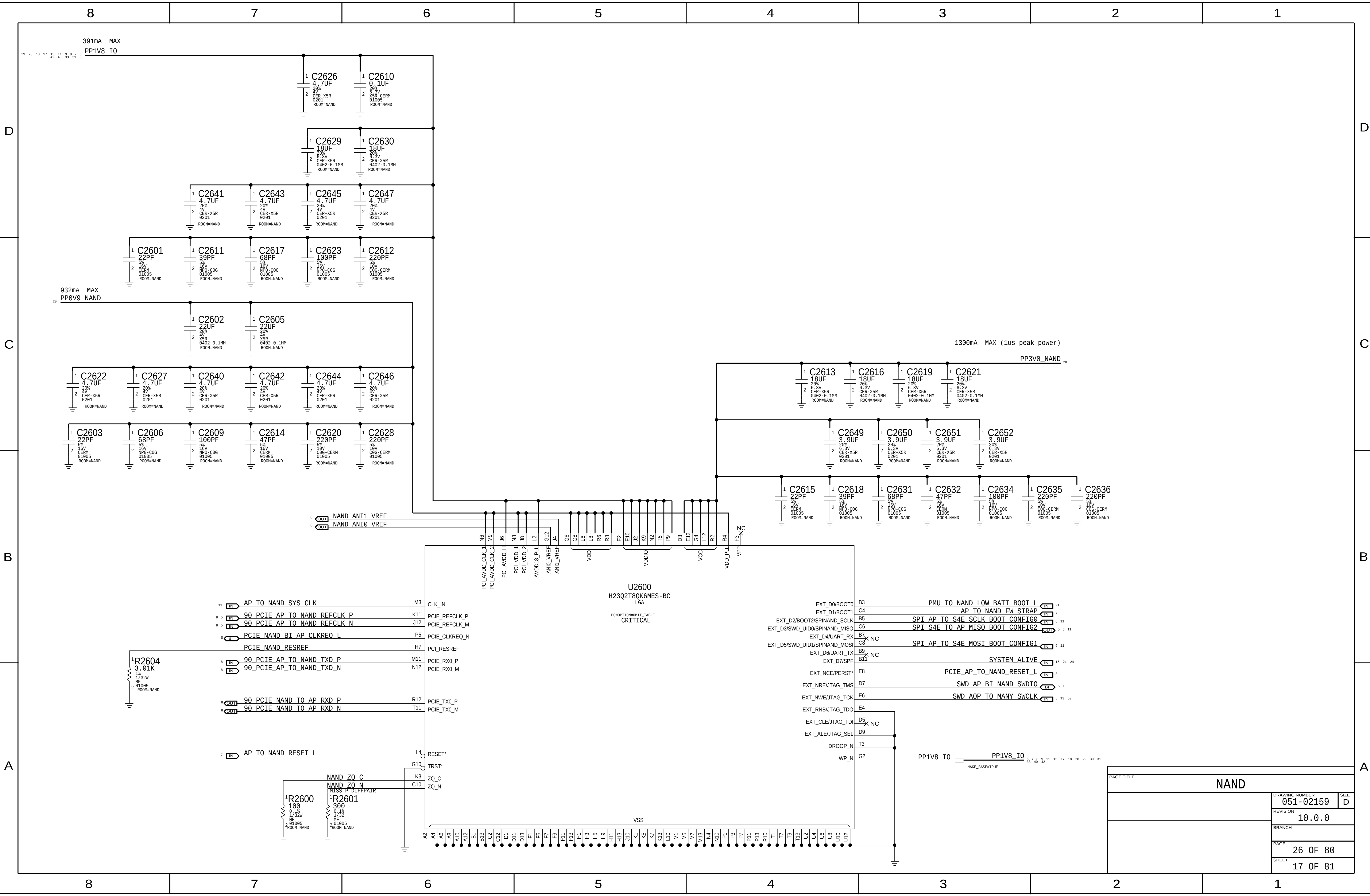


PAGE TITLE		
SOC: Power (2/3)		
DRAWING NUMBER	051-02159	SIZE
REVISION	10.0.0	D
BRANCH		
PAGE	18 OF 80	
SHEET	15 OF 81	

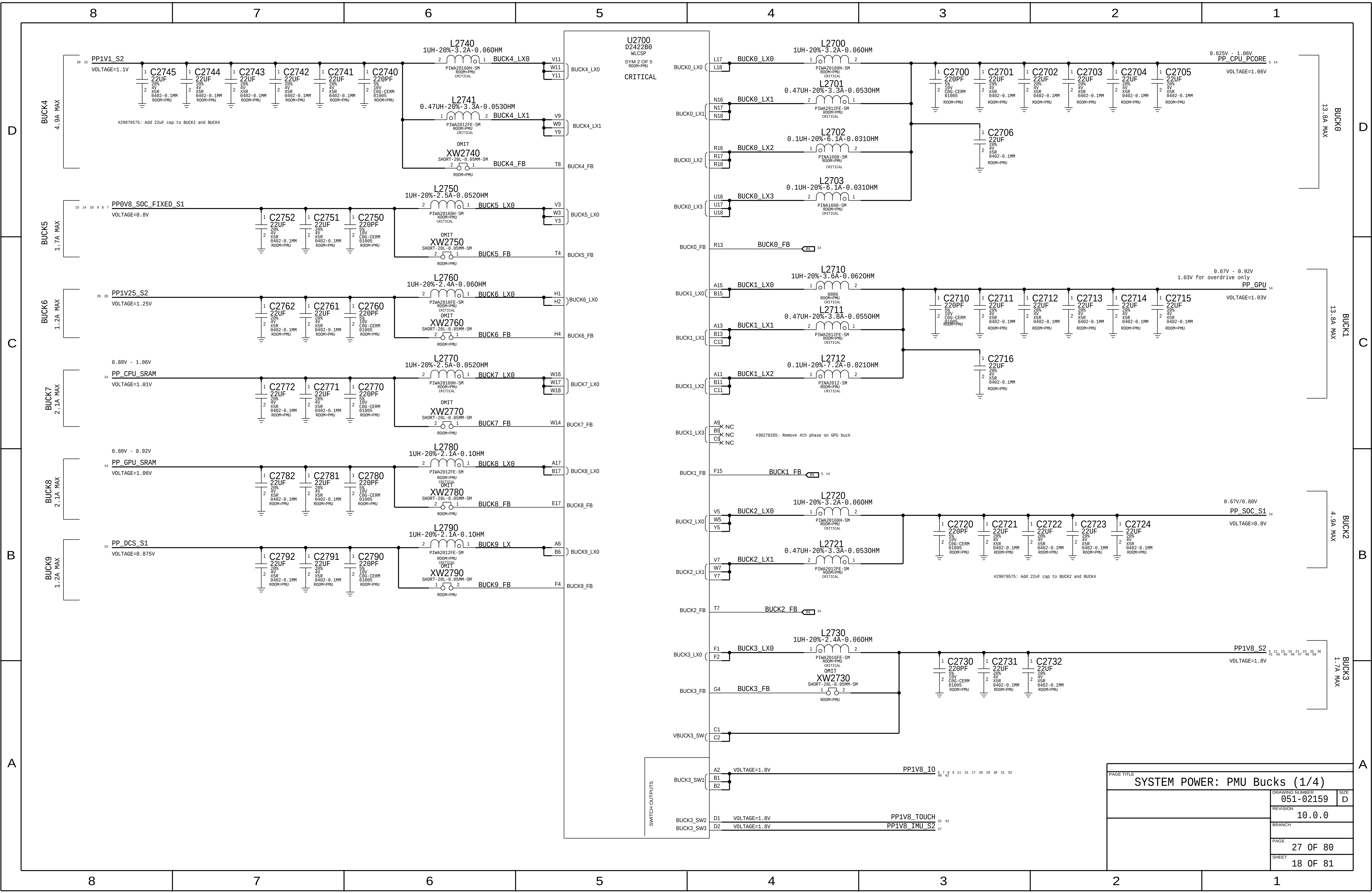
SOC - POWER SUPPLIES



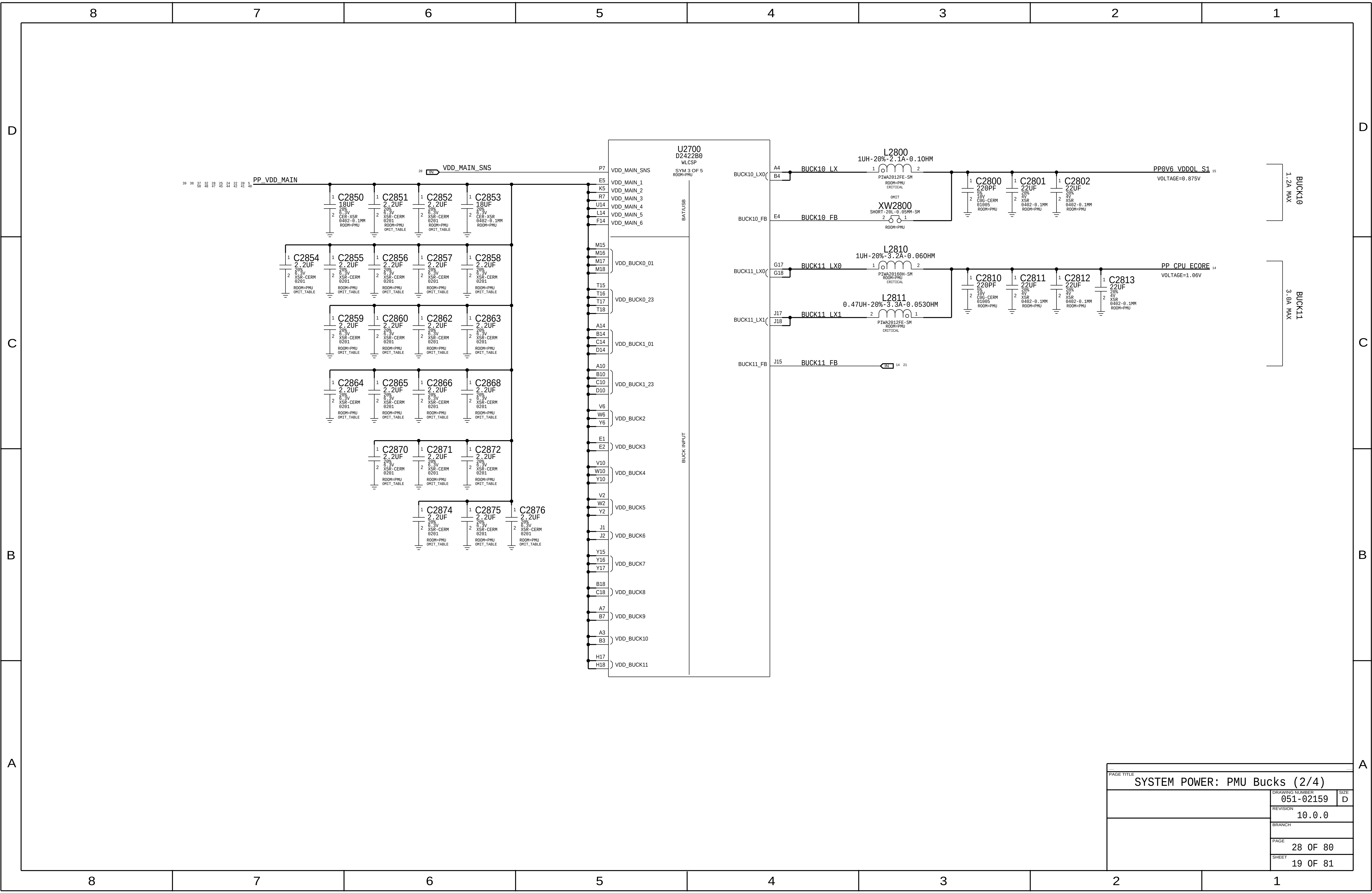
PAGE TITLE		
SOC: Power (3/3)		
	DRAWING NUMBER	051-02159
	REVISION	10.0.0
	BRANCH	
	PAGE	19 OF 80
	SHEET	16 OF 81



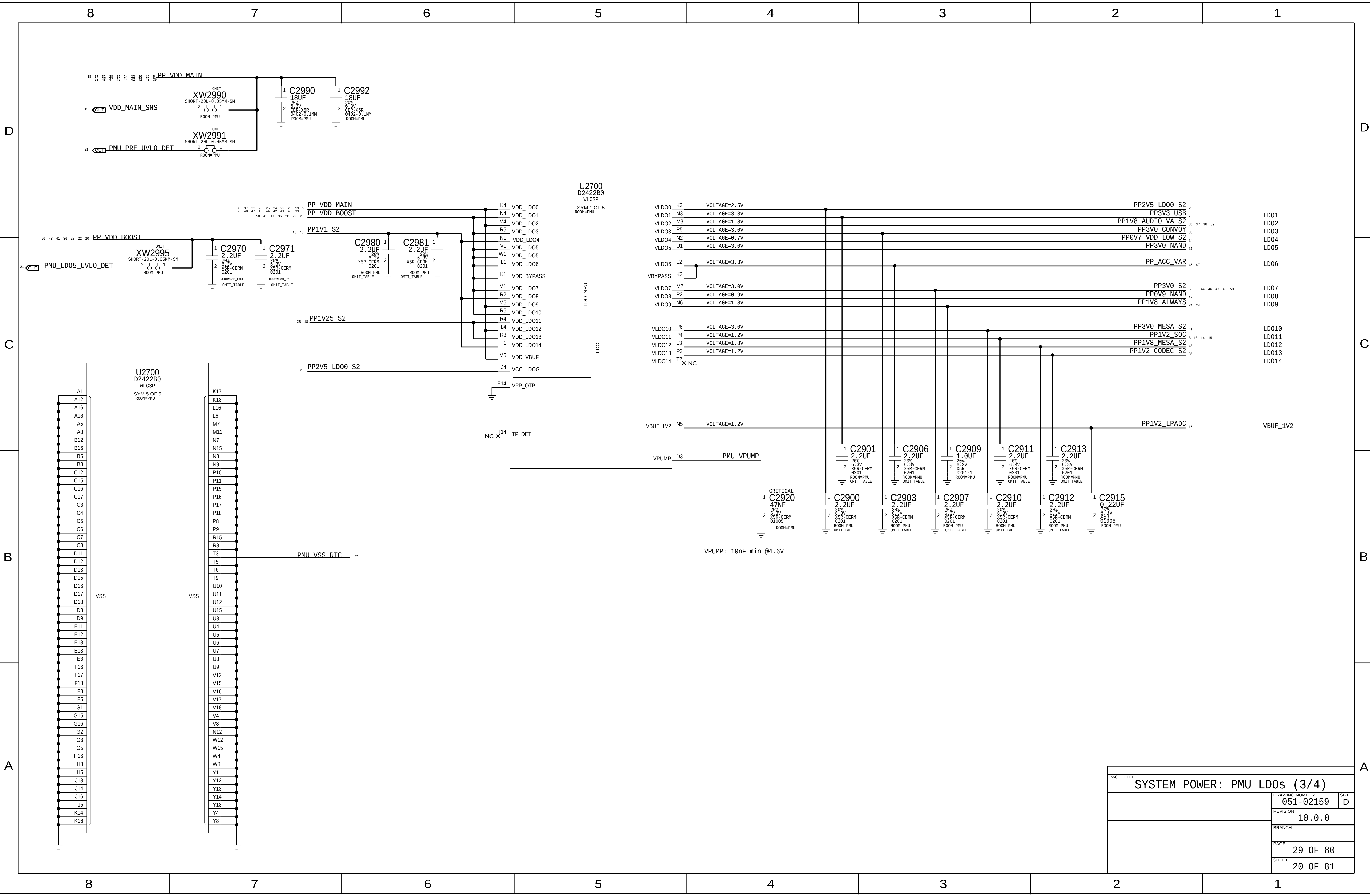
PAGE TITLE		
NAND		
DRAWING NUMBER	051-02159	SIZE
REVISION	10.0.0	D
BRANCH		
PAGE	26 OF 80	
SHEET	17 OF 81	



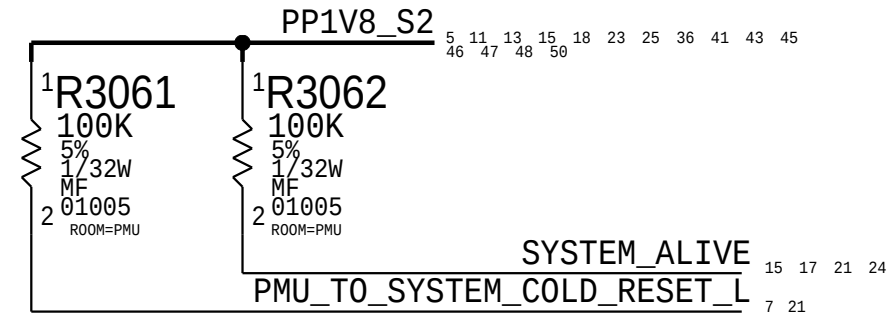
PAGE TITLE		
SYSTEM POWER: PMU Bucks (1/4)		
	DRAWING NUMBER	051-02159
	REVISION	10.0.0
	BRANCH	
	PAGE	27 OF 80
	SHEET	18 OF 81



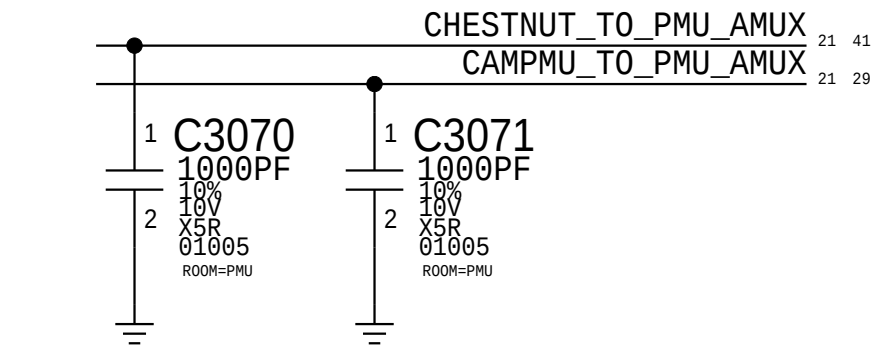
PAGE TITLE		
SYSTEM POWER: PMU Bucks (2/4)		
	DRAWING NUMBER	051-02159
	SIZE	D
	REVISION	10.0.0
	BRANCH	
	PAGE	28 OF 80
	SHEET	19 OF 81



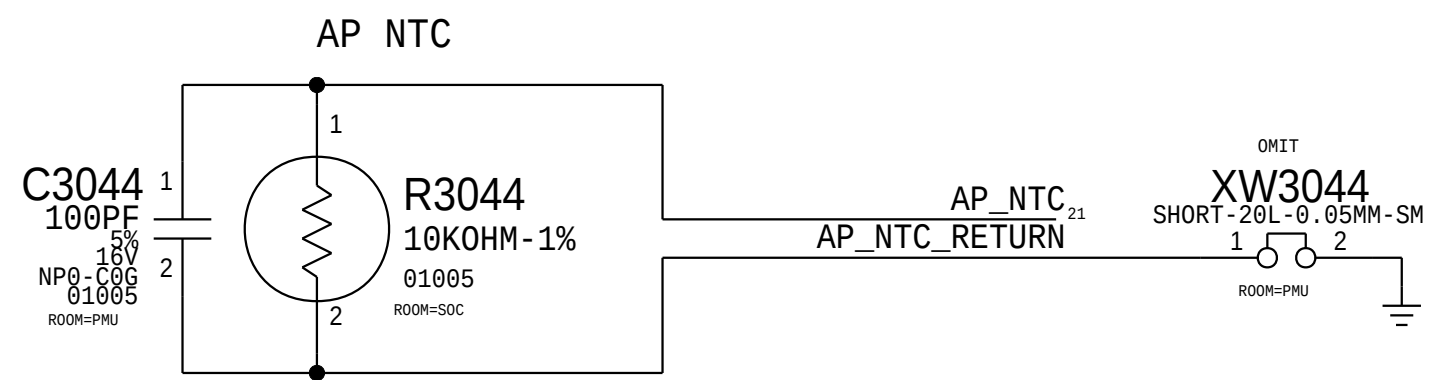
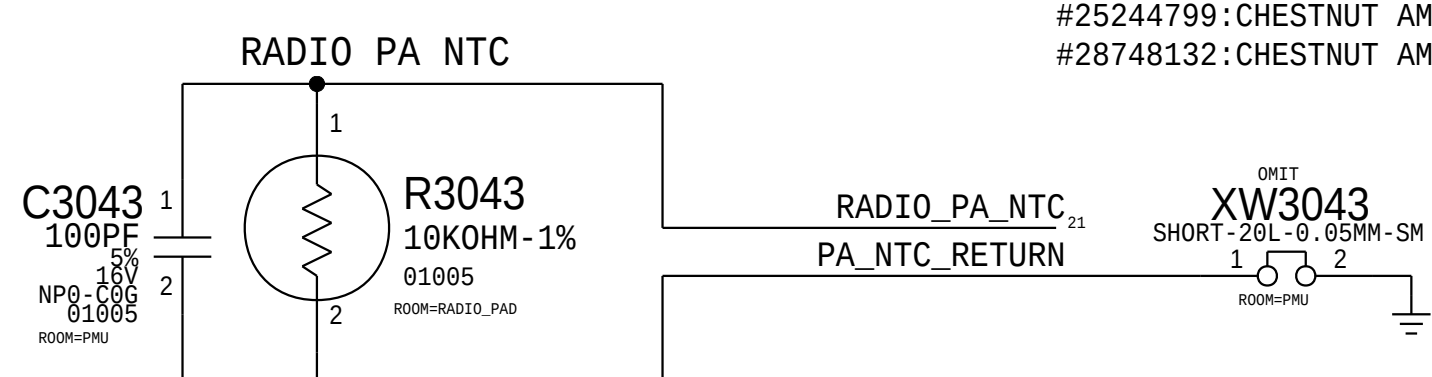
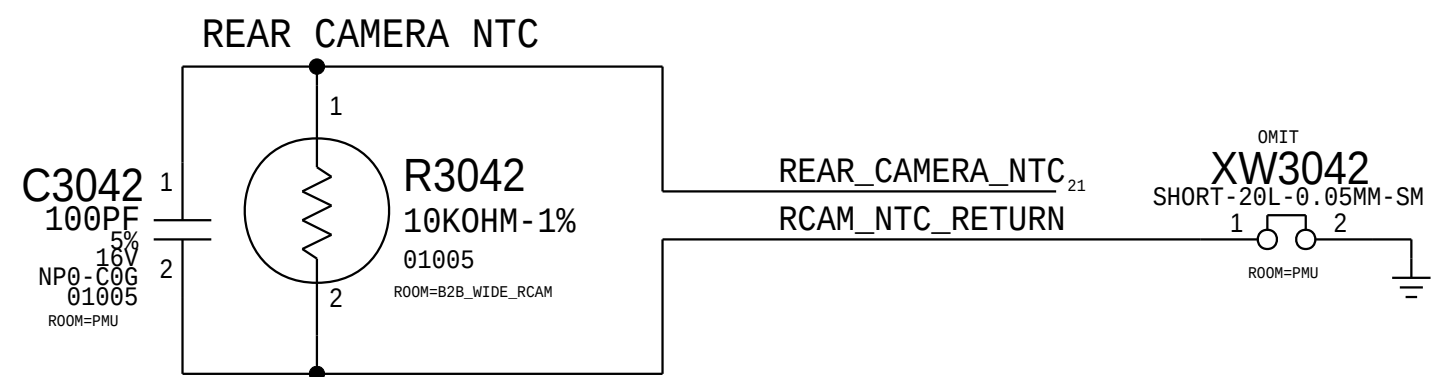
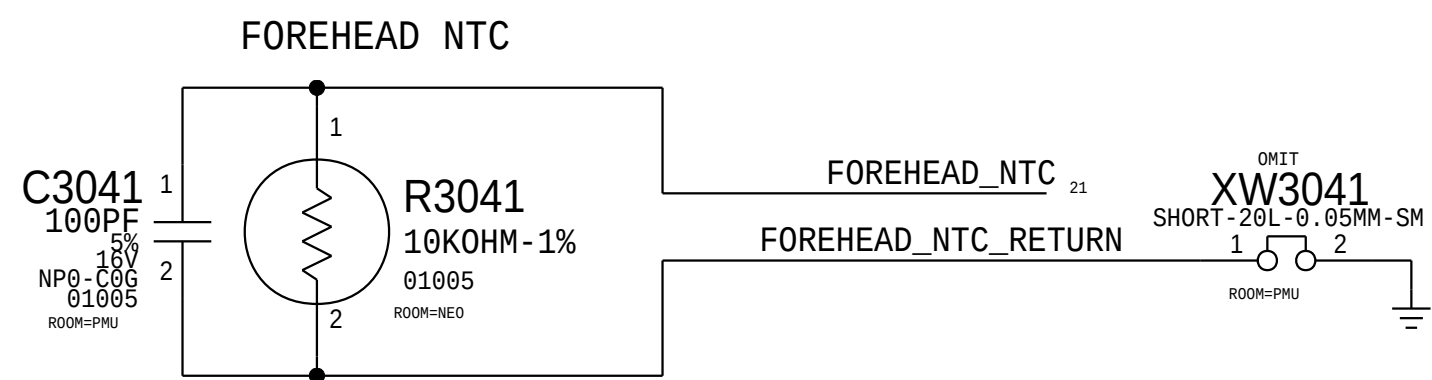
COLD_RESET & SYSTEM_ALIVE



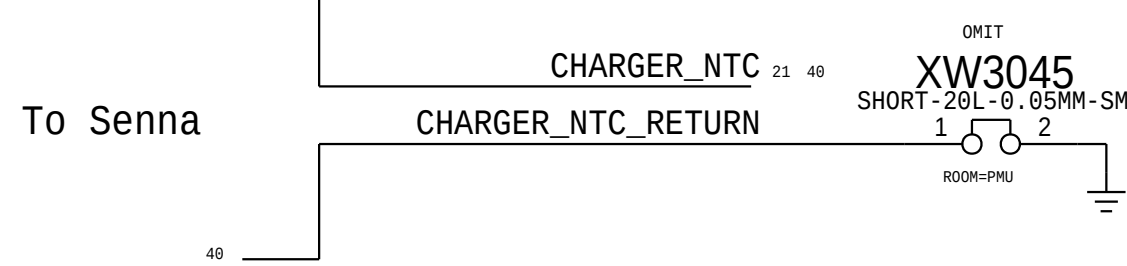
Chestnut / Ansel AMUX



NTCs



CHARGER NTC (IN SIP)



NOTE:100PF CAPS ARE THE SAMPLING CAPS FOR PMU ADC

#29404057:Damping R for ACTIVE_RDY

#29300440:Connect PMU crash to SoC

#27852679:No SDA R, PoR is SPMI

#25244799:200K pull-down

#25244799:CHESTNUT AMUX AND PWR_EN ON SAME CHANNEL
#28748132:CHESTNUT AMUX AND PWR_EN SWAP BETWEEN A6/B6

#25244799:0TP-AJ Connects XTAL1 to GND

#29312542:REPLACE PMU XTAL WITH XO

AP_TO_PMU_WDOG_RESET
HYDRA_TO_PMU_HOST_RESET
AP_TO_PMU_SOCHOT_L
PMU_TO_SYSTEM_COLD_RESET_L

PMU_TO_AP_HYDRA_ACTIVE_READY
PMU_TO_AOP_CLK32K
PMU_TO_WLAN_CLK32K
SYSTEM_ALIVE
DISPLAY_TO_MANY_BSYN
PMU_TO_AOP_IRQ_L
I2C0_AP_SCL
I2C0_AP_SDA
SPMI_PMGR_TO_PMU_SCLK
SPMI_PMU_BI_PMGR_SDATA

AP_TO_PMU_AMUX_OUT
PMU_ADC_IN
CAMPMU_TO_PMU_AMUX
AP_TO_PMU_AMUX_SYNC
HYDRA_TO_PMU_USB_BRICK_ID
DISPLAY_TO_CHESTNUT_PWR_EN
ACC_BUCK_TO_PMU_AMUX

IKTARA_TO_PMU_OVP
AP_TO_PMU_TEST_CLKOUT
PMU_TO_WLAN_CLK32K
CHESTNUT_TO_PMU_AMUX
PMU_AMUX_BY
FOREHEAD_NTC
REAR_CAMERA_NTC
RADIO_PA_NTC
AP_NTC
CHARGER_NTC
PMU_TCAL

PMU_AMUX_AY
PMU_AMUX_BY
FOREHEAD_NTC
REAR_CAMERA_NTC
RADIO_PA_NTC
AP_NTC
CHARGER_NTC
PMU_TCAL

PMU_XTAL2
PMU_VDD_RTC
PMU_VDD_REF

PMU_VDD_TCX0
PMU_VSS_RTC

RESET_IN1
RESET_IN2
RESET_IN3
RESET*
SHDN

SLEEP_32K
OUT_32K
SYS_ALIVE
FORCE_SYNC
DBL_CLICK_DET
IRQ*

AMUX_A0
AMUX_A1
AMUX_A2
AMUX_A3
AMUX_A4
AMUX_A5
AMUX_A6
AMUX_A7
AMUX_AY
AMUX_B0
AMUX_B1
AMUX_B2
AMUX_B3
AMUX_B4
AMUX_B5
AMUX_B6
AMUX_B7
AMUX_BY

TDEV1
TDEV2
TDEV3
TDEV4
TDEV5
TCAL

XTAL1
XTAL2

VDD_RTC
VDD_REF

FAULT_OUT*

PMU_IREF
PMU_VREF

PMU_TO_AP_PRE_UVLO_L
PMU_TO_AP_THROTTLE_PCORE_L
PMU_TO_AP_THROTTLE_GPU0_L
PMU_TO_AP_THROTTLE_ECORE_L
AP_CPU_PCORE_SENSE
AP_VDD_GPU_SENSE
BUCK11_FB
PMU_PRE_UVLO_DET
PMU_LD05_UVLO_DET

HYDRA_TO_PMU_USB_BRICK_ID
PMU_ADC_IN
BUTTON_VOL_DOWN_L
BUTTON_POWER_KEY_L
BUTTON_VOL_UP_L
BUTTON_RINGER_A
PMU_TO_AP_BUTTON_VOL_DOWN_L
PMU_TO_AP_BUTTON_POWER_KEY_L

PMU_TO_NFC_EN
PMU_TO_AP_THROTTLE_GPU1_L
TIGRIS_TO_PMU_INT_L
WLAN_TO_PMU_HOST_WAKE
NFC_TO_PMU_HOST_WAKE
PMU_TO_GNSS_EN
PMU_TO_BT_REG_ON
BT_TO_PMU_HOST_WAKE
CODEC_TO_PMU_WAKE_L

PMU_TO_WLAN_REG_ON
PMU_TO_BB_USB_VBUS_DETECT
PMU_TO_AP_FORCE_DFU_R
PMU_TO_CCG2_RESET_L
PMU_TO_BBPMU_RESET_R_L
PMU_TO_NAND_LOW_BATT_BOOT_L
BB_TO_PMU_PCTE_HOST_WAKE_L
PMU_TO_IKTARA_EN_EXT_1P8V
PMU_TO_BOOST_EN
PMU_TO_DISPLAY_PANICB
I2C0_AP_SCL

PMU_TO_IKTARA_RESET_L

PMU_HYDRA_TO_AP_FORCE_DFU
PMU_TO_BBPMU_RESET_L

PMU_TO_BT_REG_ON
BT_TO_PMU_HOST_WAKE
CODEC_TO_PMU_WAKE_L

PMU_TO_WLAN_REG_ON
PMU_TO_BB_USB_VBUS_DETECT
PMU_TO_AP_FORCE_DFU_R
PMU_TO_CCG2_RESET_L
PMU_TO_BBPMU_RESET_R_L
PMU_TO_NAND_LOW_BATT_BOOT_L
BB_TO_PMU_PCTE_HOST_WAKE_L
PMU_TO_IKTARA_EN_EXT_1P8V
PMU_TO_BOOST_EN
PMU_TO_DISPLAY_PANICB
I2C0_AP_SCL

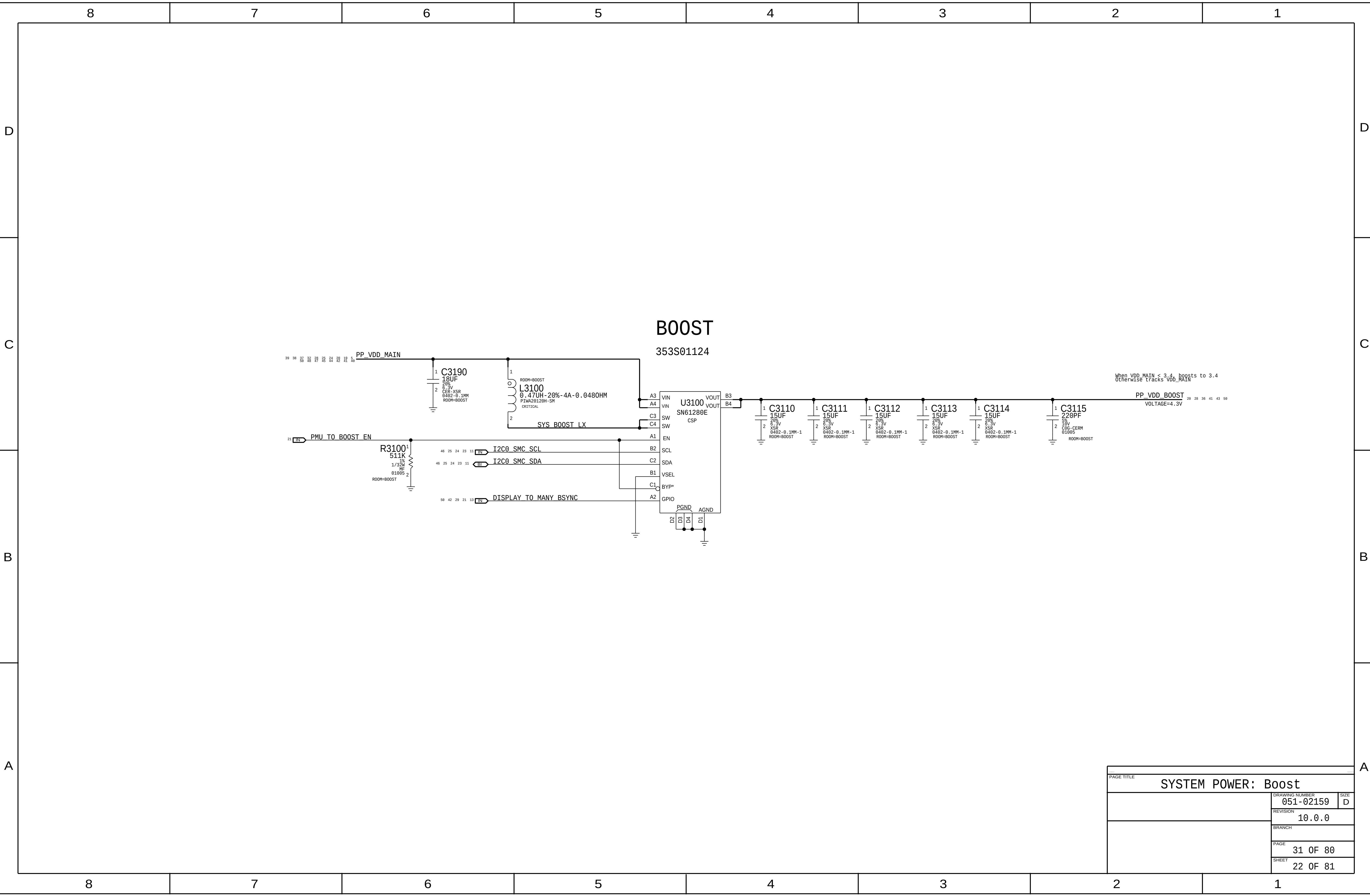
PMU_TO_IKTARA_RESET_L

PMU_TO_WLAN_REG_ON
PMU_TO_BB_USB_VBUS_DETECT
PMU_TO_AP_FORCE_DFU_R
PMU_TO_CCG2_RESET_L
PMU_TO_BBPMU_RESET_R_L
PMU_TO_NAND_LOW_BATT_BOOT_L
BB_TO_PMU_PCTE_HOST_WAKE_L
PMU_TO_IKTARA_EN_EXT_1P8V
PMU_TO_BOOST_EN
PMU_TO_DISPLAY_PANICB
I2C0_AP_SCL

PMU_TO_IKTARA_RESET_L

PMU_TO_IKTARA_RESET_L

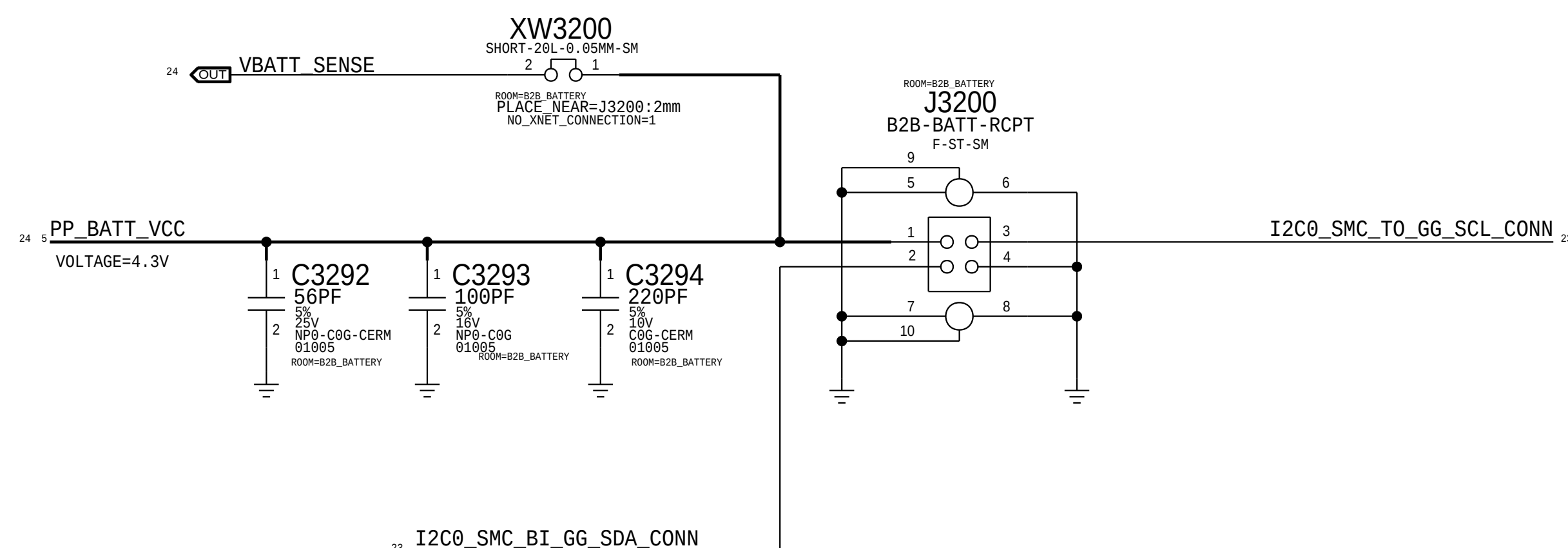
PAGE TITLE		
SYSTEM POWER: PMU (4/4)		
DRAWING NUMBER	051-02159	SIZE
REVISION	10.0.0	D
BRANCH		
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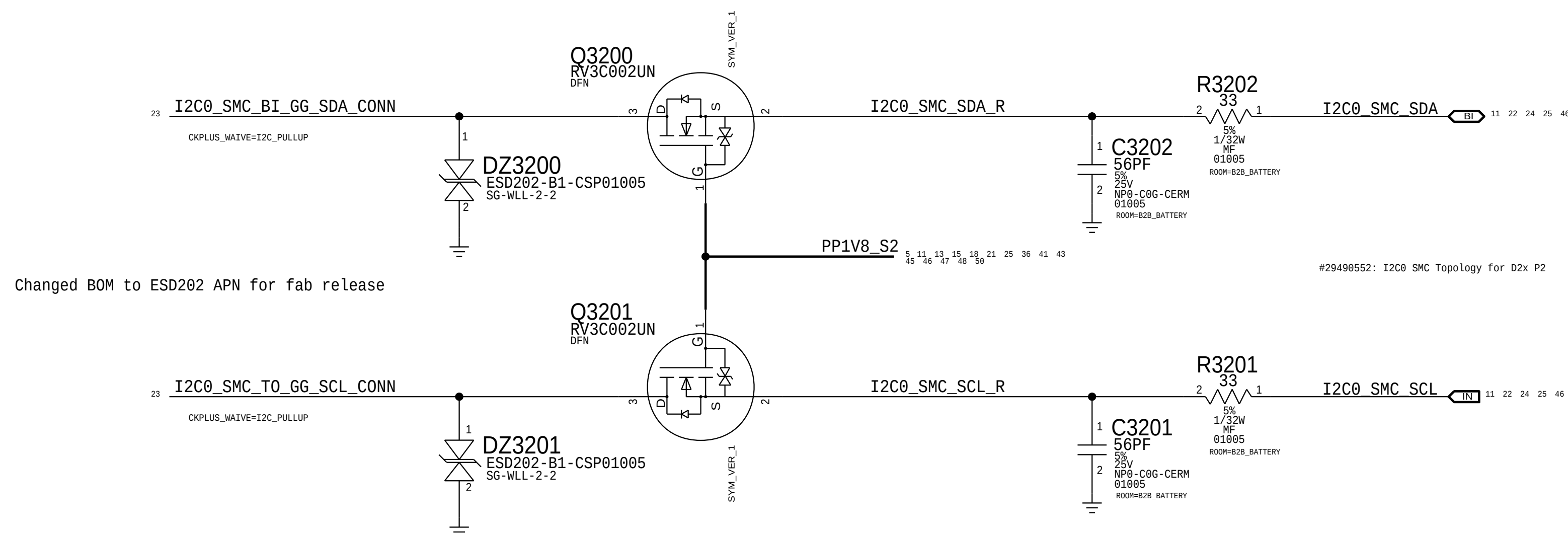
PAGE TITLE		
SYSTEM POWER: Boost		
	DRAWING NUMBER	051-02159
	REVISION	10.0.0
	BRANCH	
	PAGE	31 OF 80
	SHEET	22 OF 81

Battery Connector

THIS ONE ON MLB ---> 516S00232

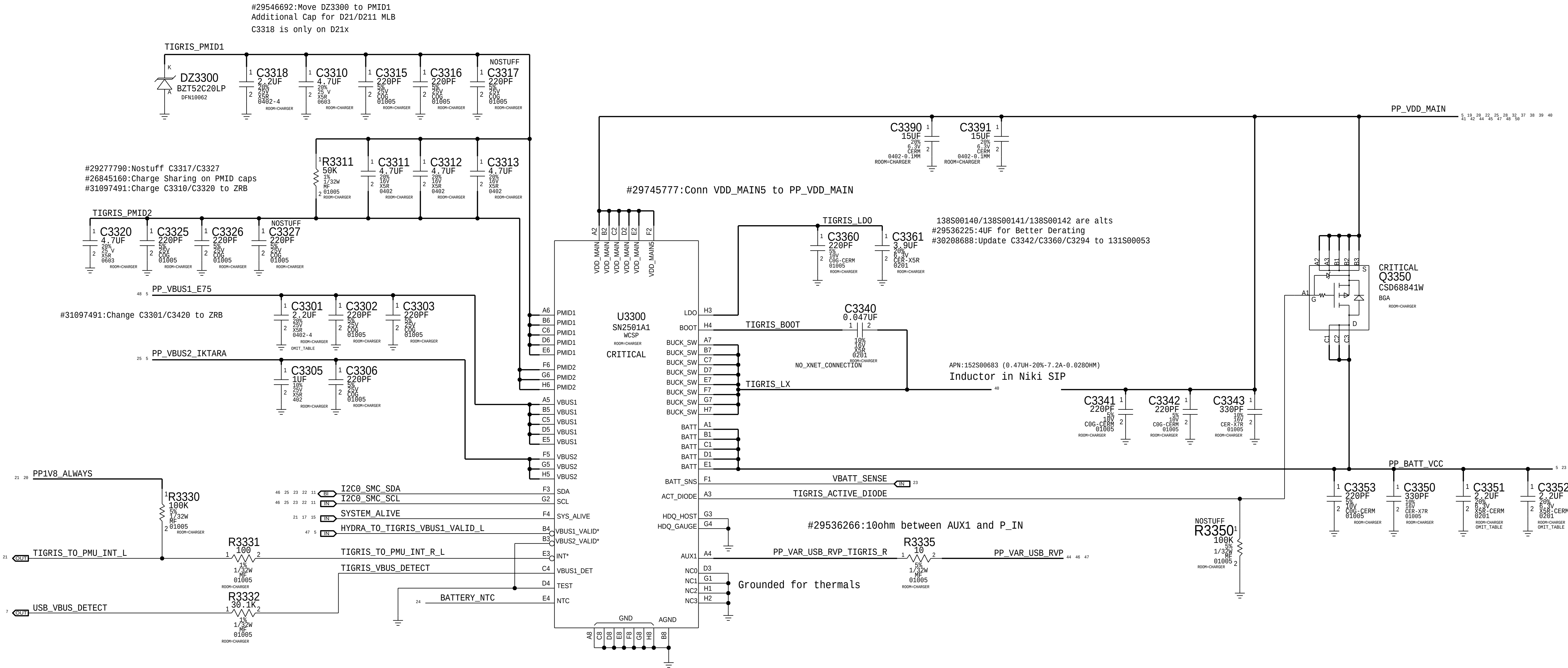


```
#32515053:Add DZ3200/DZ3201 on I2C CONN with 377S00001
#31734657:Replace battery I2C FETs with 376S00134
Gas gauge I2C level translator
```



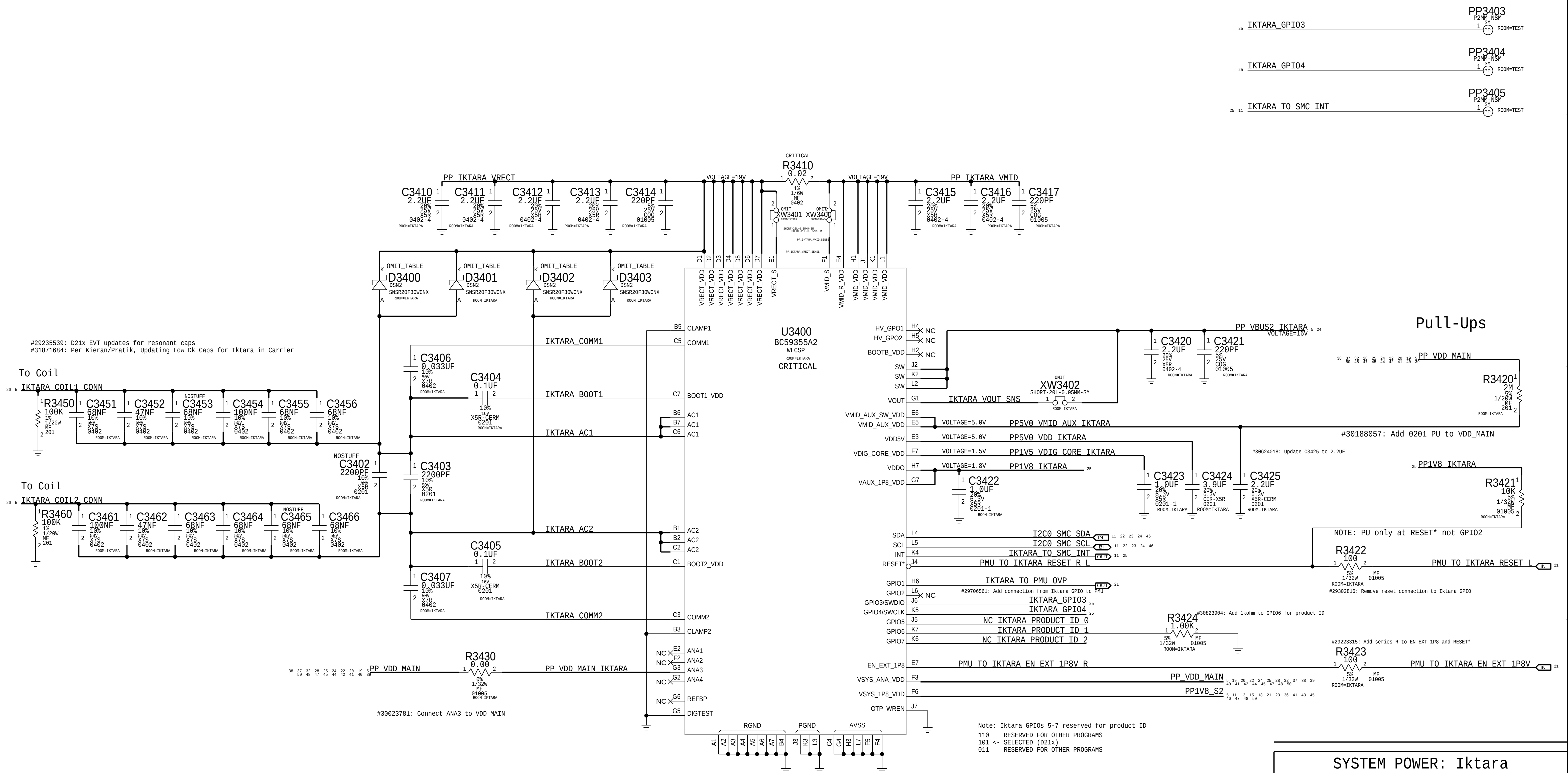
Tigris 2

APN:343S00170



Iktara

Iktara Debug



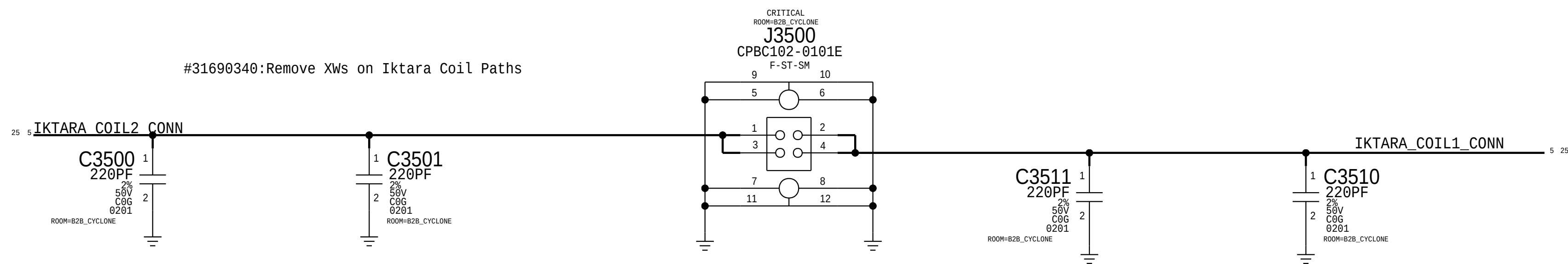
SYSTEM POWER: Iktara

D

D

Cyclone connector

THIS ONE ON MLB ---> 516S00311



B

B

A

A

SYSTEM POWER: B2B Cyclone

D

C



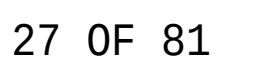
D

C

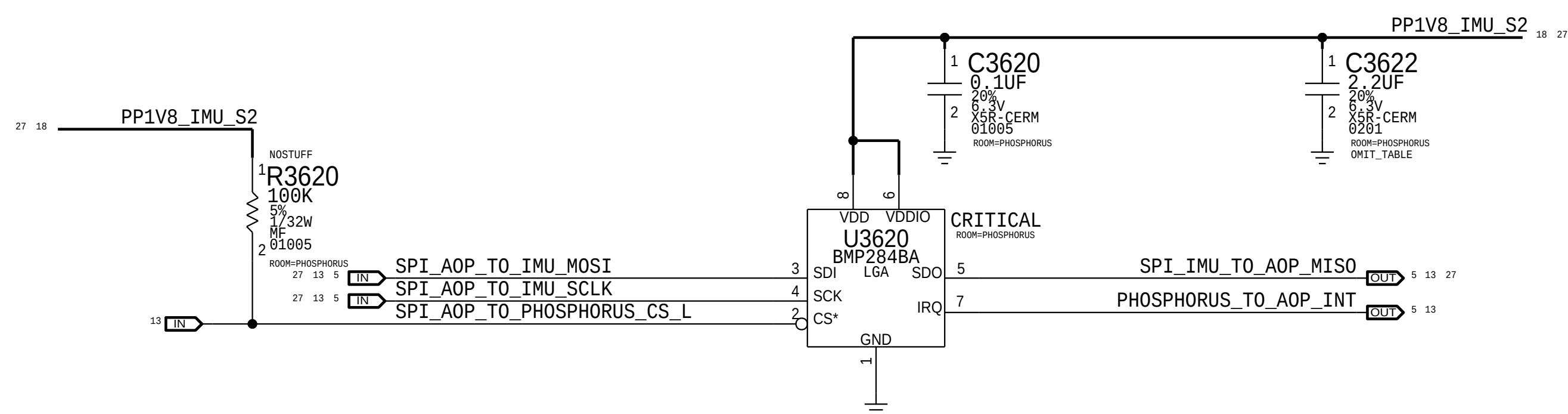
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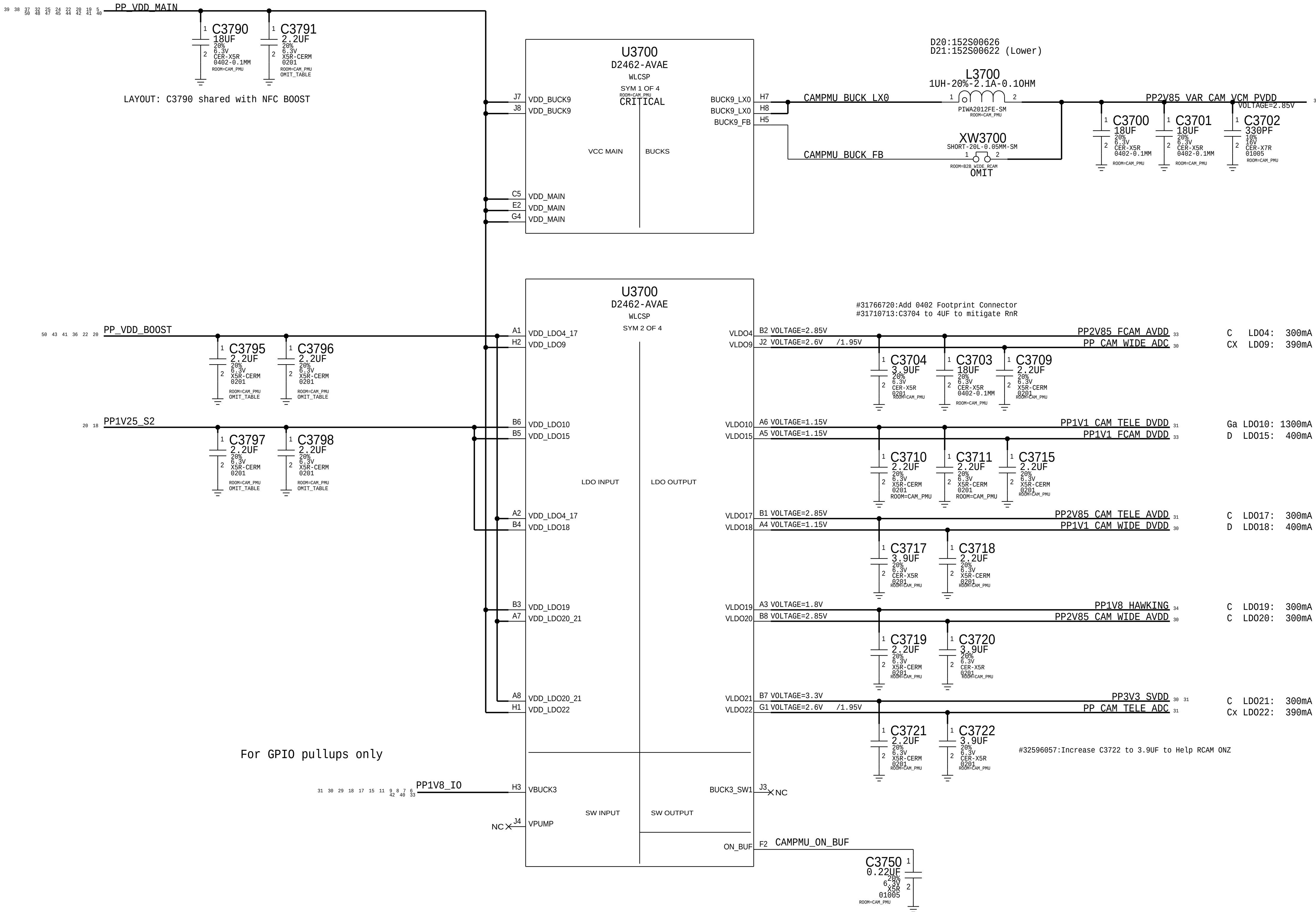
A



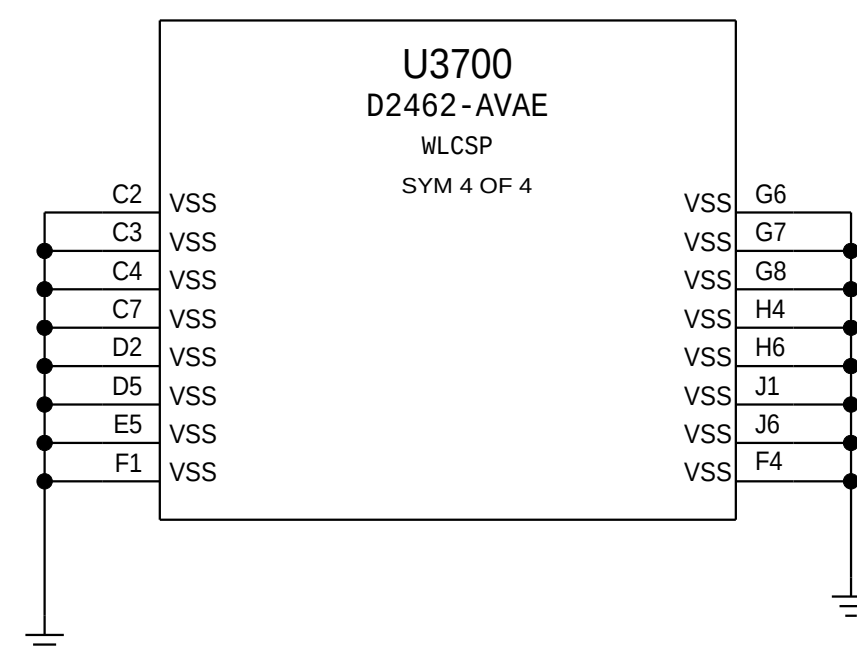
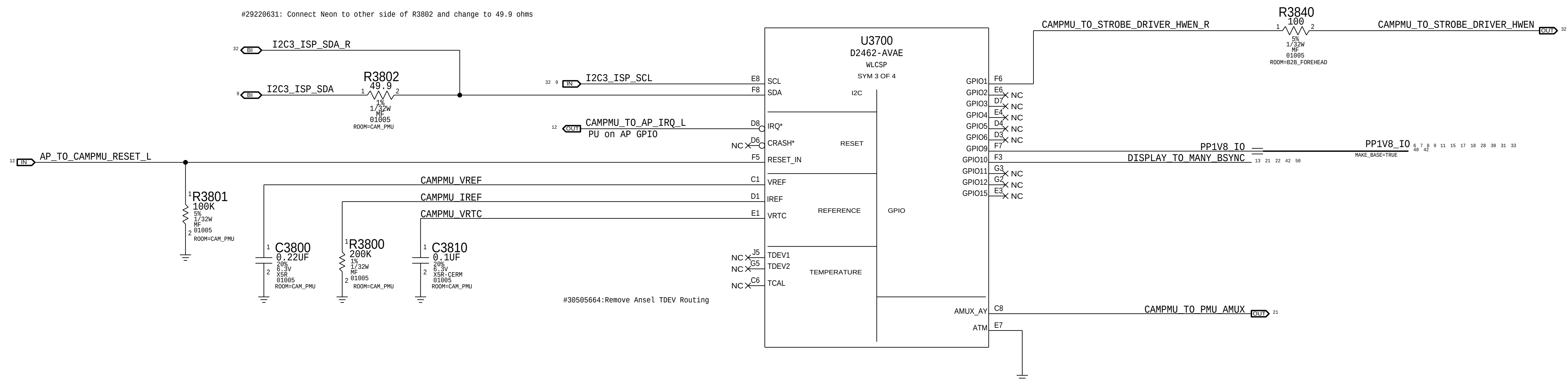
APN: 338S00334



Camera PMU

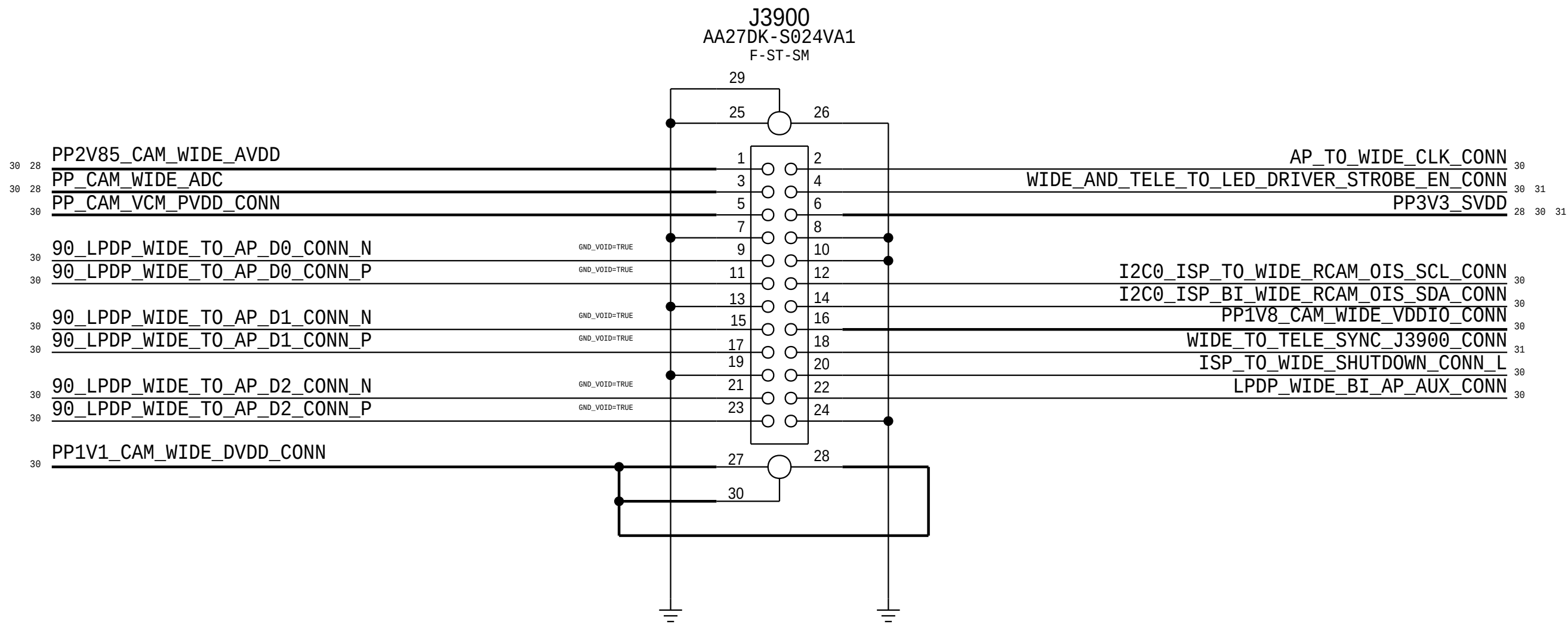


CAMERA: PMU (1/2)

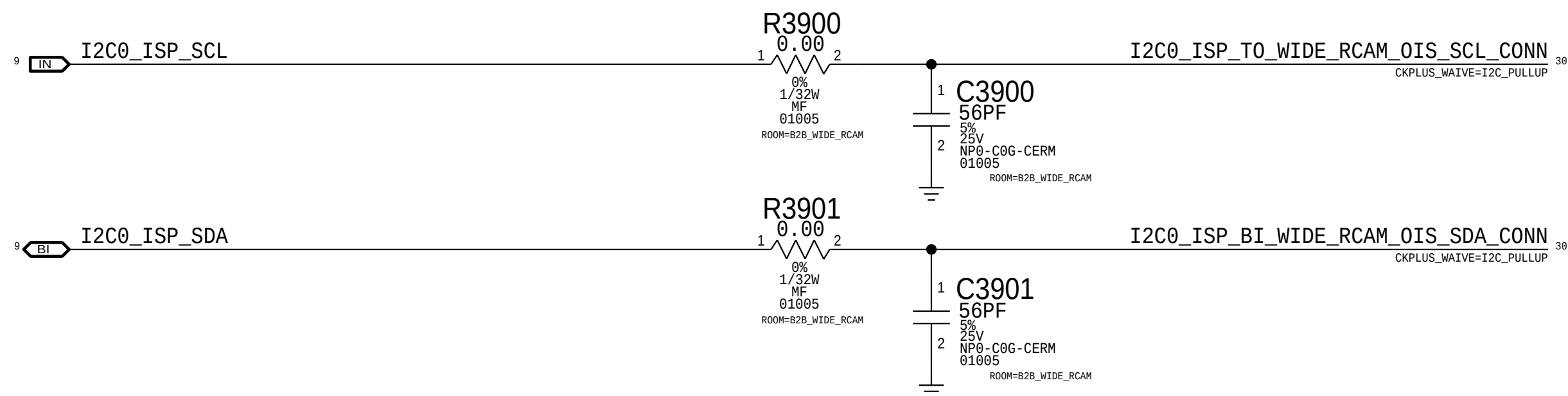


Wide Camera Connector

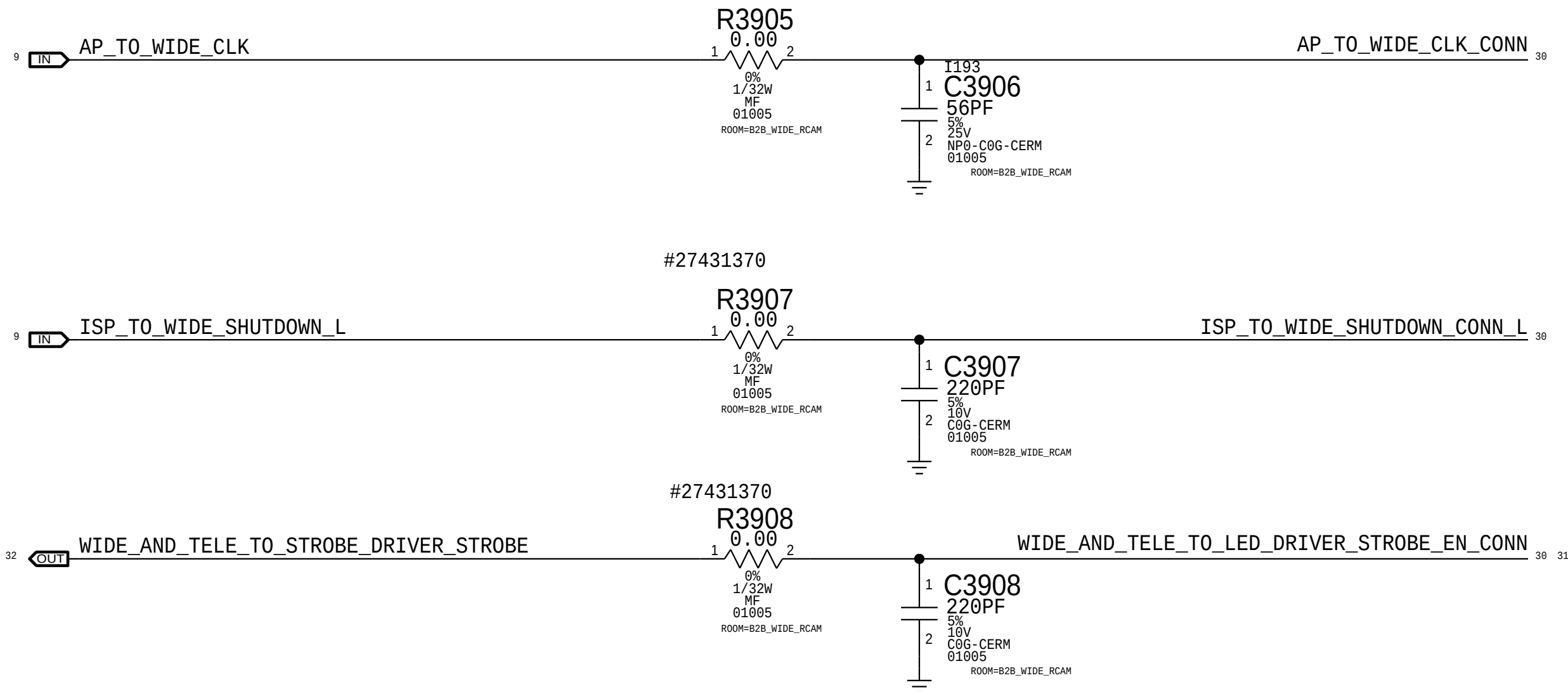
THIS ONE ----> 516S00310/P2 RCPT (USED ON MLB)
516S00309/P2 PLUG (USED ON FLEX)



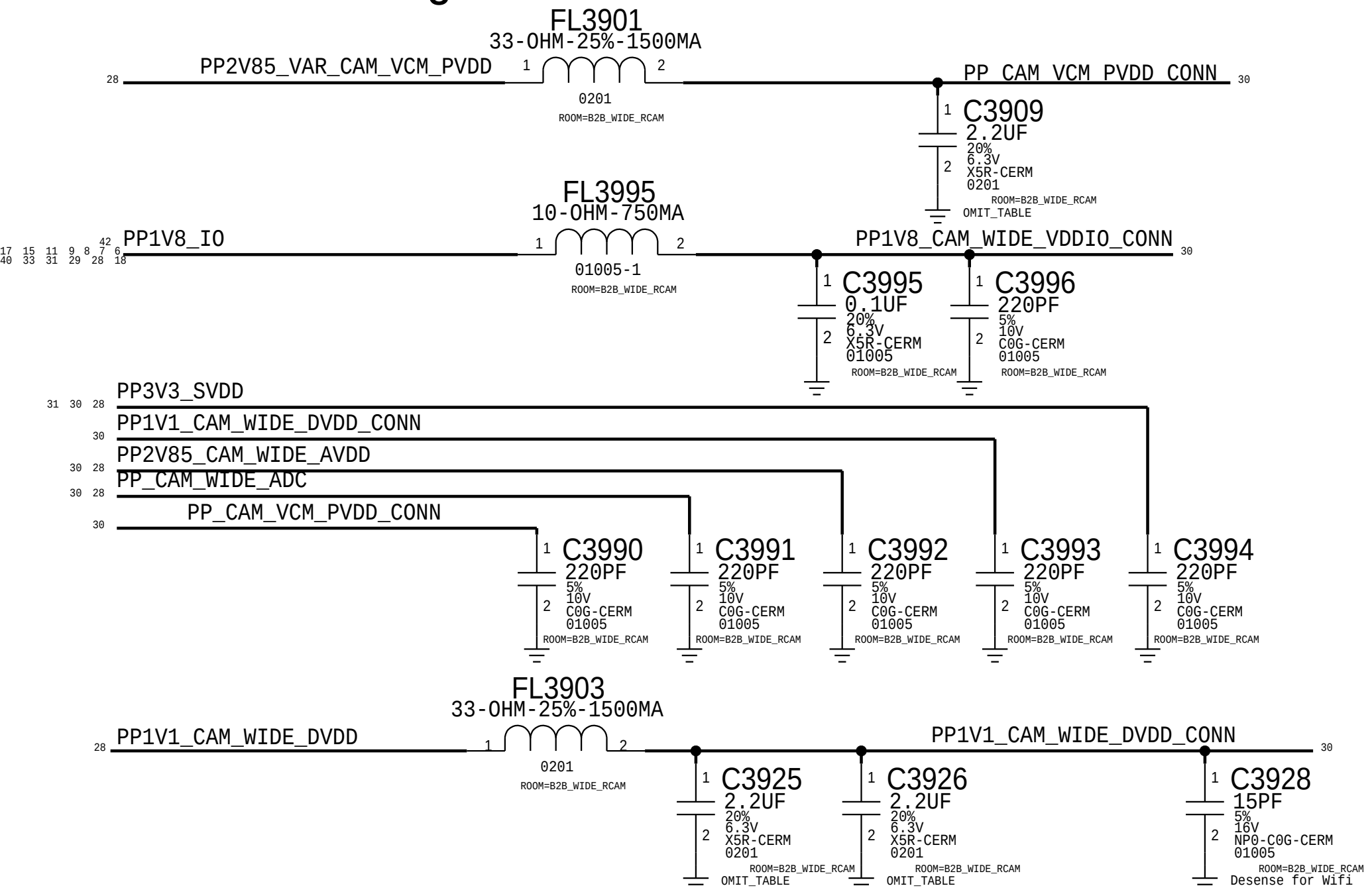
ISP I2C



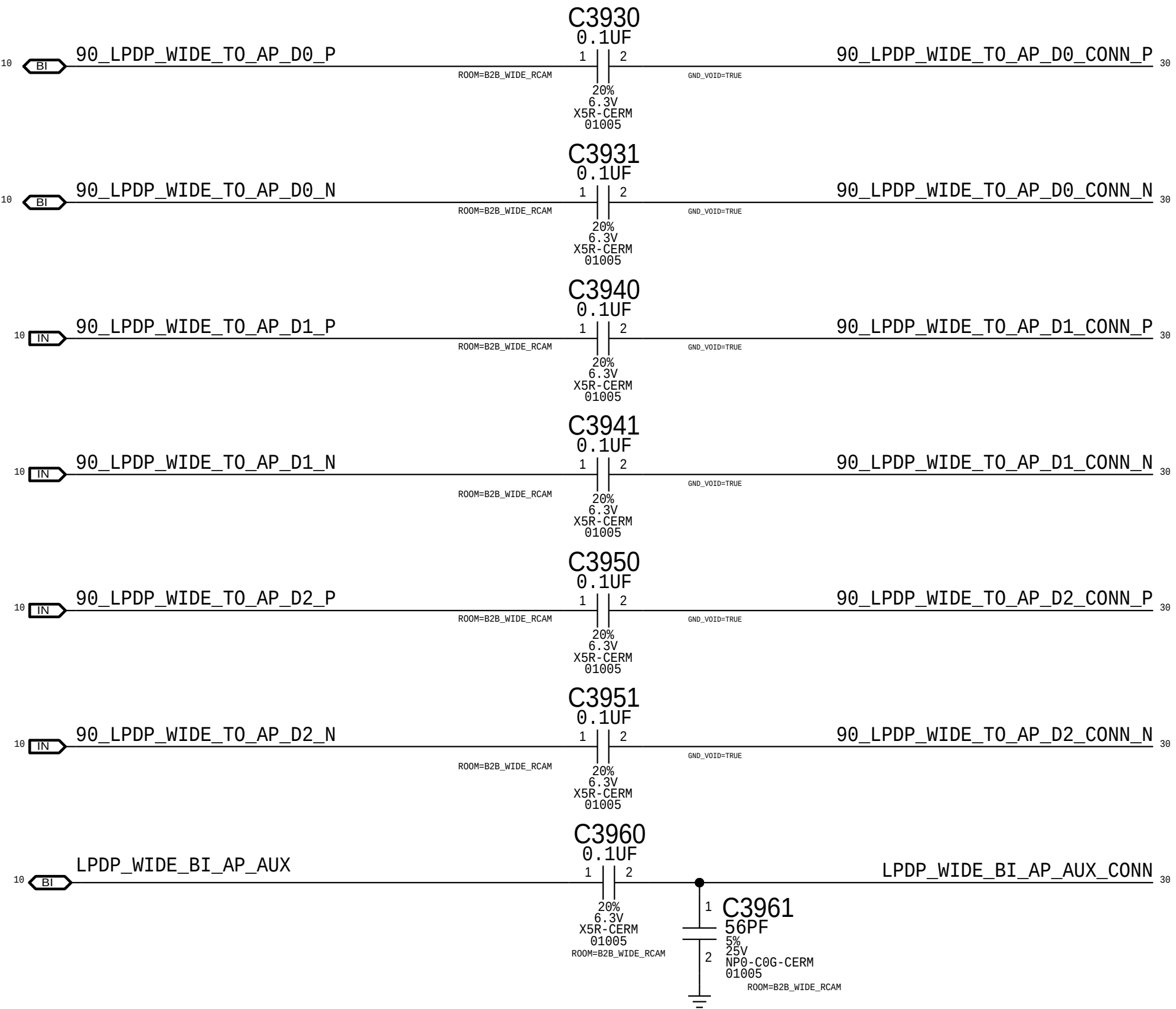
I0 Filters



Power Filtering



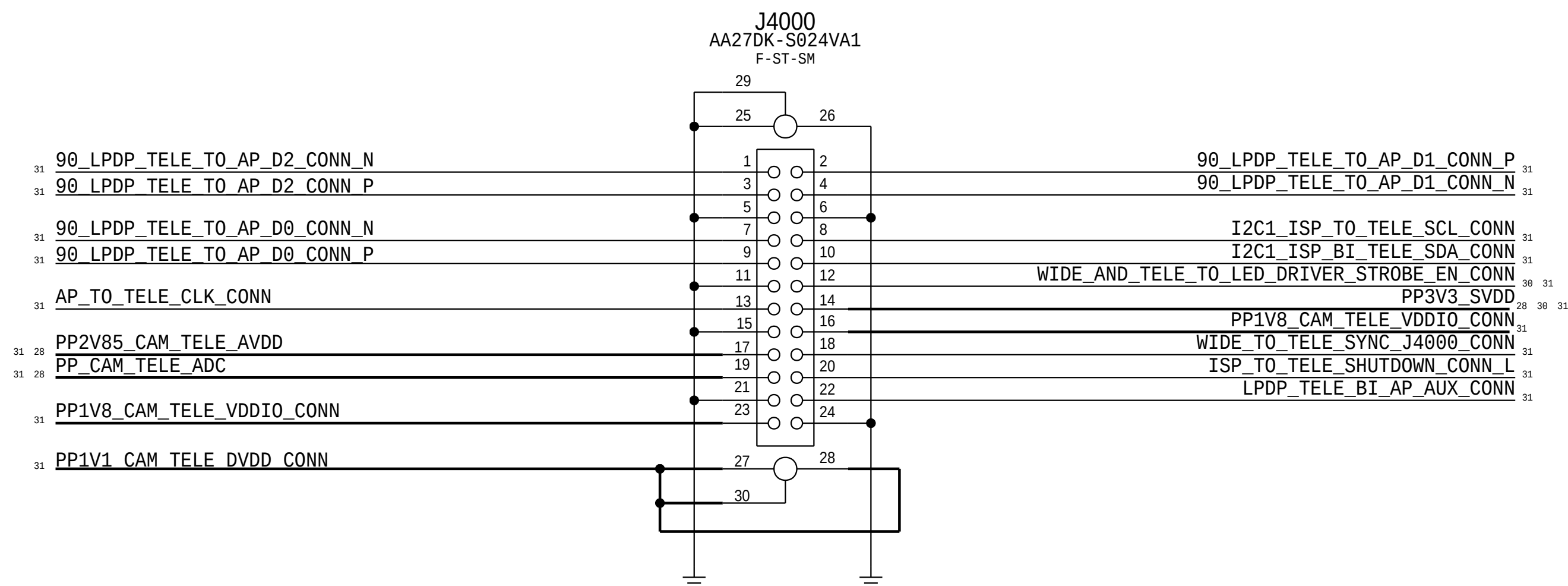
LPDP Filters



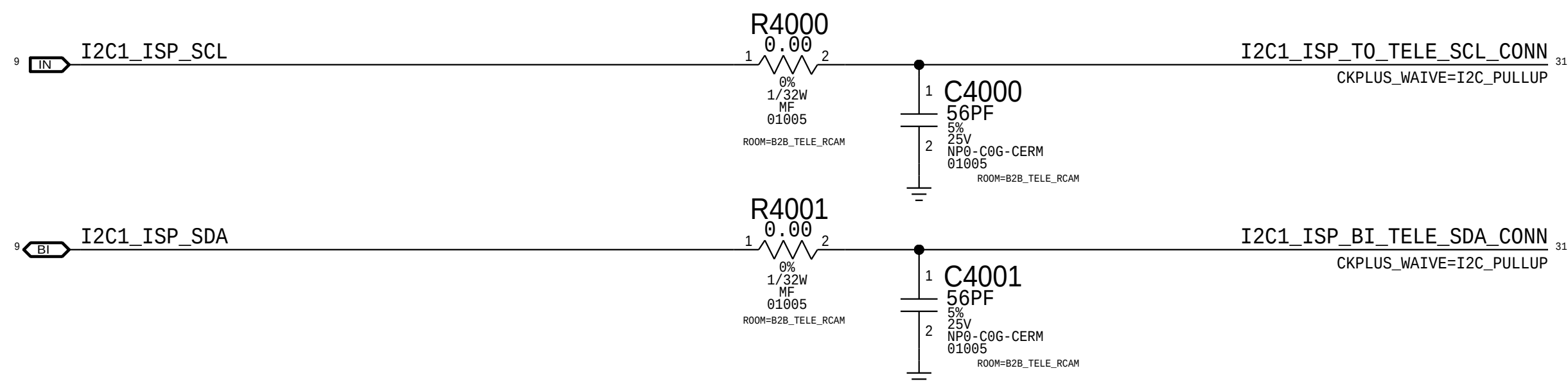
CAMERA: B2B Wide (KY)



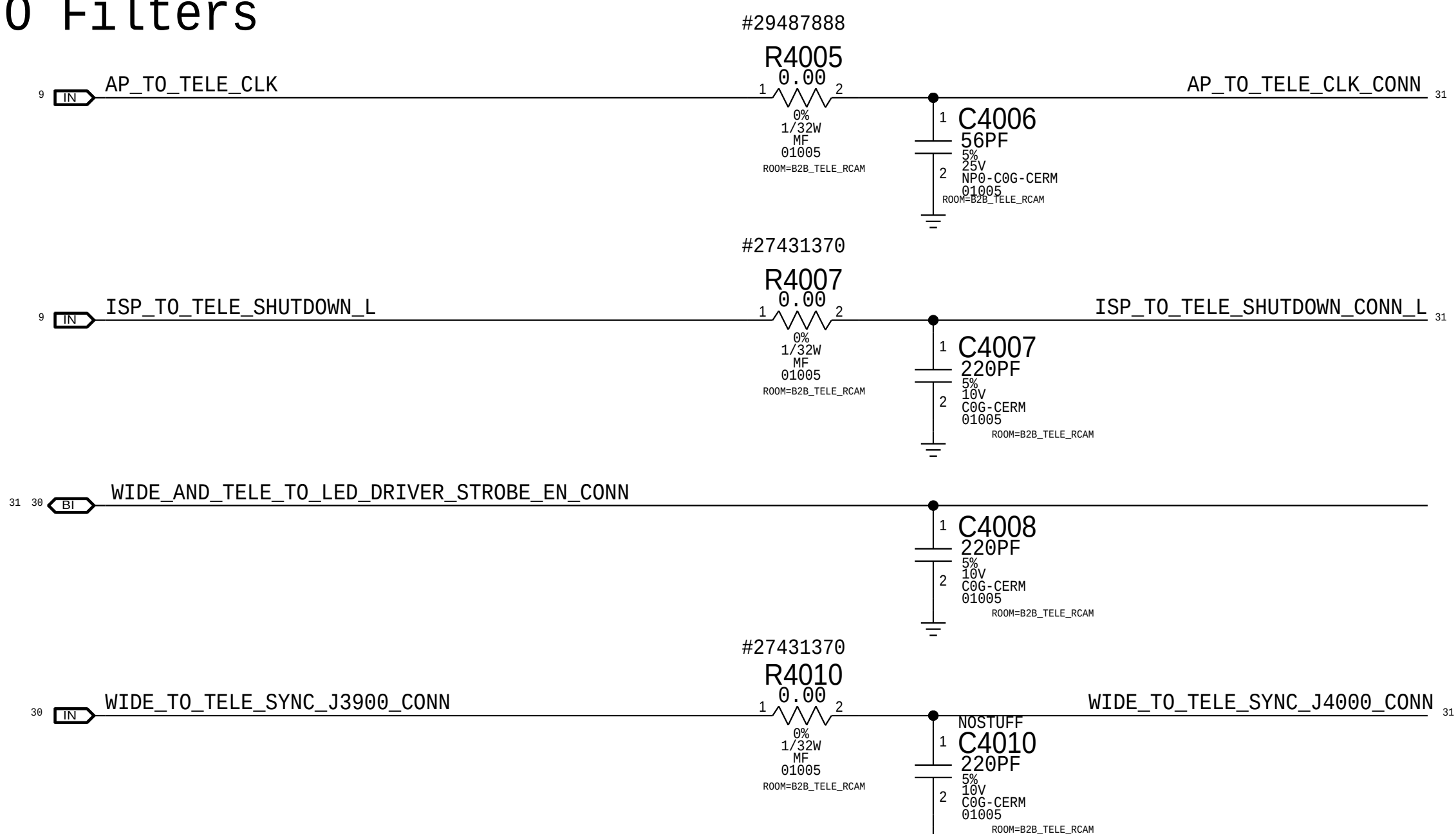
THIS ONE ---> 516S00310/P2 RCPT (USED ON MLB)
516S00309/P2 PLUG (USED ON FLEX)



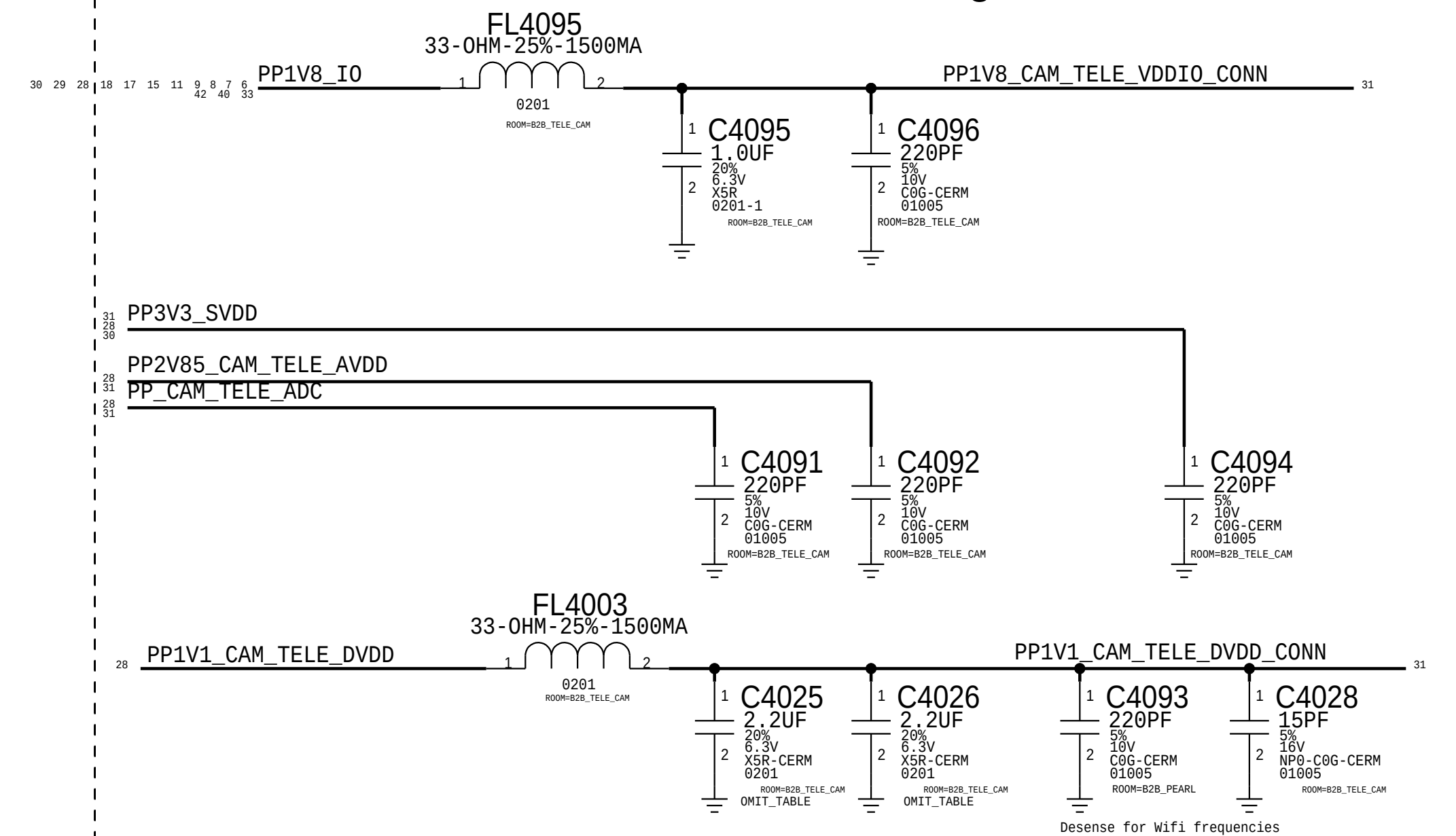
ISP I2C



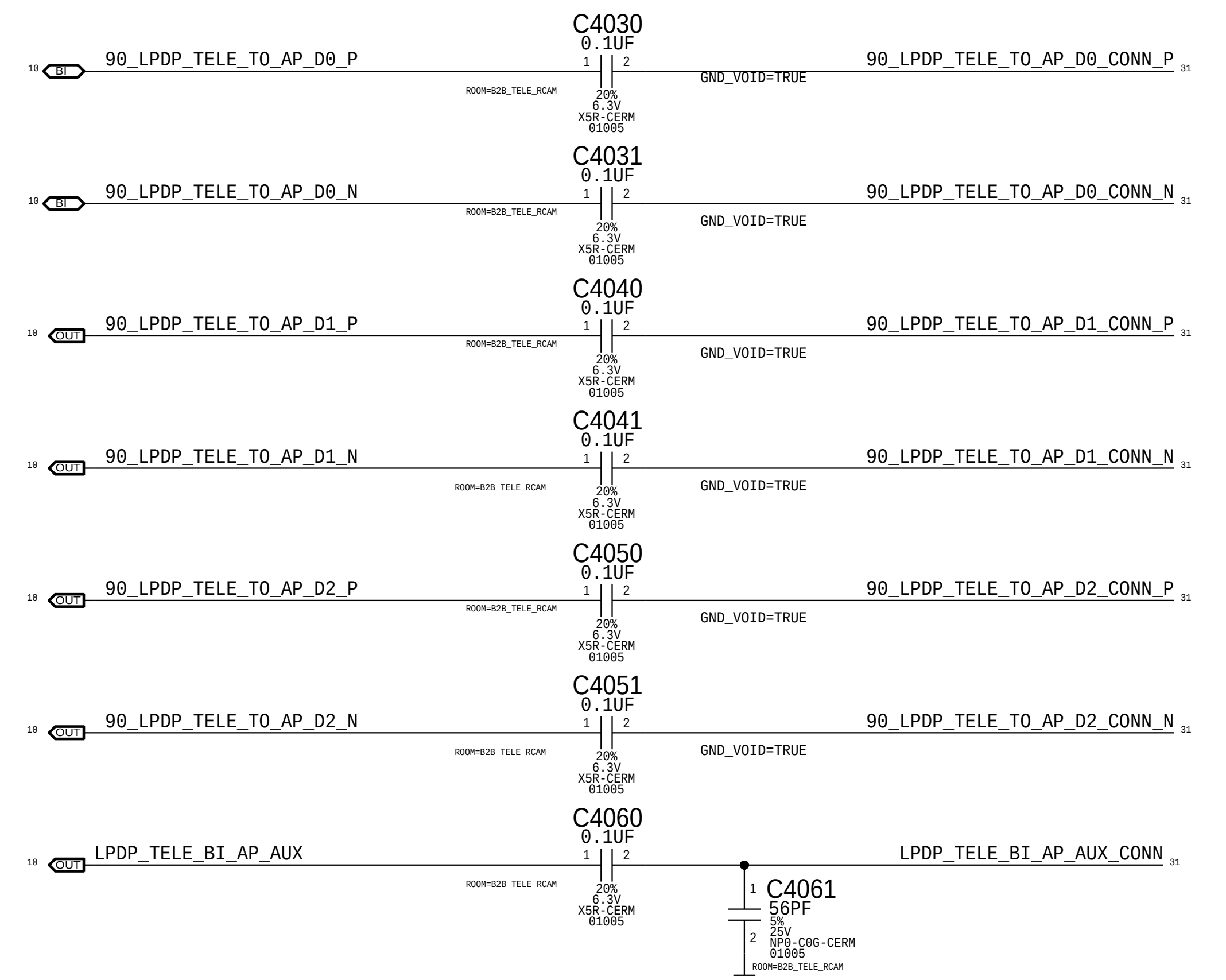
IO Filters



Power Filtering



LPDP

CAMERA: B2B Tele (TN)
$$= A$$

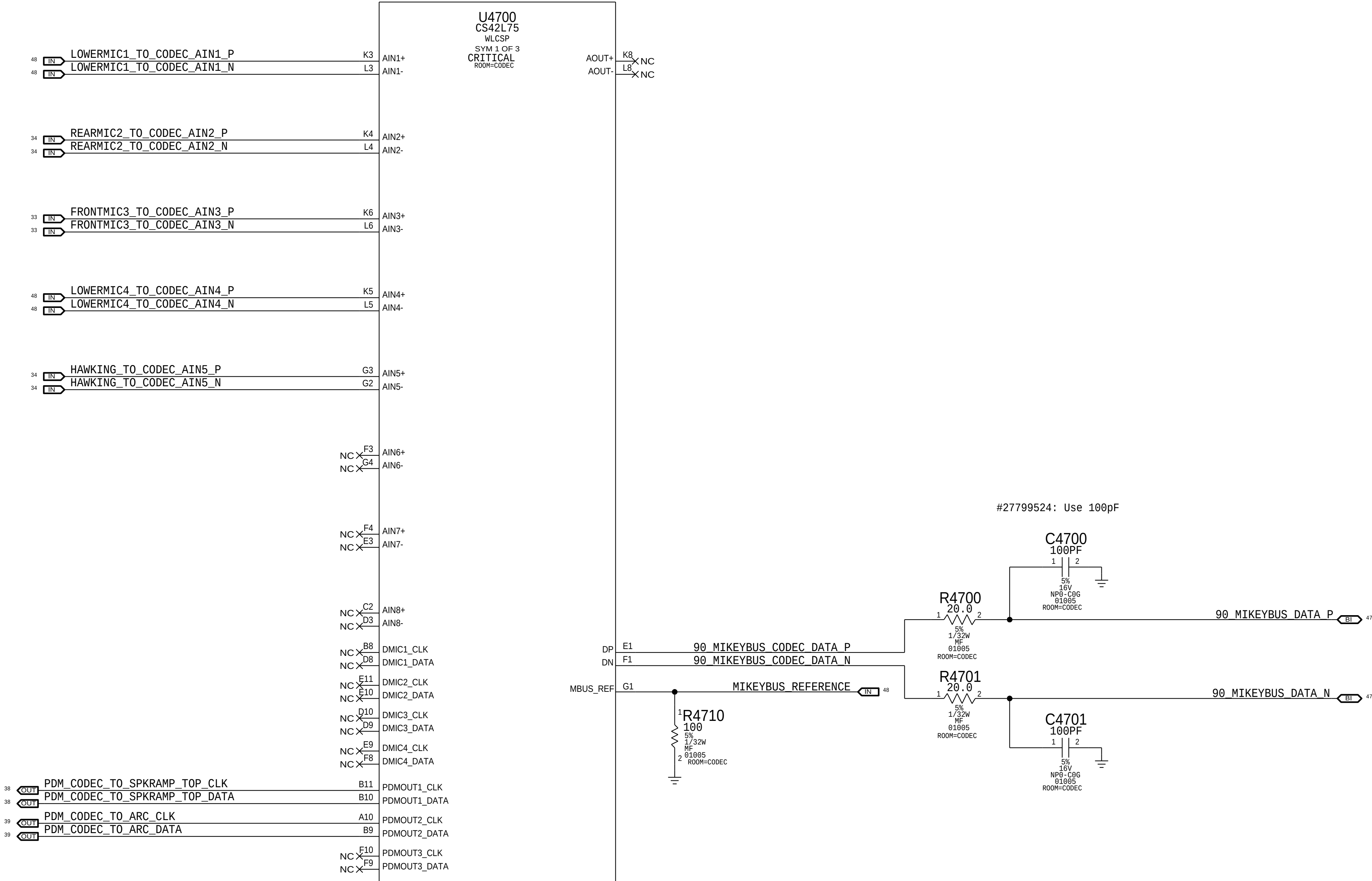
31 OF 81

D

A

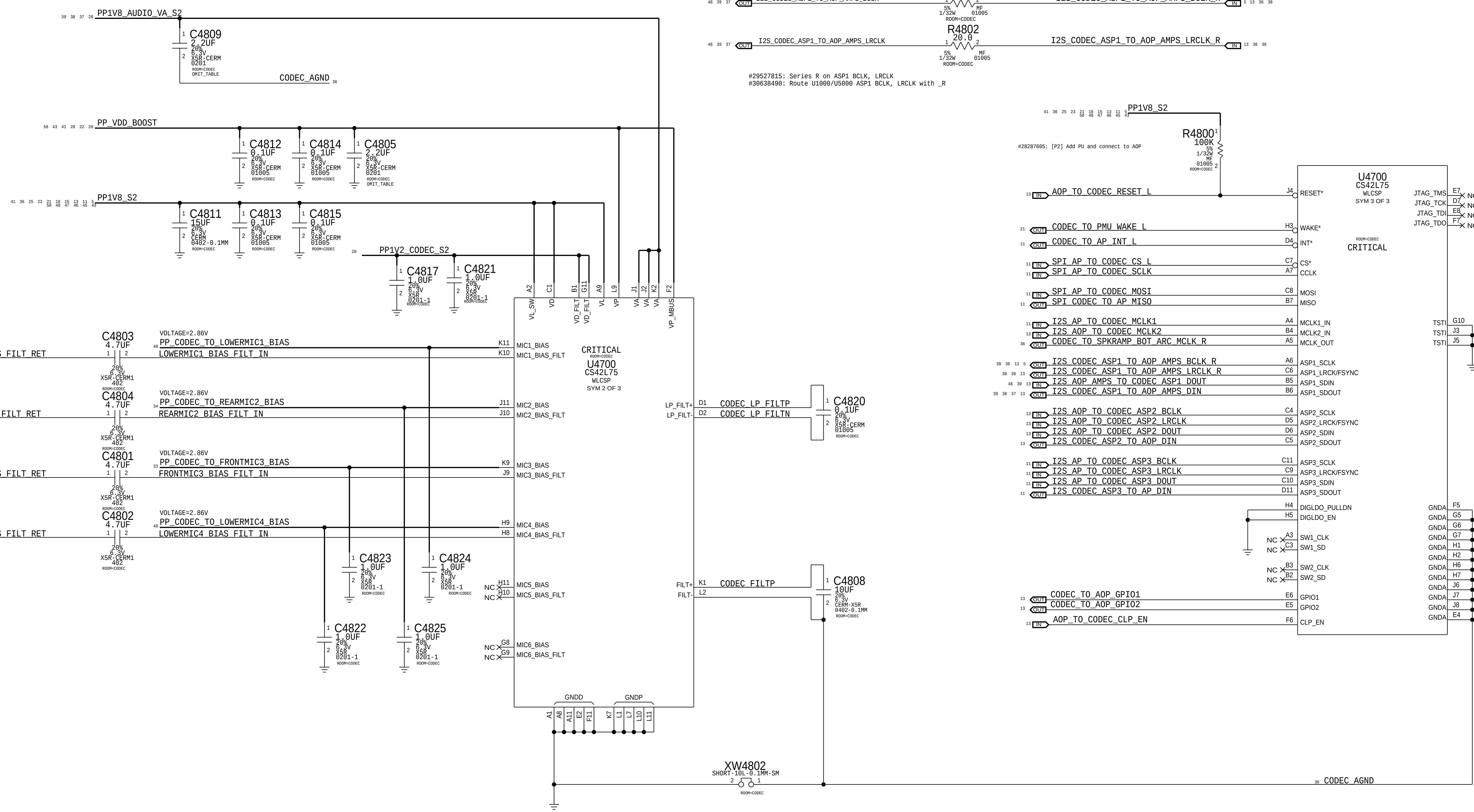
32 0F 81

CALLAN AUDIO CODEC (ANALOG INPUTS & OUTPUTS)



AUDIO: CODEC (1/2)

CALLAN AUDIO CODEC (POWER & I/O)

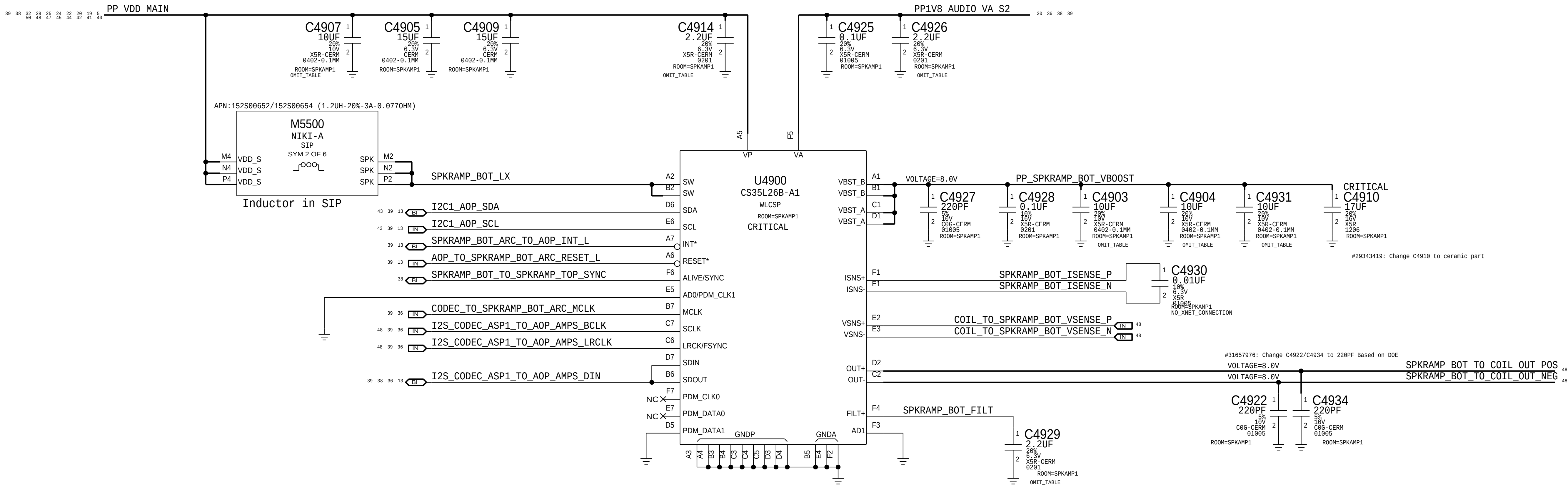


AUDIO: CODEC (2/2)

South Speaker Amplifier

APN: 338S00143

I2C ADDRESS: 1000 000x
0x80

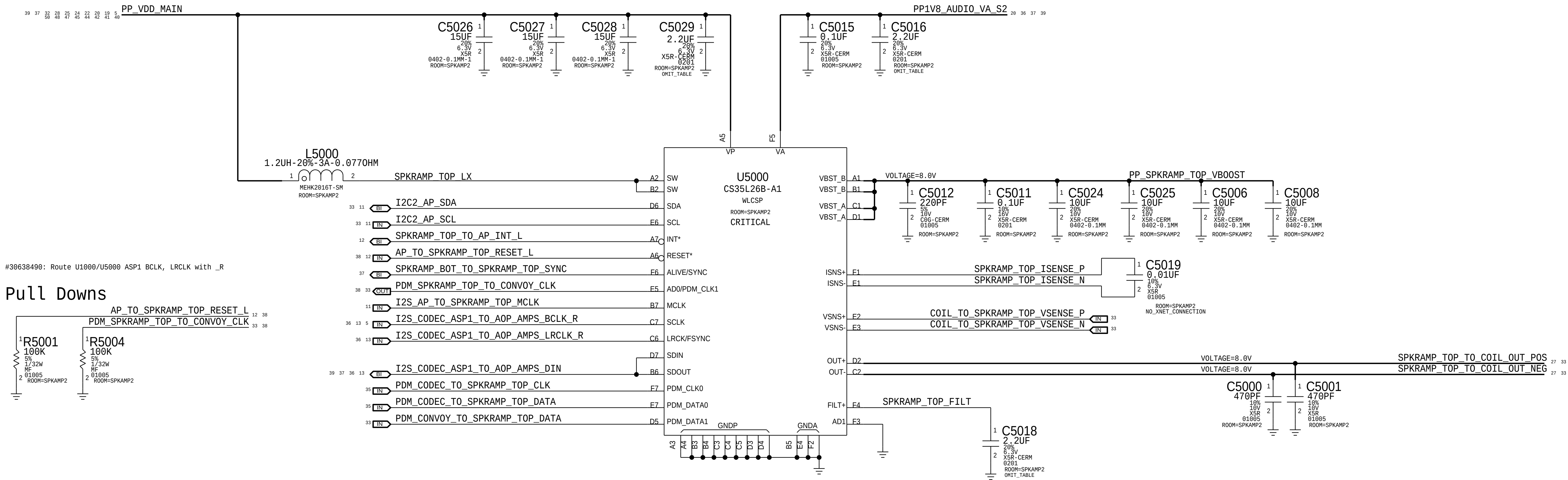


AUDIO: Speaker Amp Bottom

North Speaker Amplifier

APN: 338S00295

I2C ADDRESS: 1000 000x (0x80)



AUDIO: Speaker Amp Top

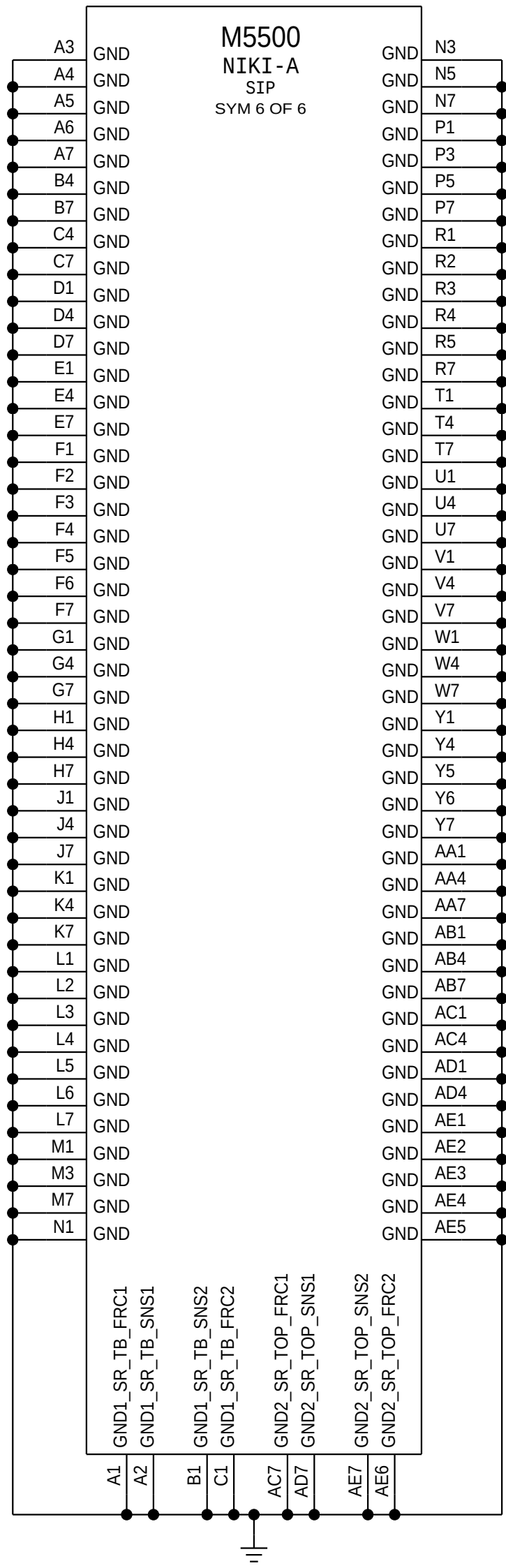
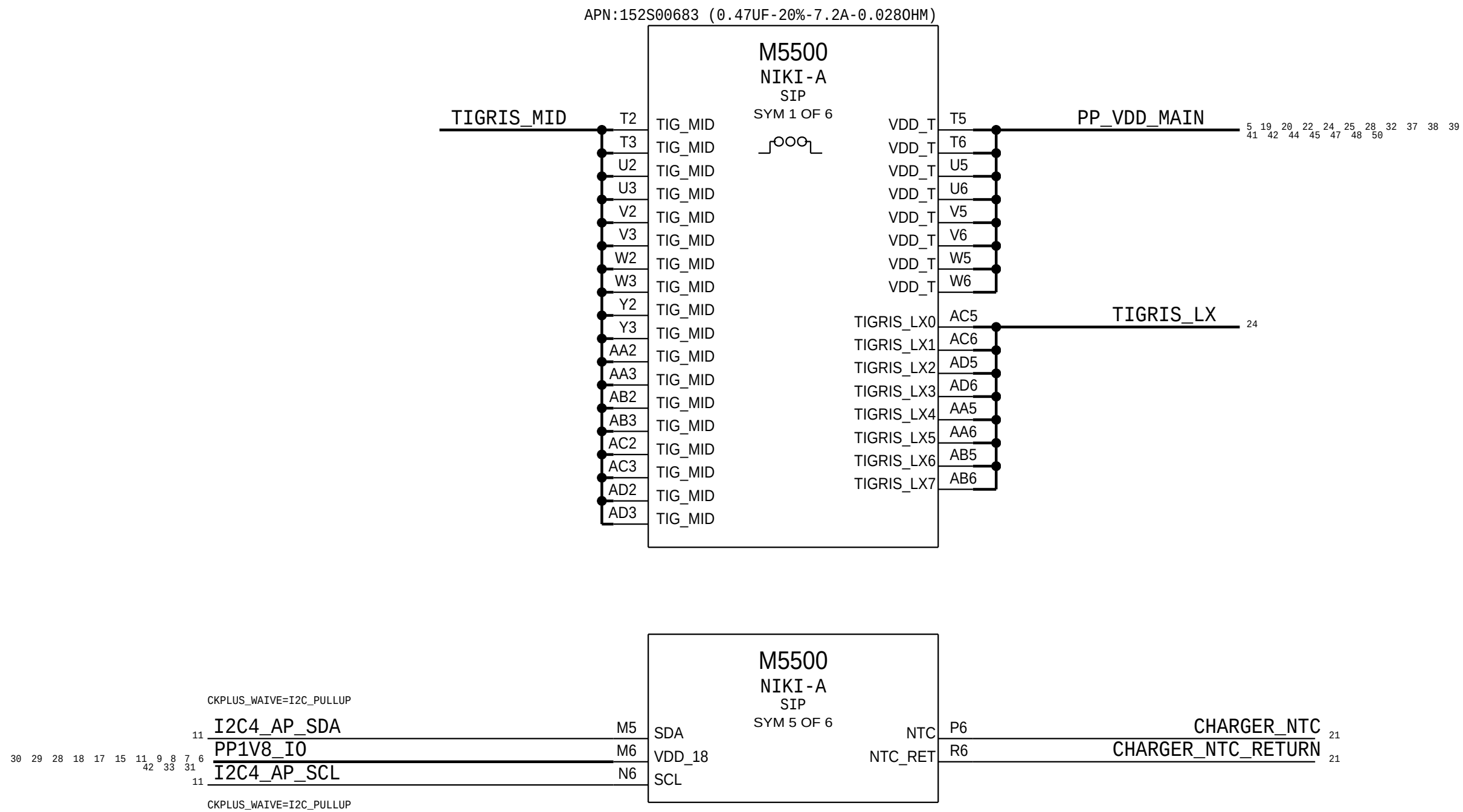
ARC D
APN: 338S00295
I2C ADDRESS: 1

I2C ADDRESS: 1000 001x (0x82)



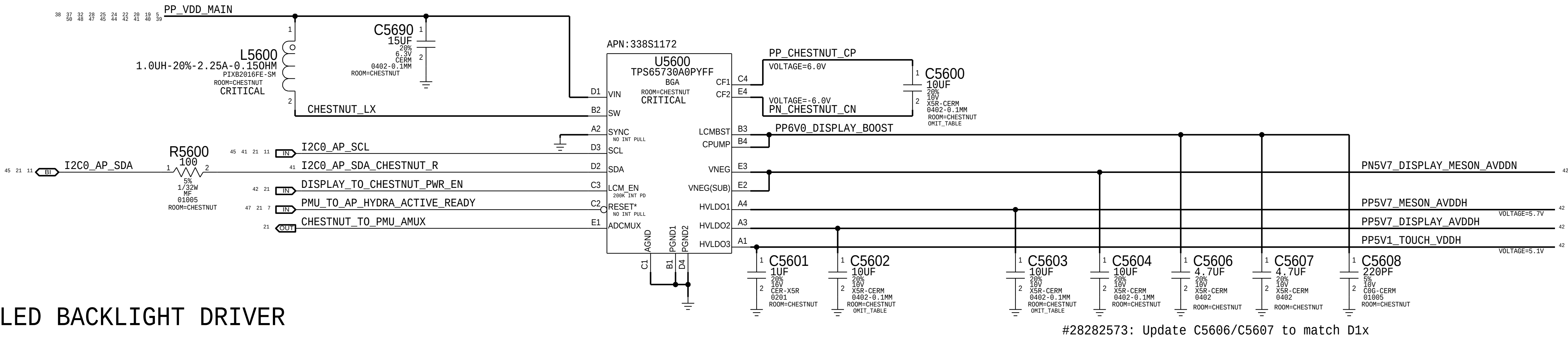
ARC: Driver	
	SIZE
	39 OF 81

Niki SIP



CG: Niki

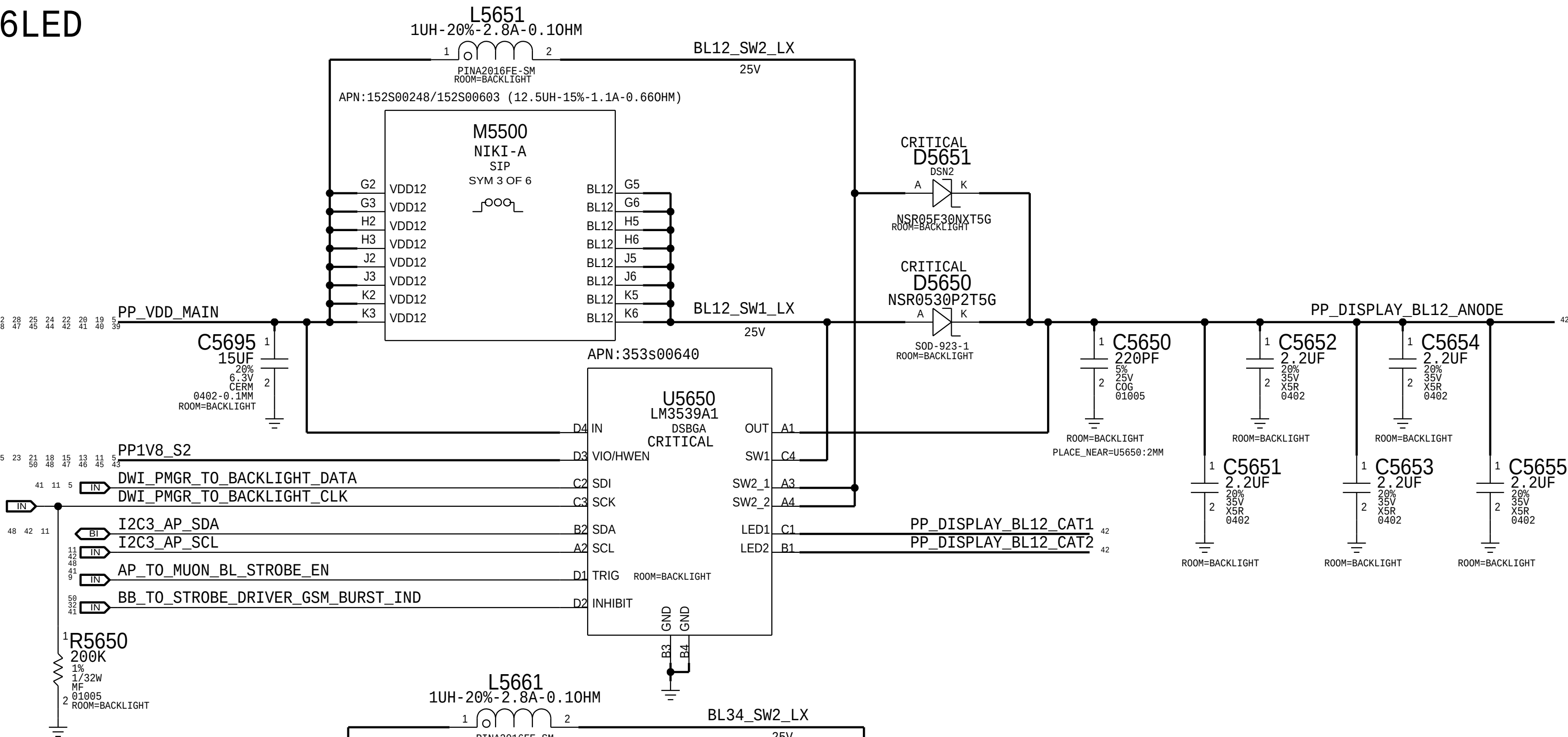
CHESTNUT DISPLAY PMU



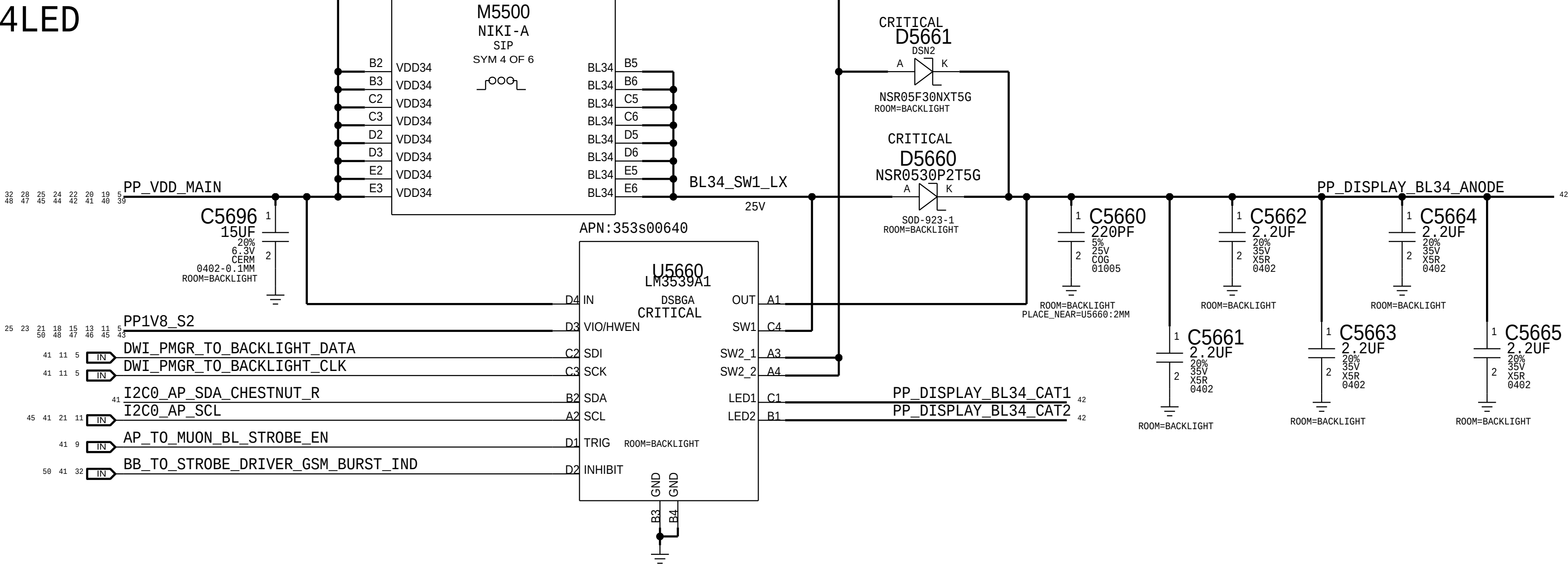
#28282573: Update C5606/C5607 to match D1x

LED BACKLIGHT DRIVER

6LED

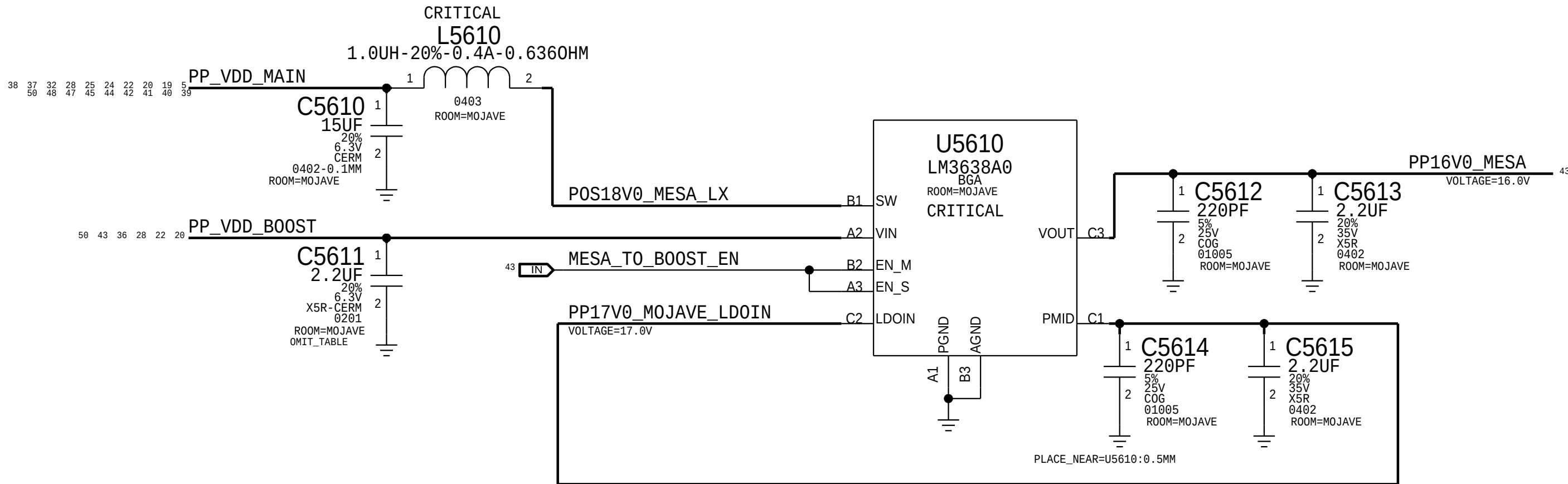


4LED

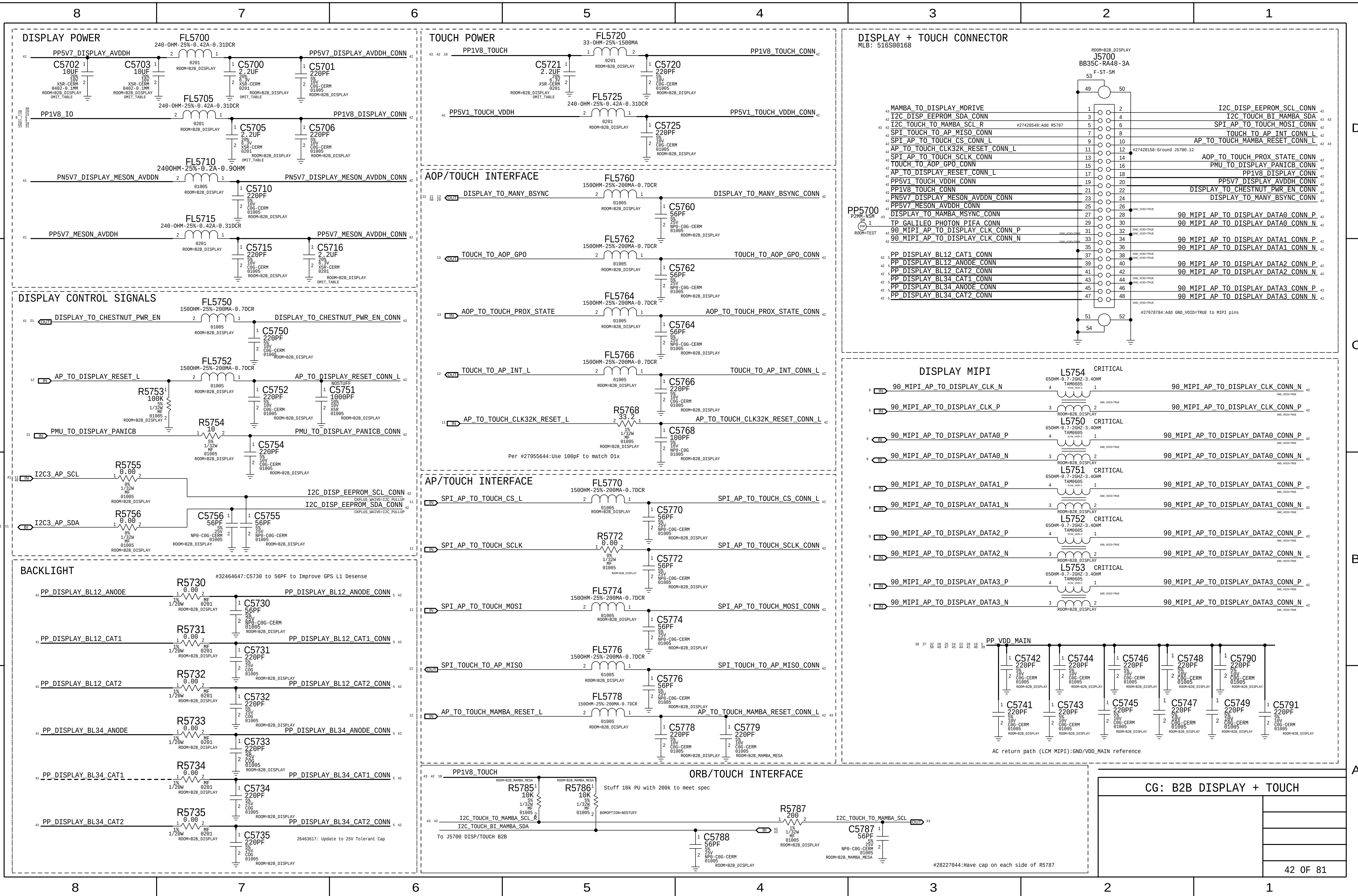


MOJAVE MESA BOOST

APN:353S00671

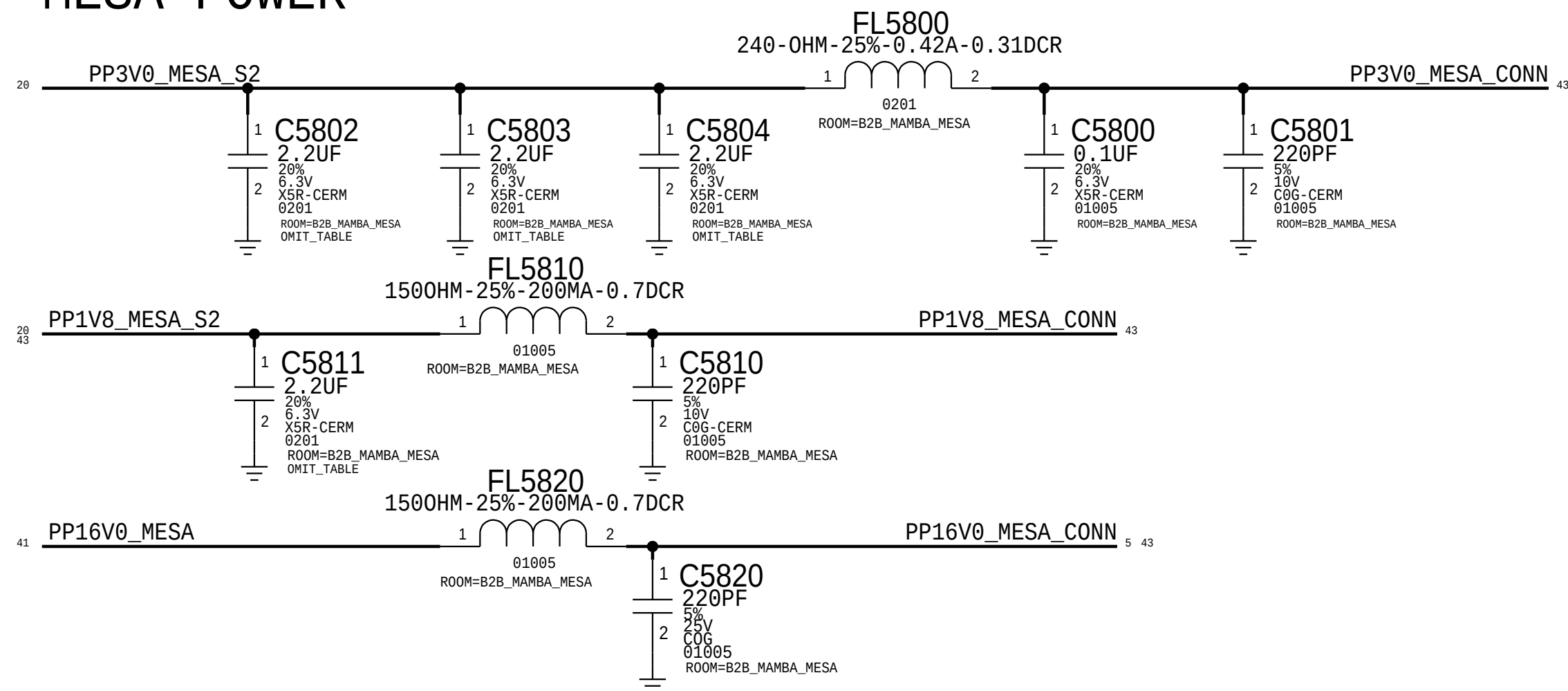


CG: PMUs



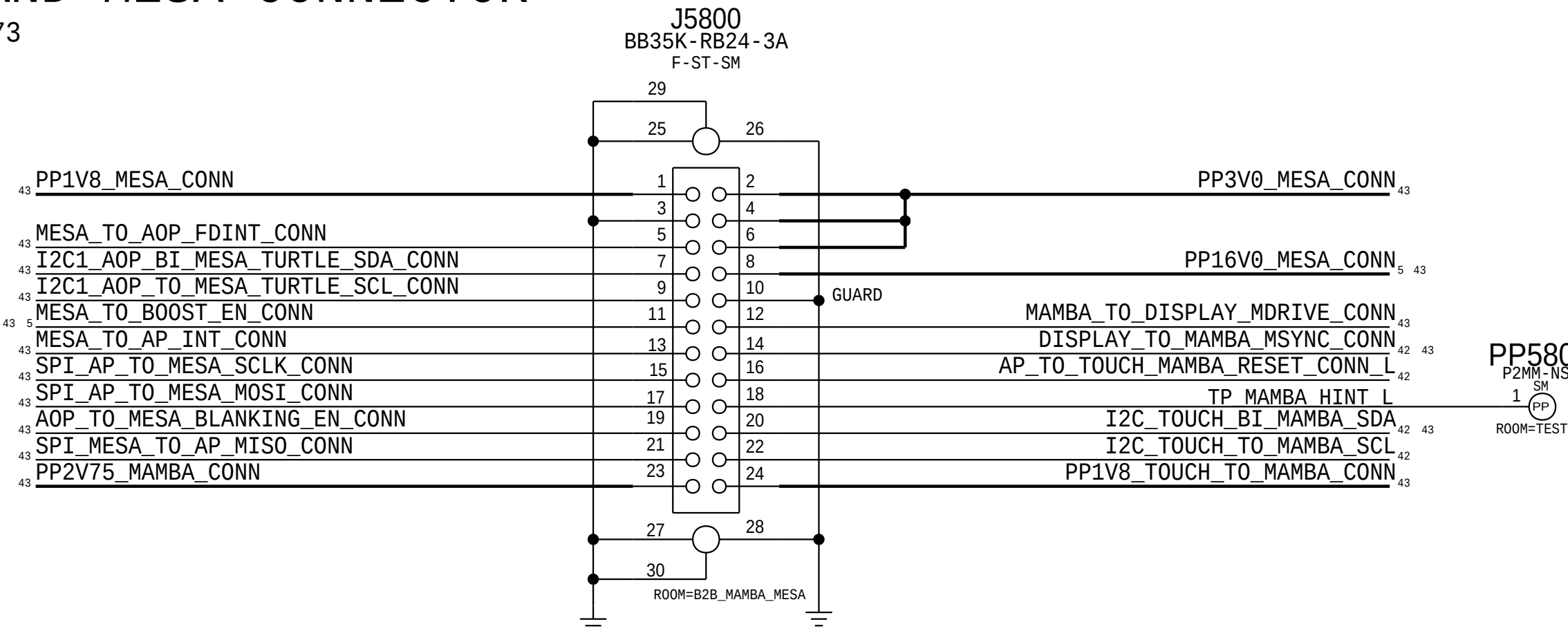
MESA POWER

#30272427: Change FL5800 to 155S00067

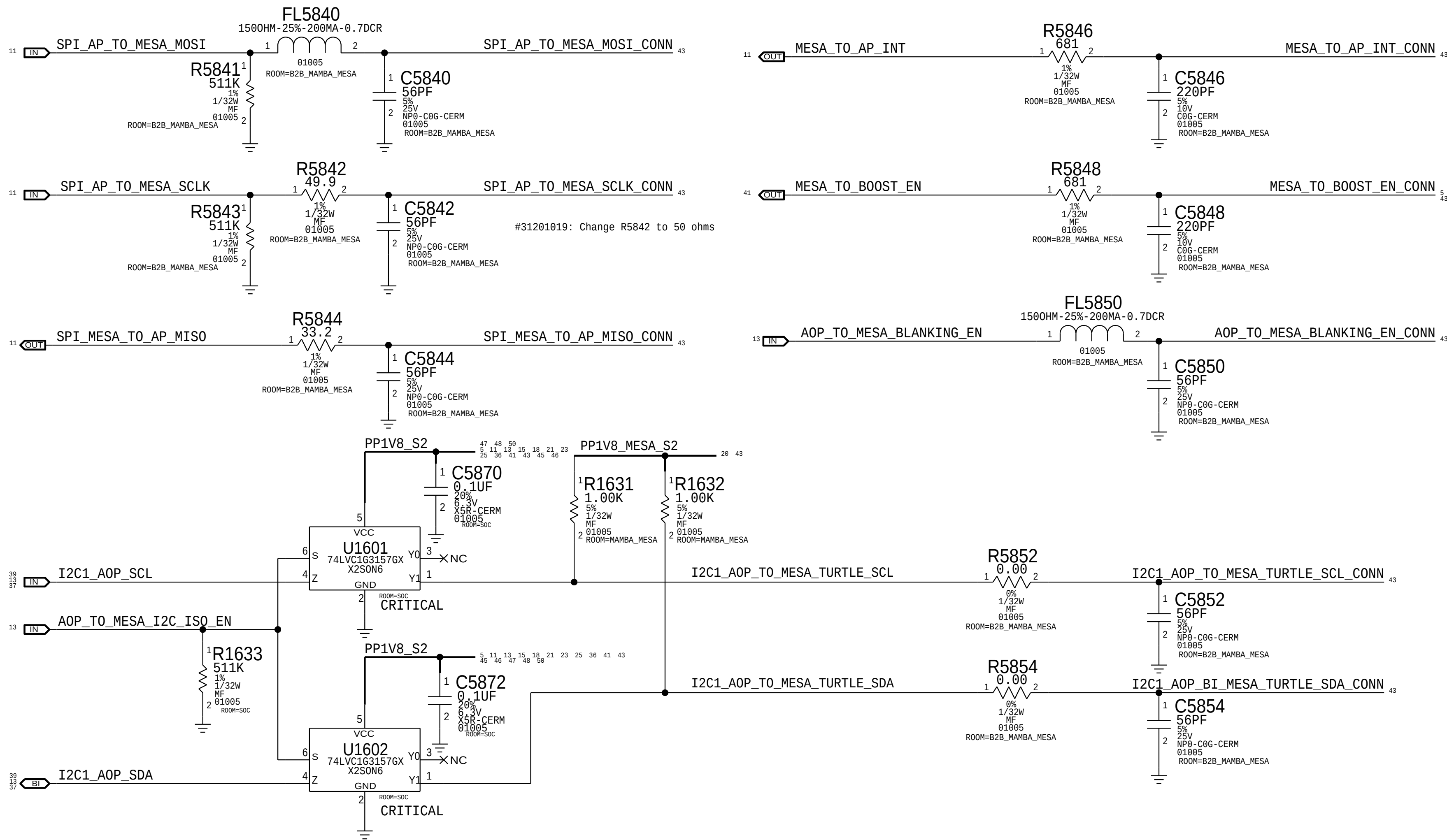


MAMBA AND MESA CONNECTOR

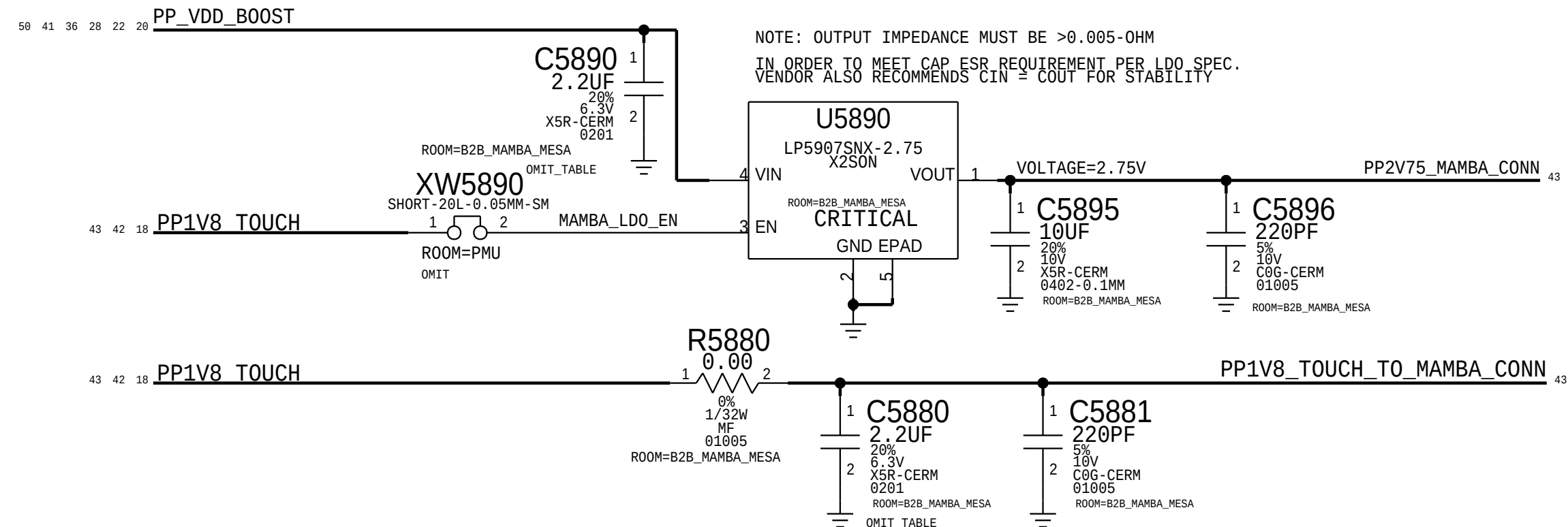
MLB: 516S00273



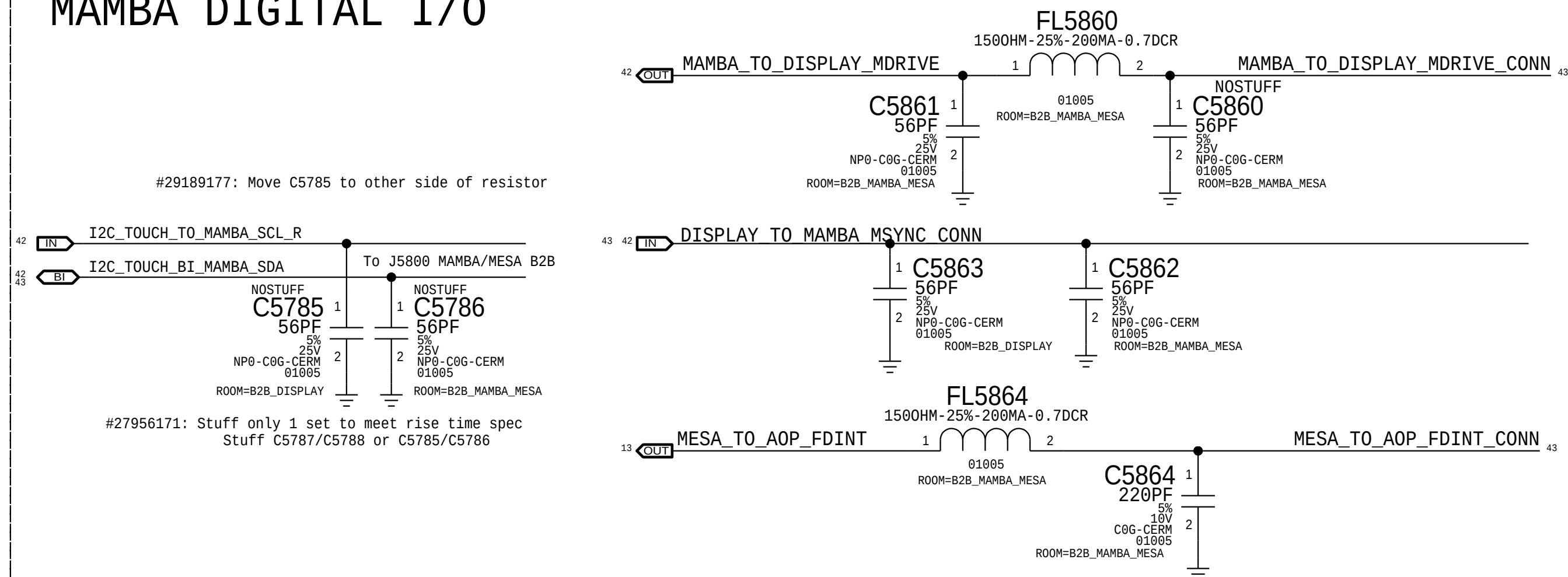
MESA DIGITAL I/O



MAMBA POWER

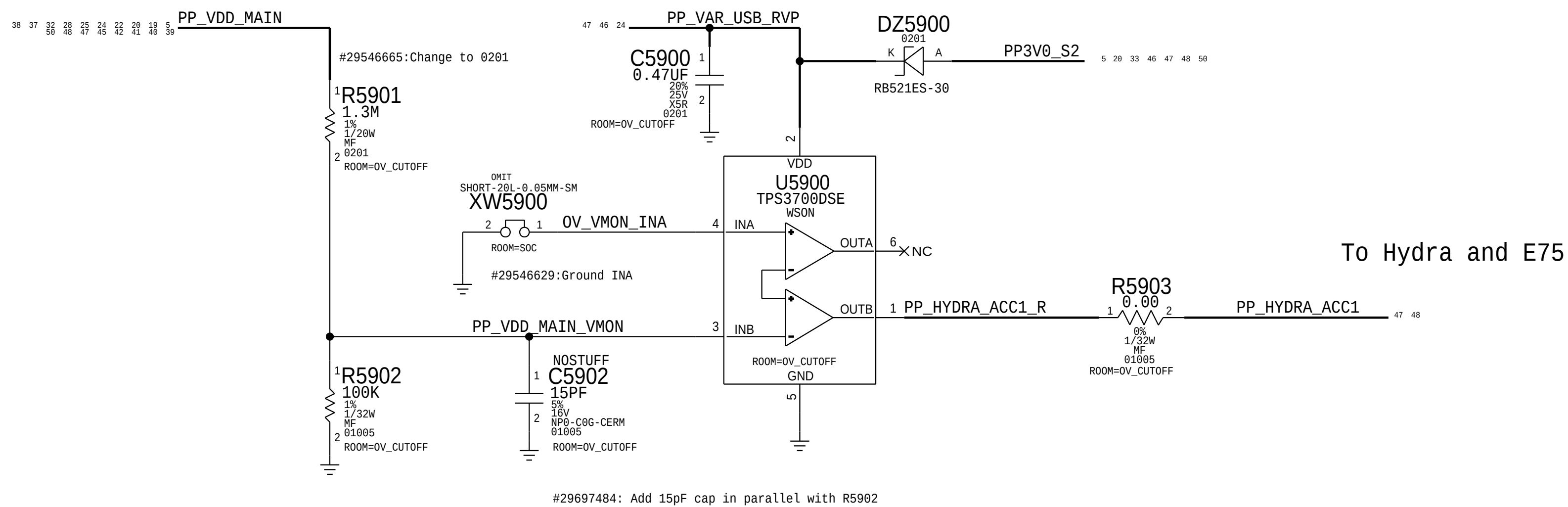


MAMBA DIGITAL I/O

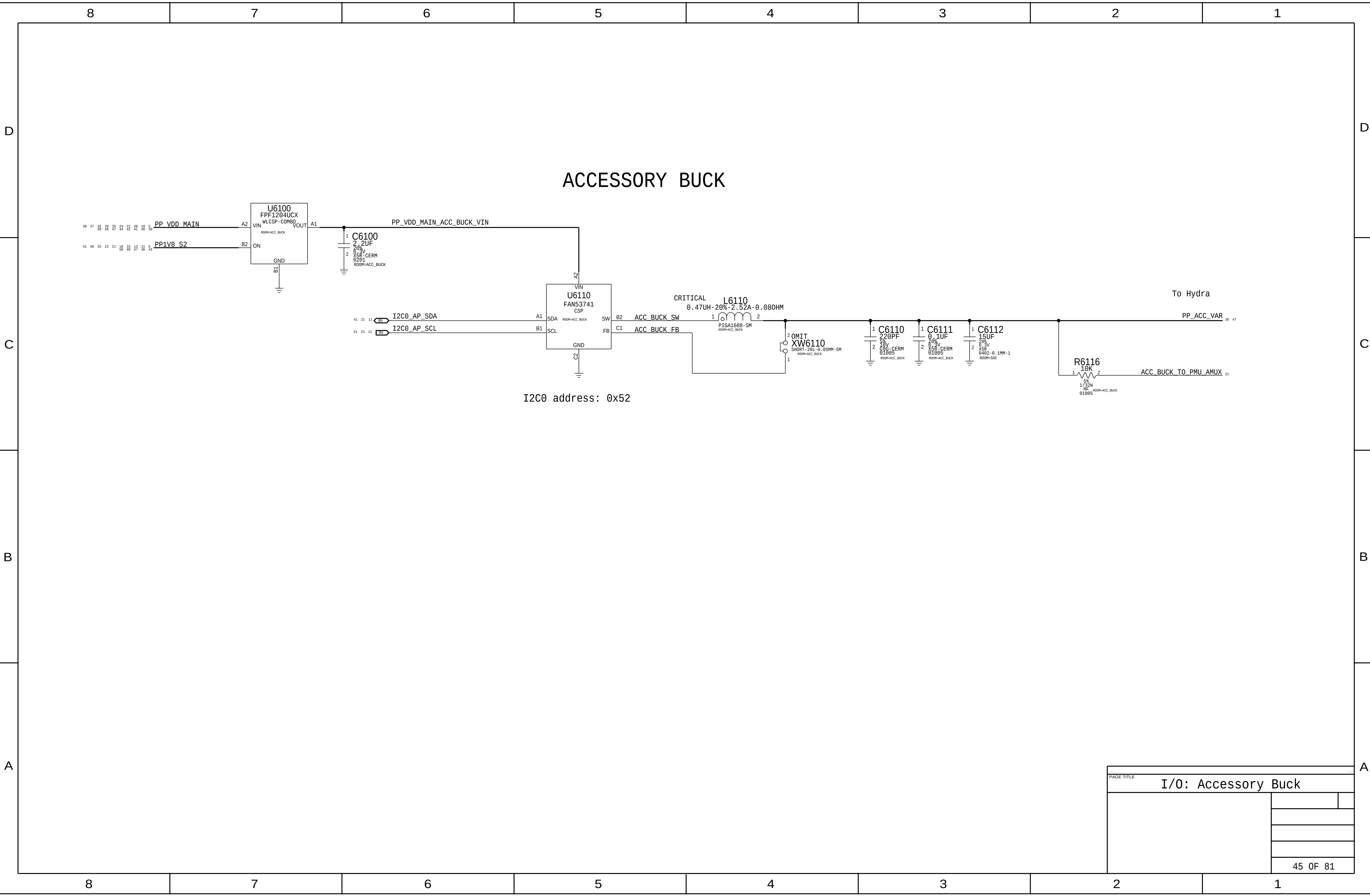


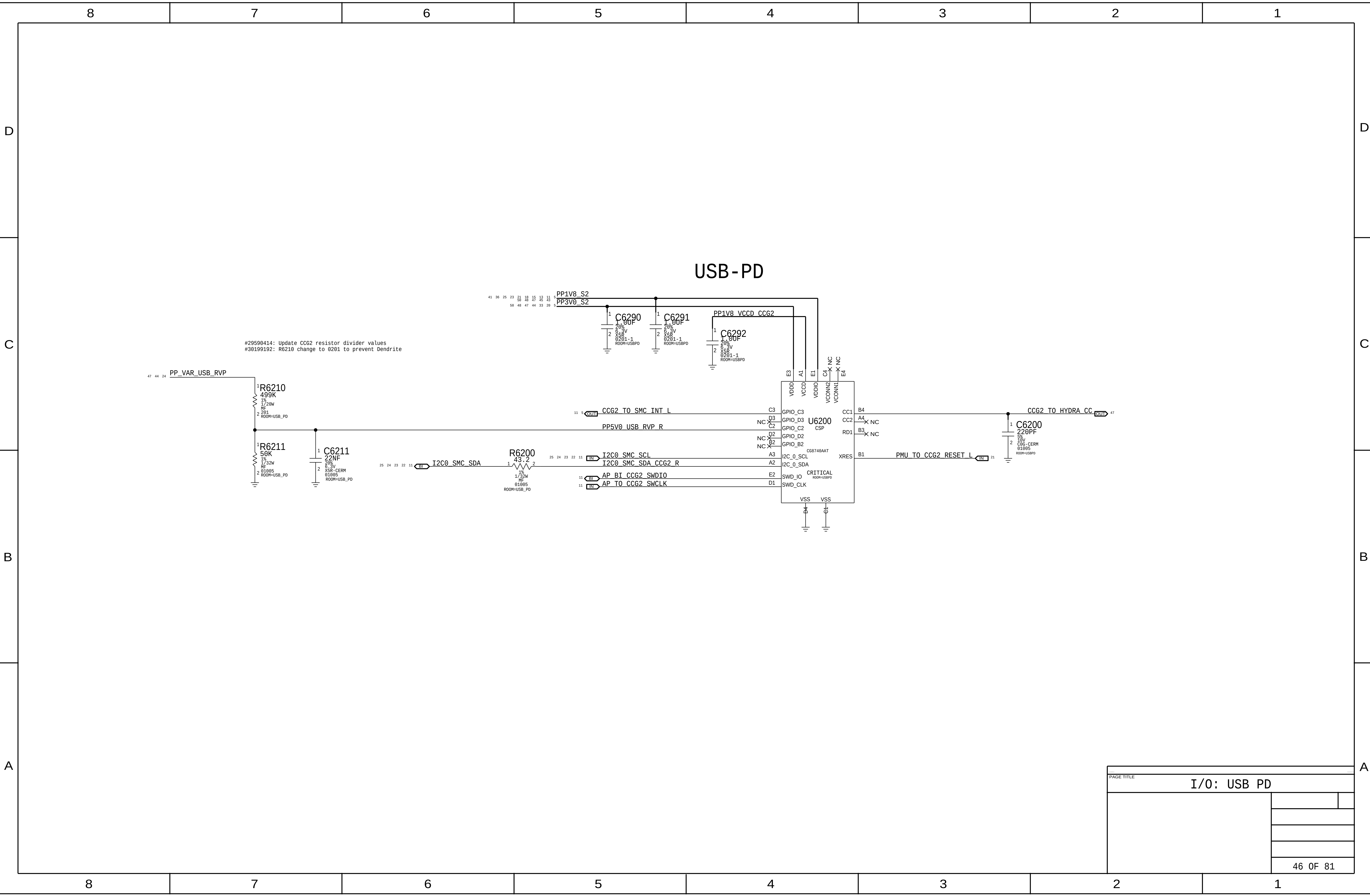
PAGE TITLE	
CG: B2B ORB + MESA	
IV ALL RIGHTS RESERVED	
43 OF 81	

VDD_MAIN OV CUT-OFF CIRCUIT



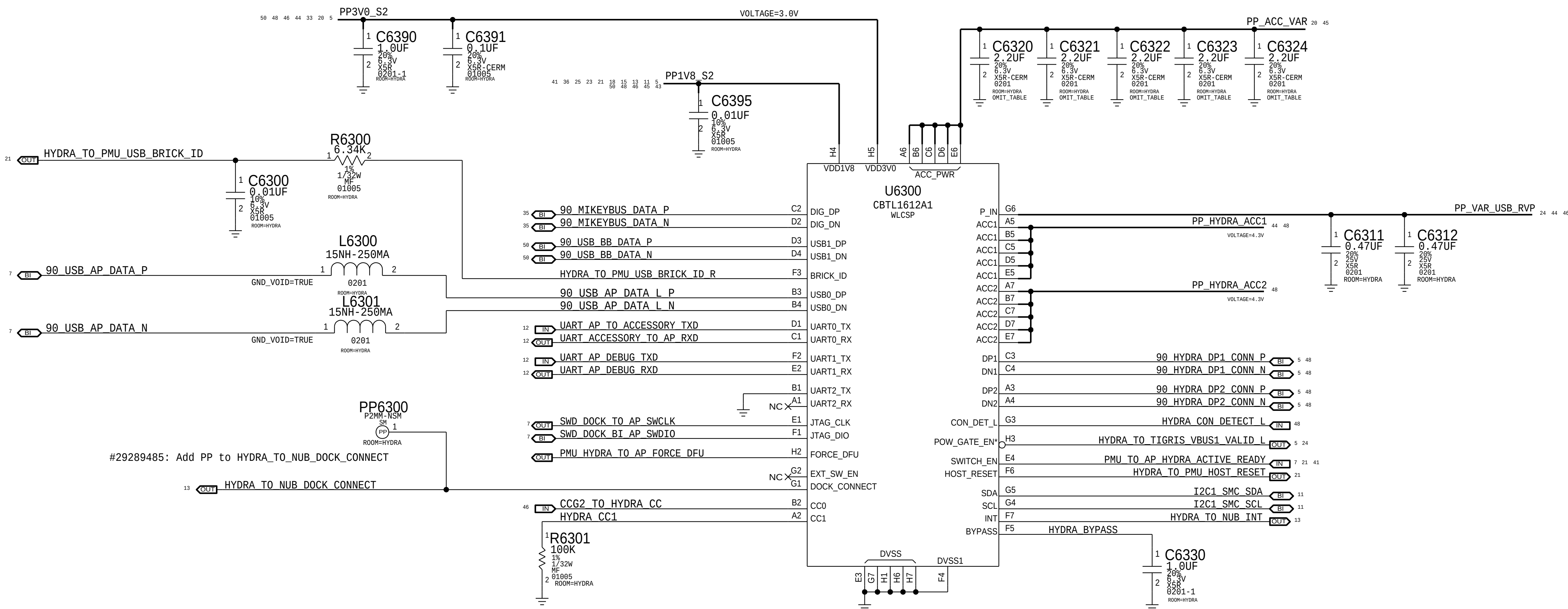
PAGE TITLE		
I/O: Overvoltage Cut-Off Circuit		
	DRAWING NUMBER	SIZE
	051-02159	D
44 OF 81		





HYDRA

12C ADDRESS: 0011010X



UART TX/RX pin names on Hydra match signal names (i.e. TX pin is input and connects to TX signal)

D20x Stiching AC Cap:

AP USBx7

BB USB

MIKEYBUS

MIPI

PP VDD MAIN

C6301
220PF
10V
C0G-CERM
01005
ROOM=HYDRA

C6331
220PF
10V
C0G-CERM
01005
ROOM=HYDRA

C6332
220PF
10V
C0G-CERM
01005
ROOM=HYDRA

C6333
220PF
10V
C0G-CERM
01005
ROOM=HYDRA

C6334
220PF
10V
C0G-CERM
01005
ROOM=HYDRA

C6335
220PF
10V
C0G-CERM
01005
ROOM=HYDRA

C6336
220PF
10V
C0G-CERM
01005
ROOM=HYDRA

C6302
220PF
10V
C0G-CERM
01005
ROOM=BASEBAND

C6303
220PF
10V
C0G-CERM
01005
ROOM=CODEC

C6304
220PF
10V
C0G-CERM
01005
ROOM=CAN_PMU

I/O: Hydra

DRAWING NUMBER

SIZE

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D

C

B

A

D

C

B

A

8

7

6

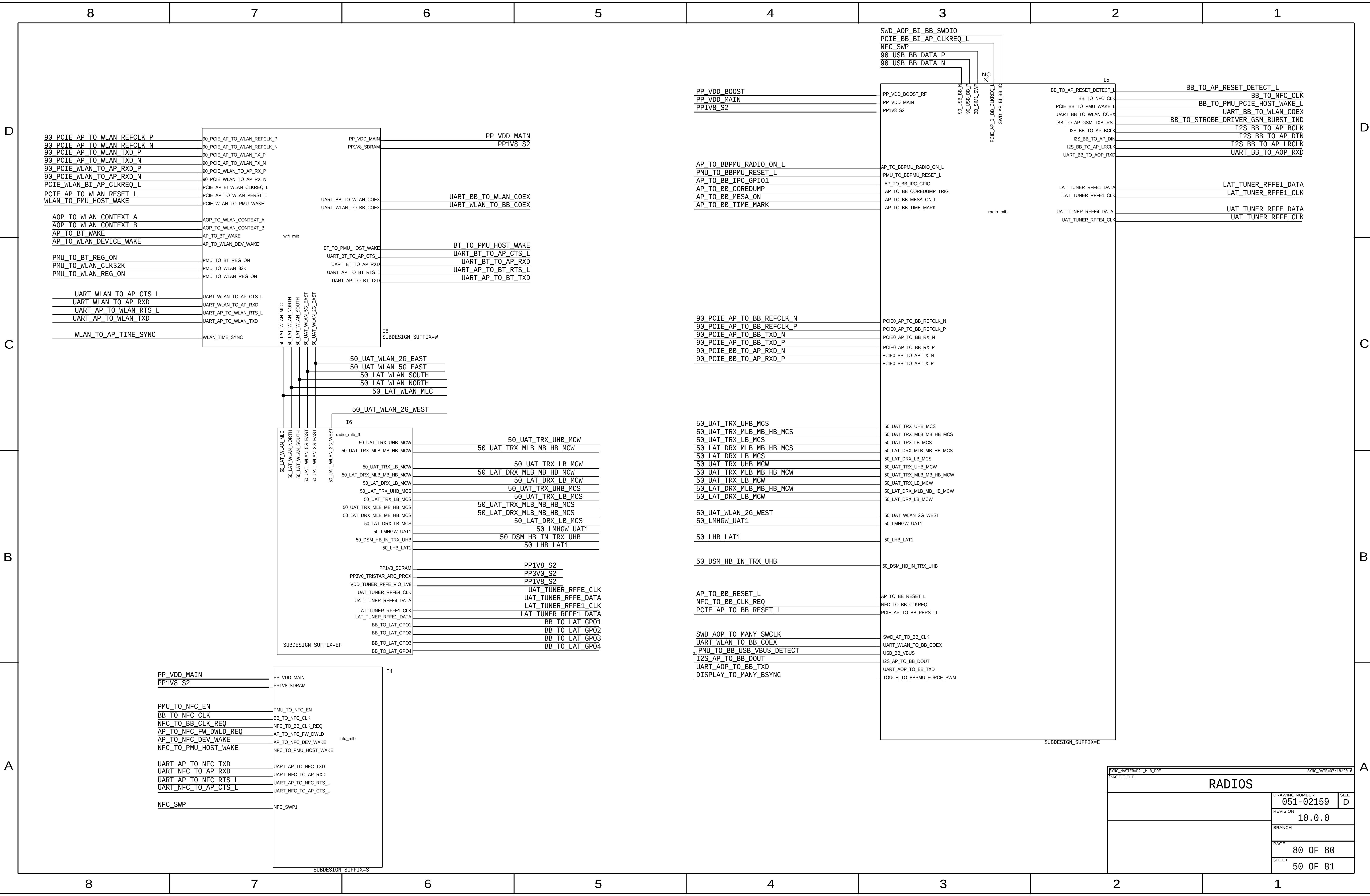
5

4

3

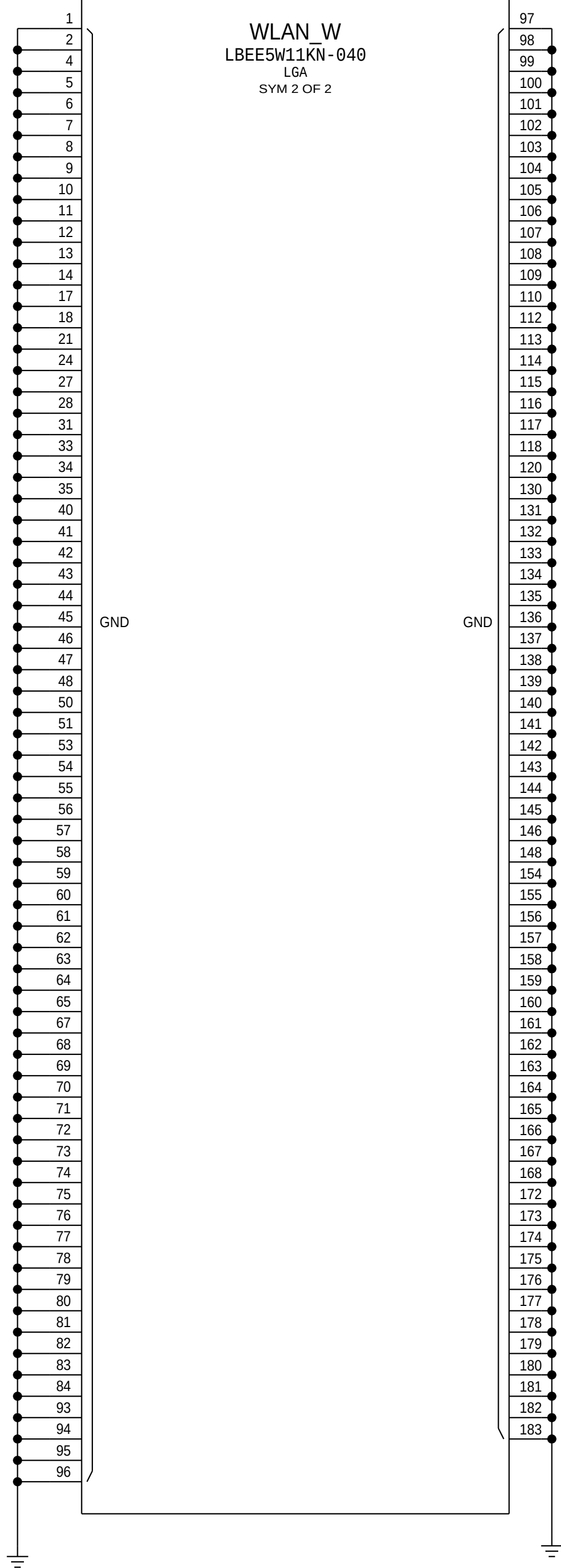
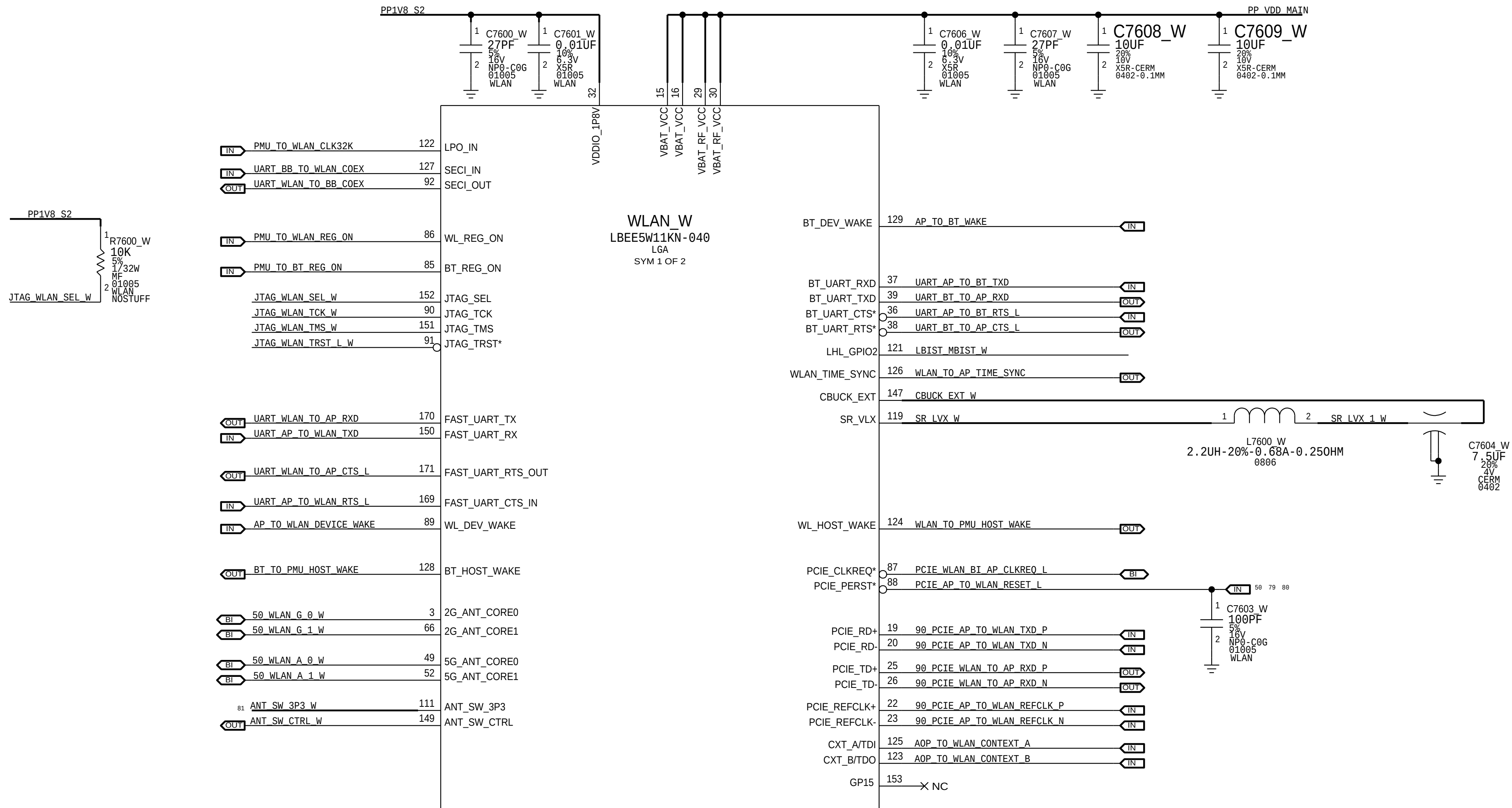
2

1



RADIO5		
051-02159		
10.0.0		
80 OF 80		
50 OF 81		

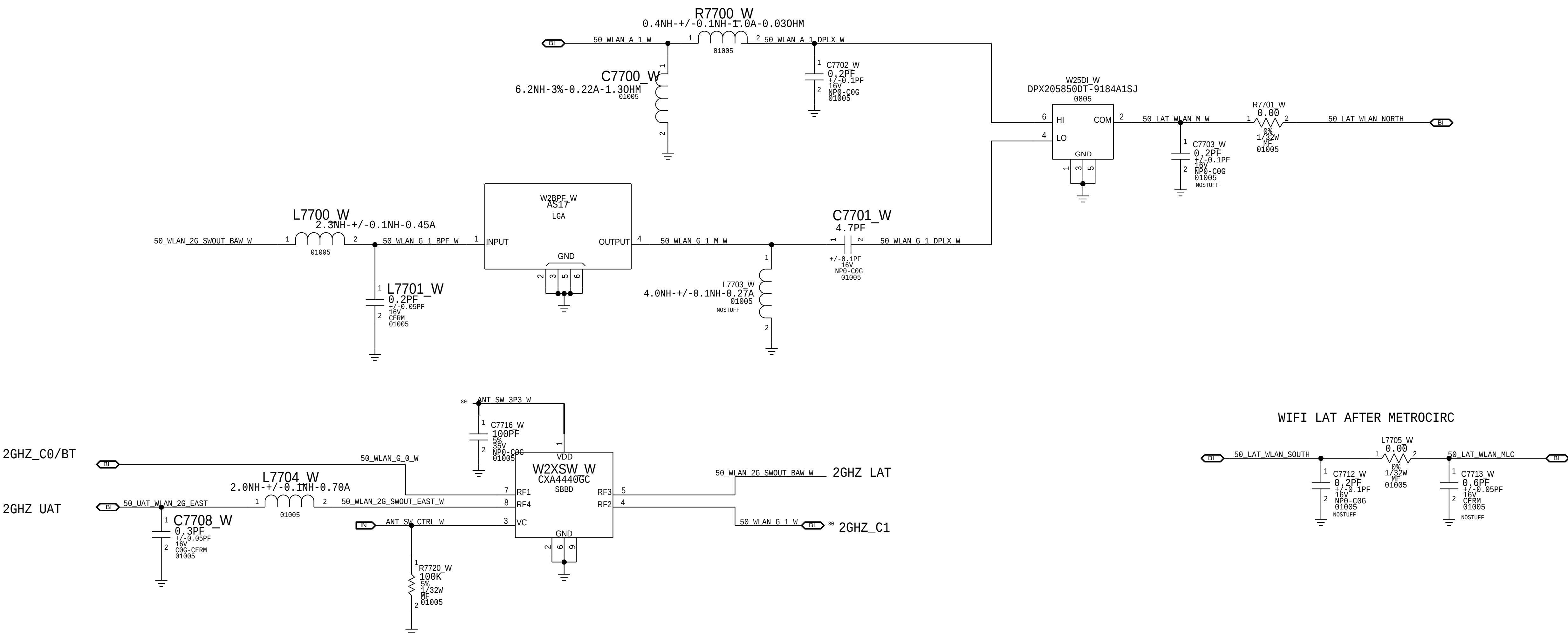
WIFI/BT



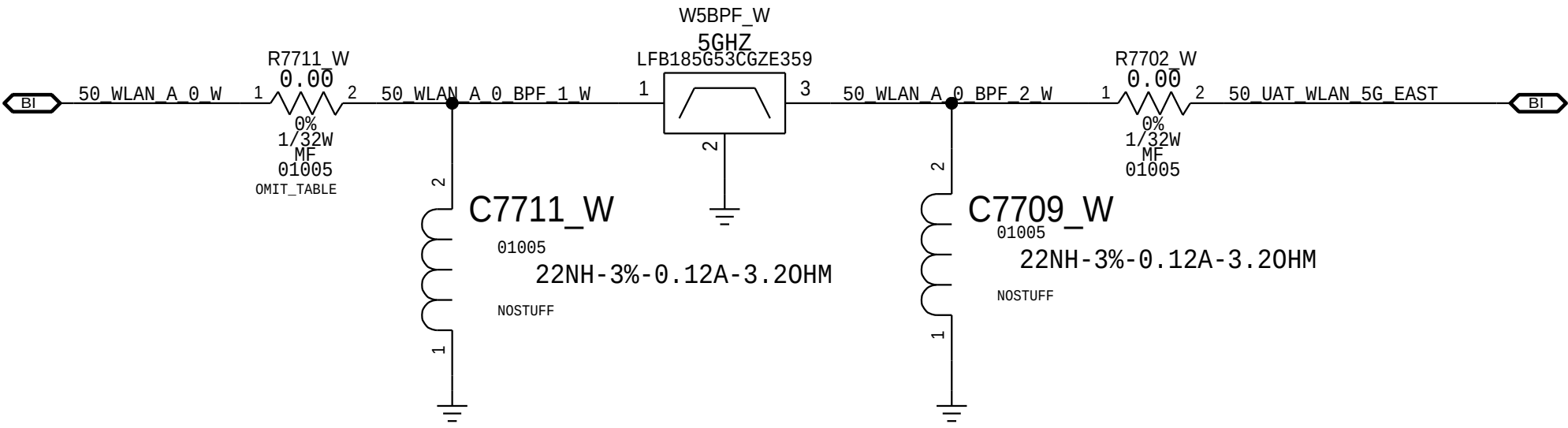
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
339508397	339508399	BOM_TABLE_ALTS	WLAN_W	ALT WIFI/BT MODULE

SYNC_MASTER=WIFI			SYNC_DATE=11/28/2016		
PAGE TITLE			Guinness		
			DRAWING NUMBER	051-02159	SIZE D
			REVISION	10.0.0	
			BRANCH		
			PAGE	4 OF 5	
			SHEET	80 OF 81	

WIFI LOWER ANTENNA FEED



WIFI UPPER ANTENNA FEED



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
152S00498	1	5G CORE0 BPF MATCH	R7711_W	CRITICAL	D211
117S0161	1	5G CORE0 BPF MATCH	R7711_W	CRITICAL	JPN
117S0161	1	5G CORE0 BPF MATCH	R7711_W	CRITICAL	ROW

PAGE TITLE			SYNC_DATE=11/28/2016		
\$PAGE_TITLE=Wi-FiAntFeeds			DRAWING NUMBER	051-02159	SIZE
			REVISION	10.0.0	D
			BRANCH		
			PAGE	5 OF 5	
			SHEET	81 OF 81	

87654321

page1

SCH, MLB, D21

DRAWING NUMBER

051-02159

SIZE

D

REVISION

10.0.0

BRANCH

PAGE

1 OF 75

SHEET

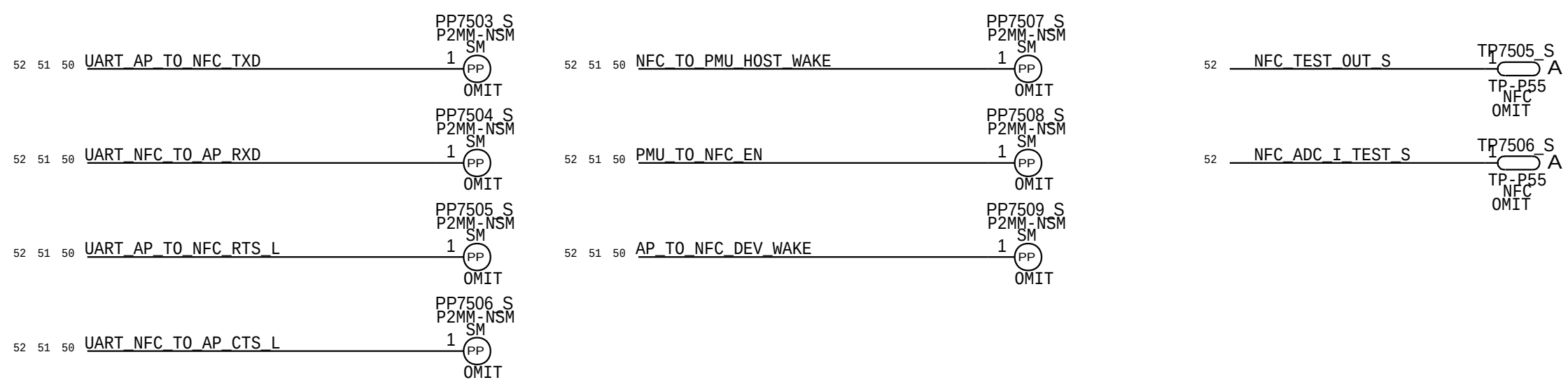
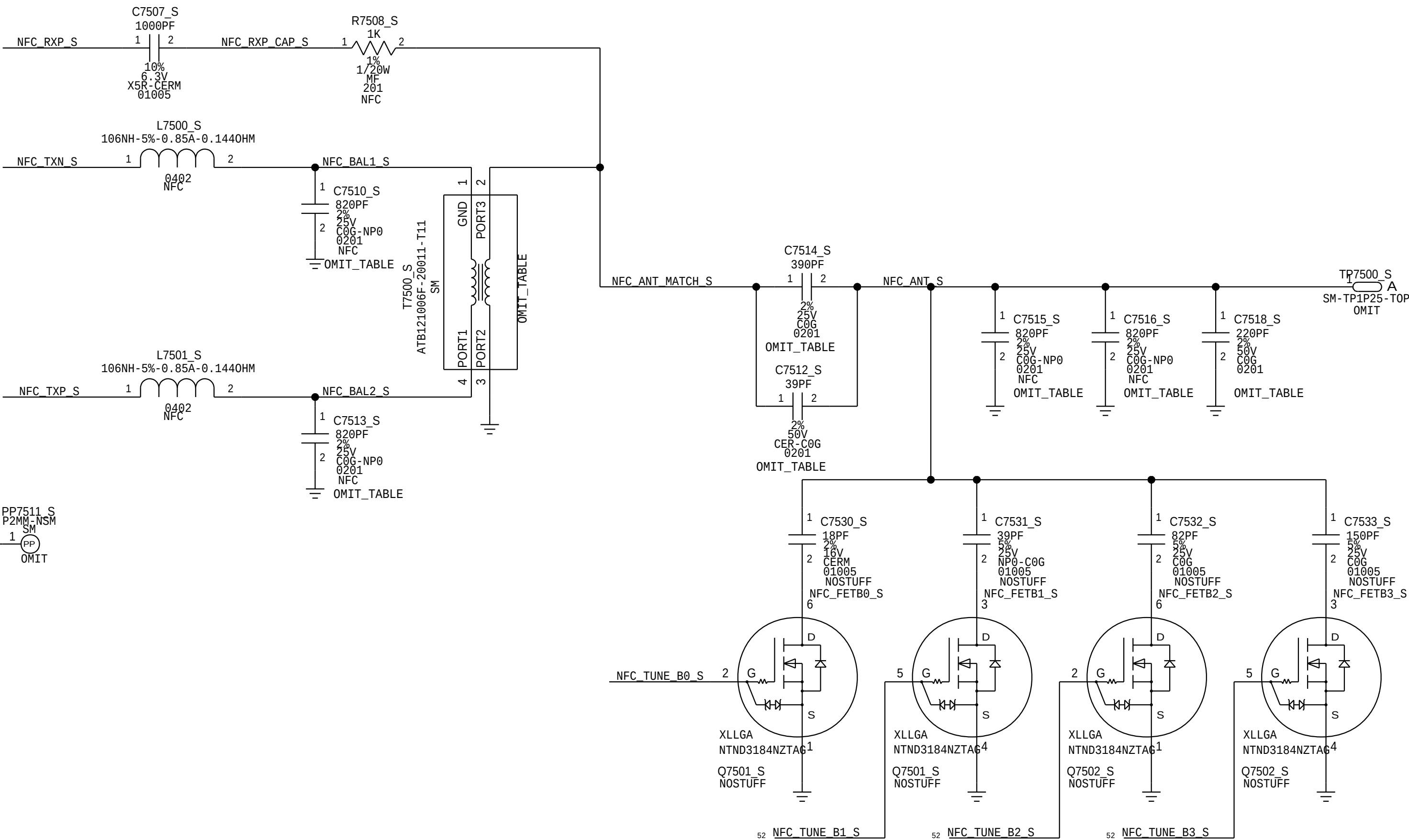
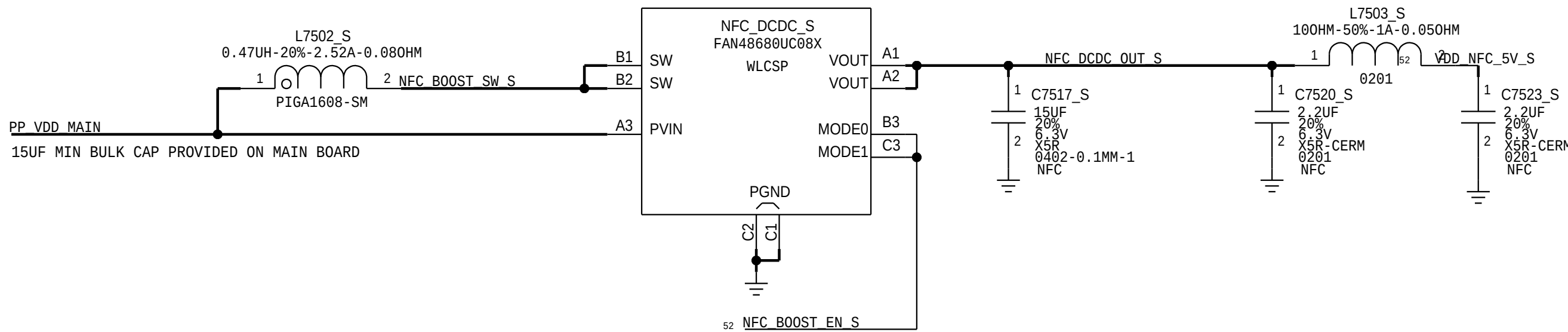
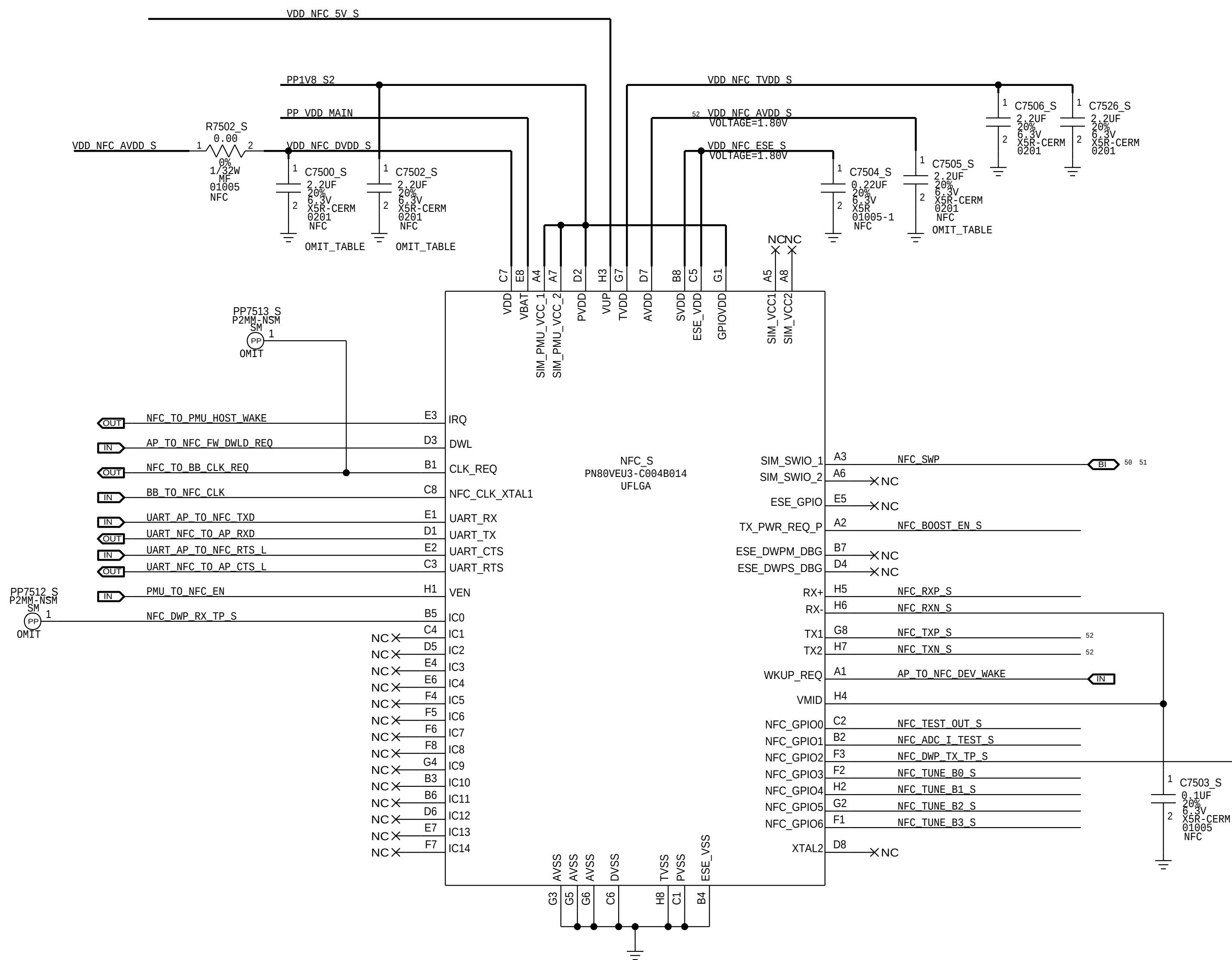
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STOCKHOLM

5V BOOSTER

NFC CONTROLLER

NFC FRONT END



PAGE TITLE		
NFC		
DRAWING NUMBER		051-02159
REVISION		10.0.0
BRANCH		
PAGE		75 OF 75
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BASEBAND

D

C

B

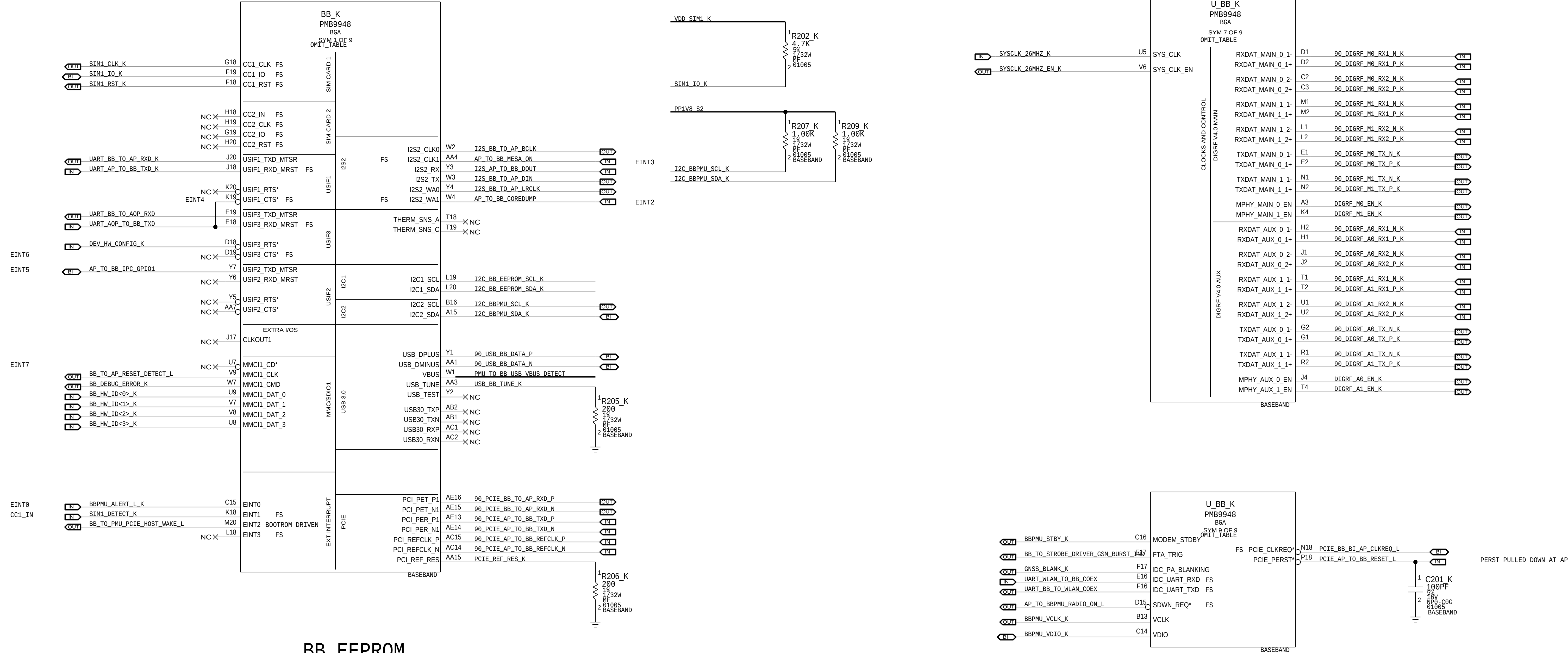
A

D

C

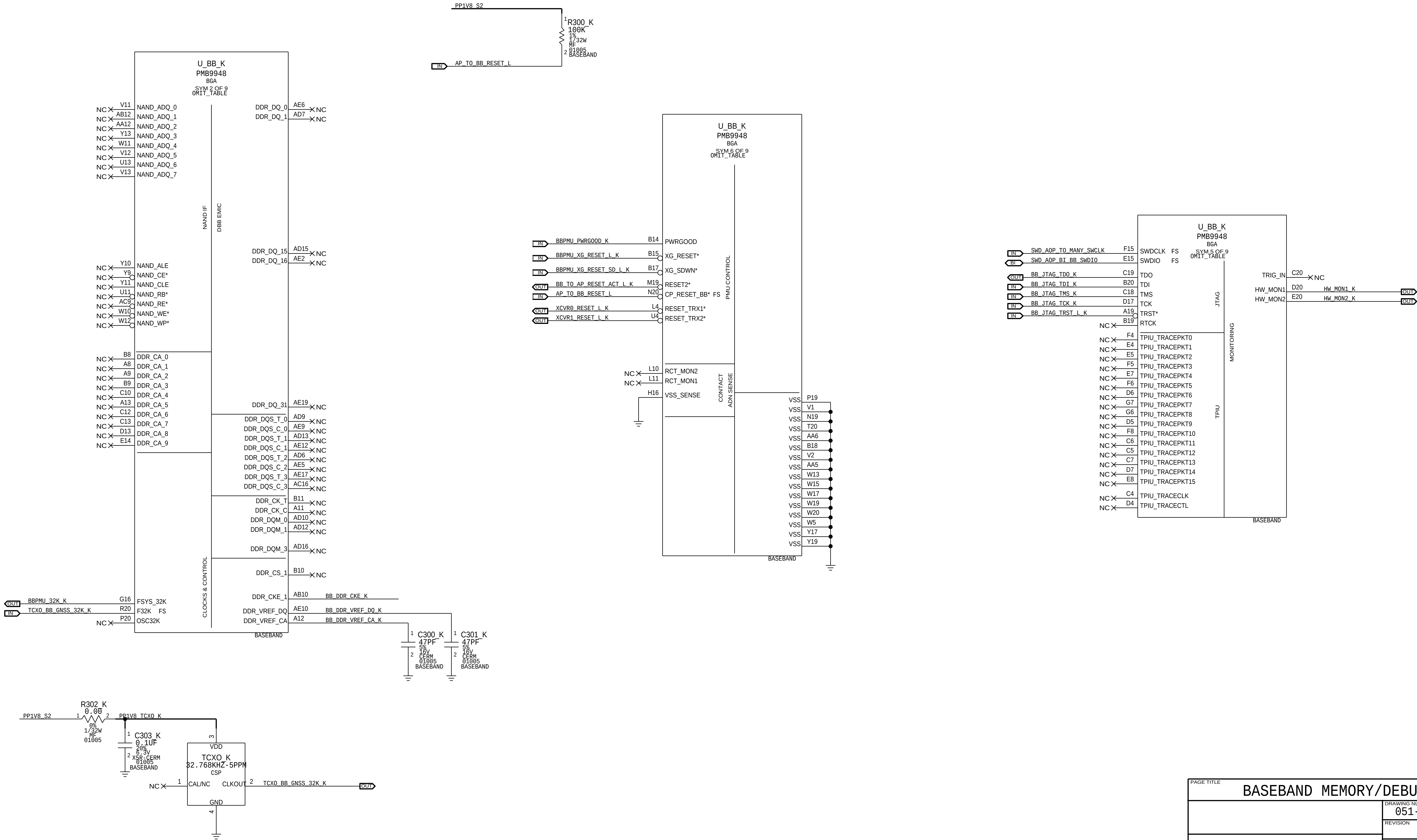
B

A



PAGE TITLE		
BASEBAND		
DRAWING NUMBER	051-02247	SIZE
REVISION	7.0.0	D
BRANCH	evt-1	
PAGE	2 OF 17	
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BASEBAND MEMORY/DEBUG



PAGE TITLE		
BASEBAND MEMORY/DEBUG		
DRAWING NUMBER	051-02247	SIZE
REVISION	7.0.0	D
BRANCH	evt-1	
PAGE	3 OF 17	
SHEET	15 OF 34	

BASEBAND POWER

D

C

B

A

D

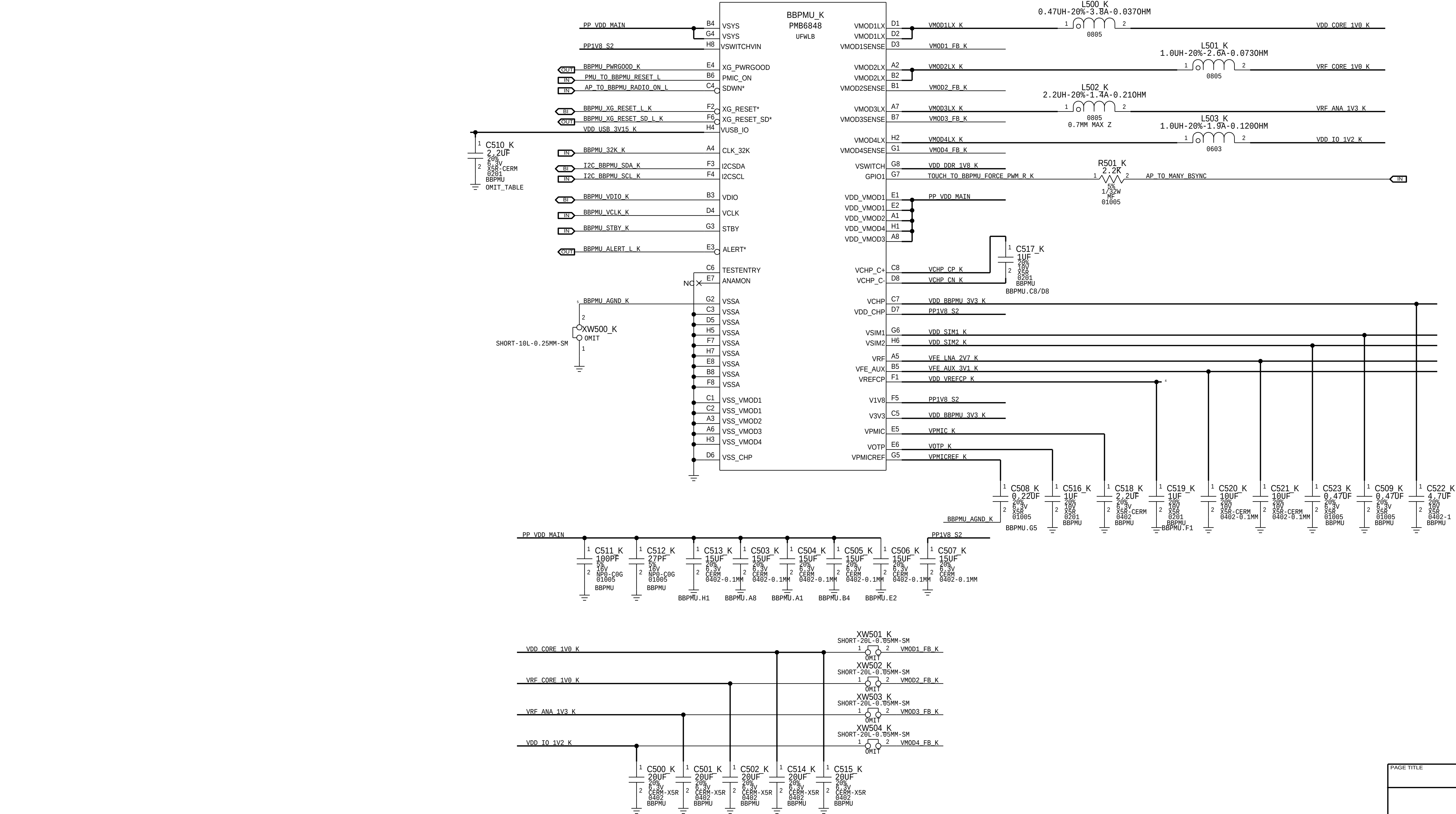
C

B

A

PAGE TITLE			
BASEBAND POWER			
DRAWING NUMBER		051-02247	SIZE D
REVISION		7.0.0	
BRANCH		evt-1	
PAGE		4 OF 17	
SHEET		16 OF 34	

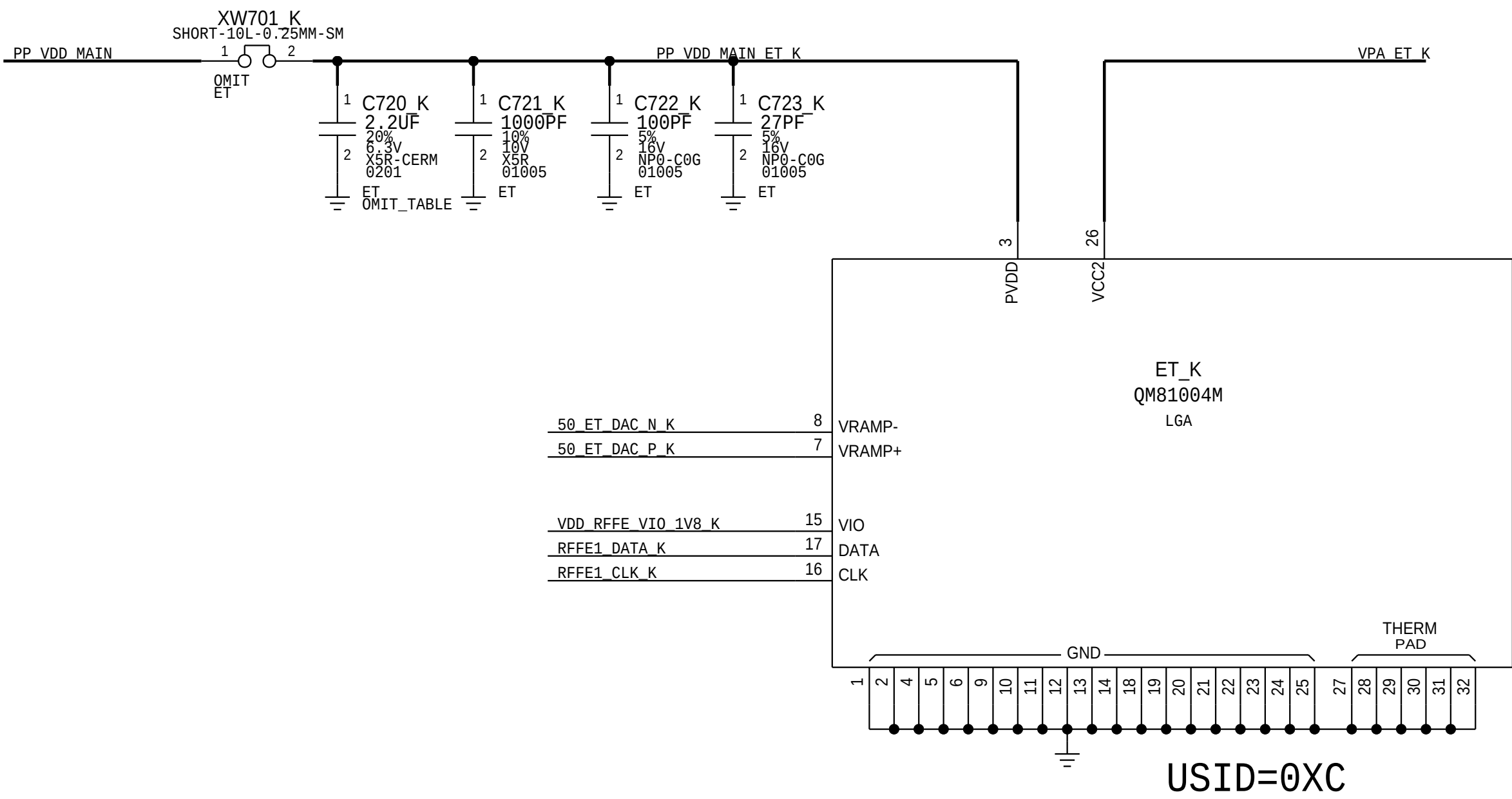
BASEBAND PMIC



PAGE TITLE		
BASEBAND PMIC		
DRAWING NUMBER	051-02247	SIZE
REVISION	7.0.0	D
BRANCH	evt-1	
PAGE	5 OF 17	
SHEET	17 OF 34	

ET MODULATOR

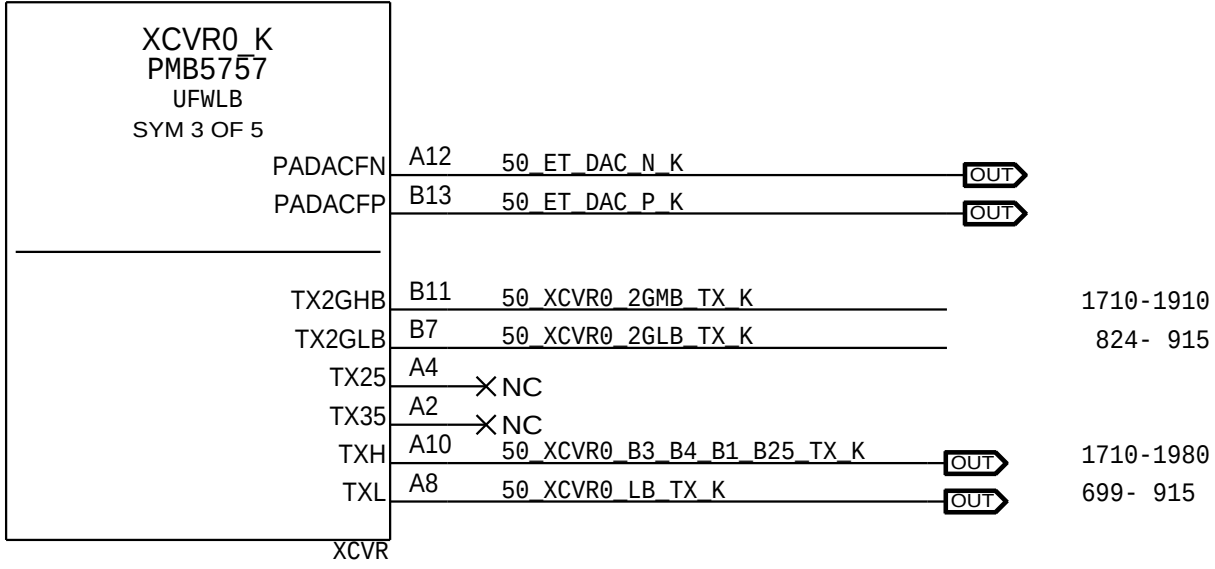
ALPES II MODULE



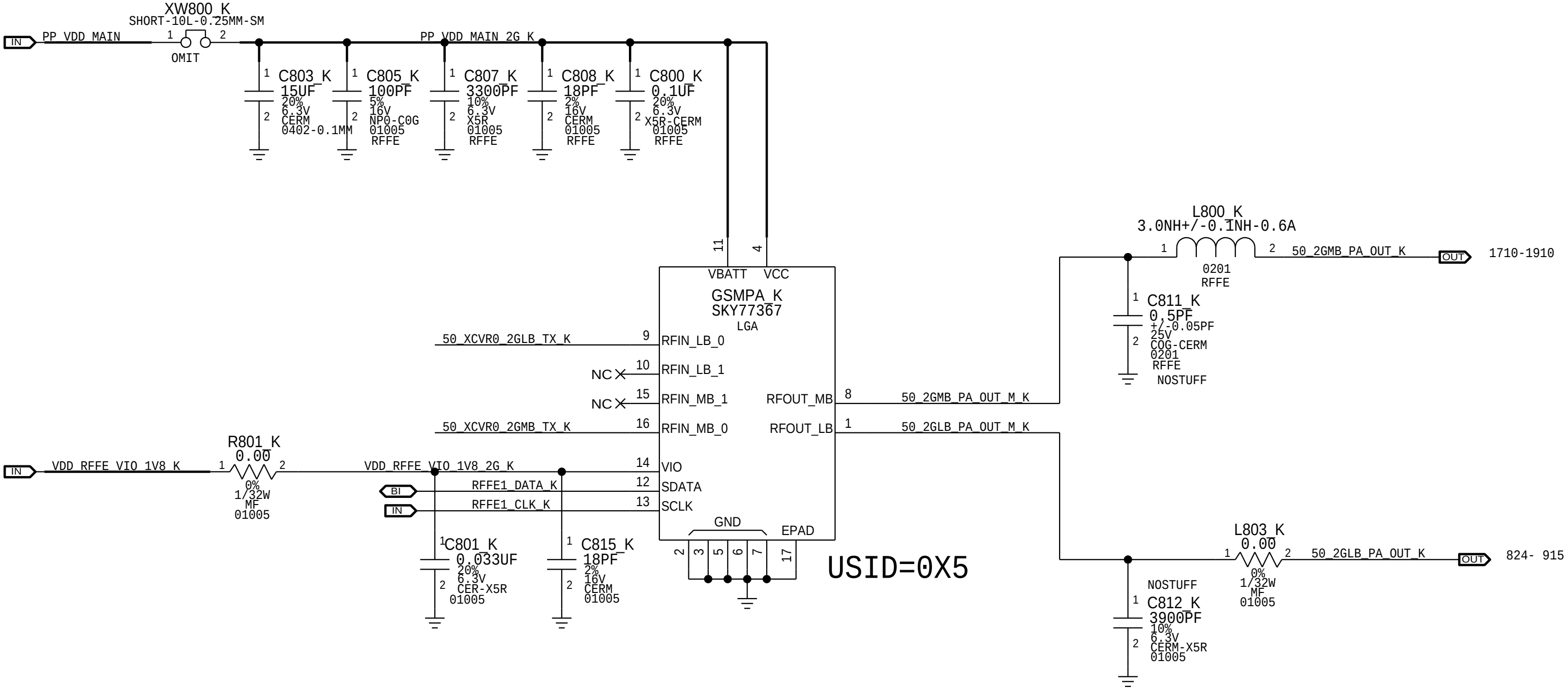
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ET MODULATOR		
	DRAWING NUMBER	051-02247
	REVISION	7.0.0
	BRANCH	evt-1
	PAGE	7 OF 17
	SHEET	19 OF 34

TDD TRANSMIT

XCVR0 TX

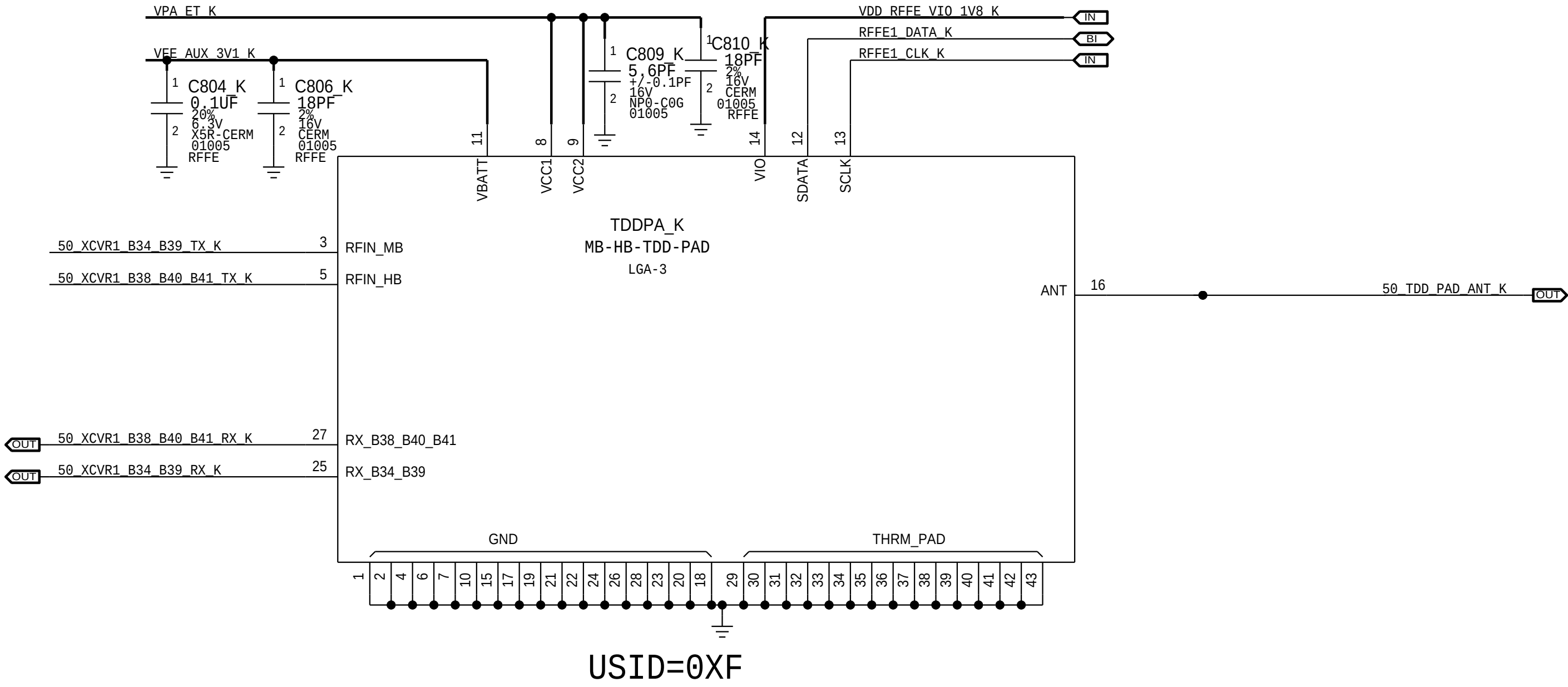
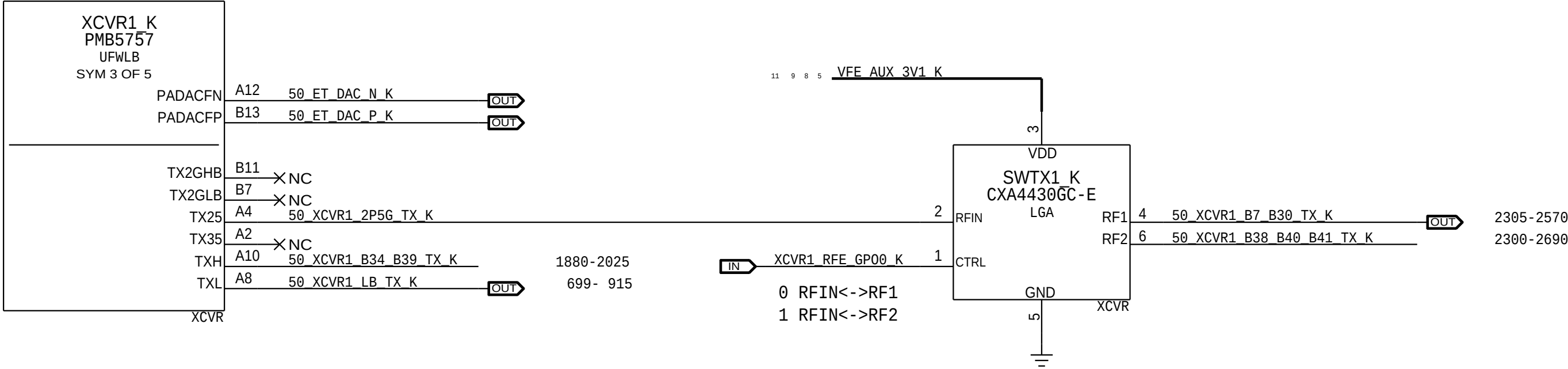


2G PA



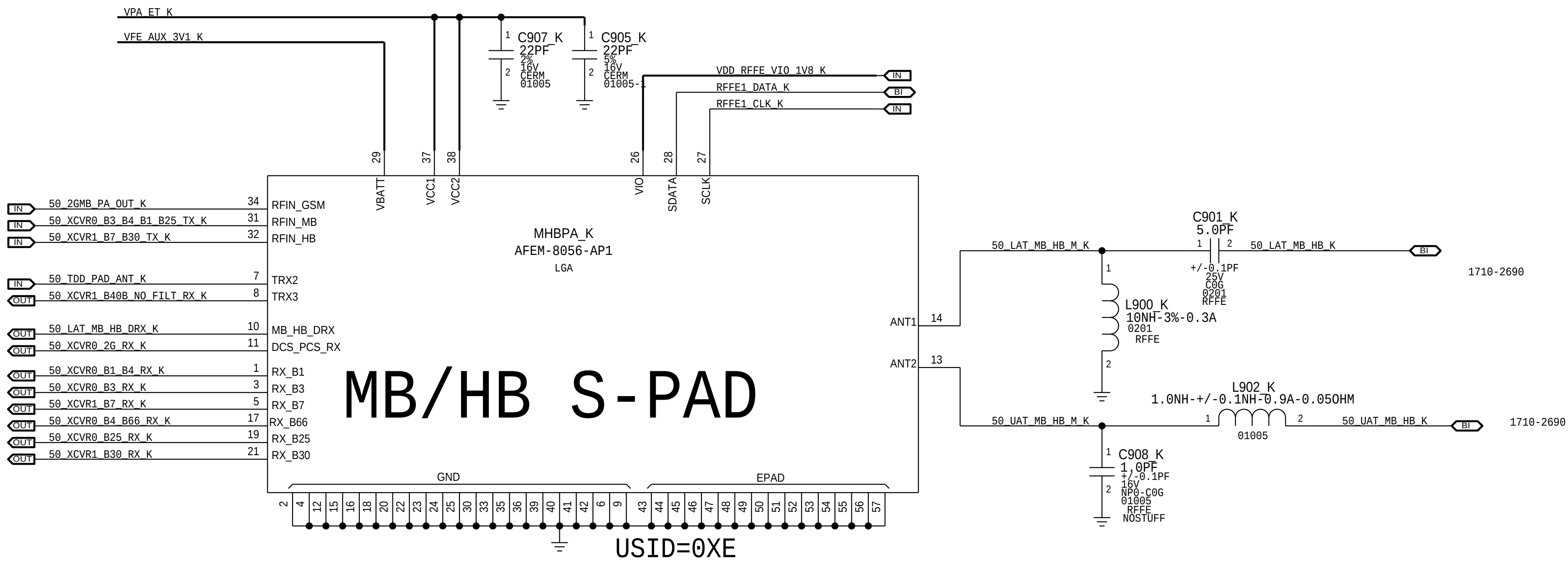
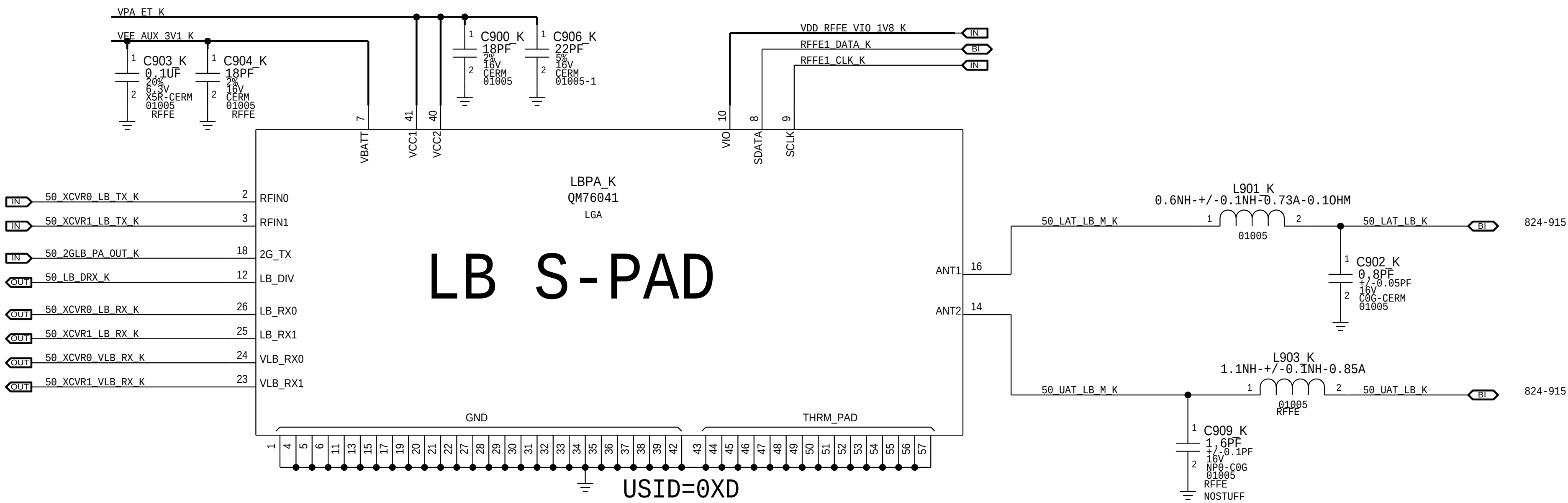
MB HB TDD S-PAD

XCVR1 TX



PAGE TITLE		
TDD TRANSMIT		
DRAWING NUMBER	051-02247	SIZE D
REVISION	7.0.0	
BRANCH	evt-1	
PAGE	8 OF 17	
SHEET	20 OF 34	

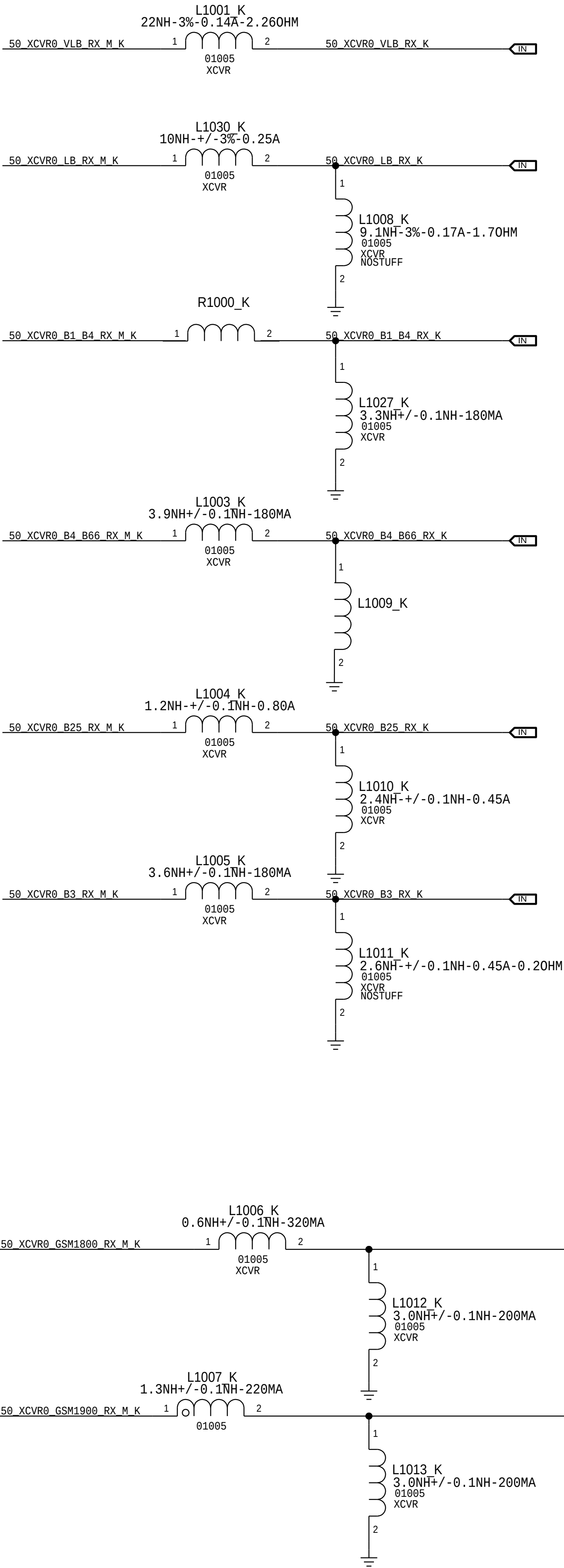
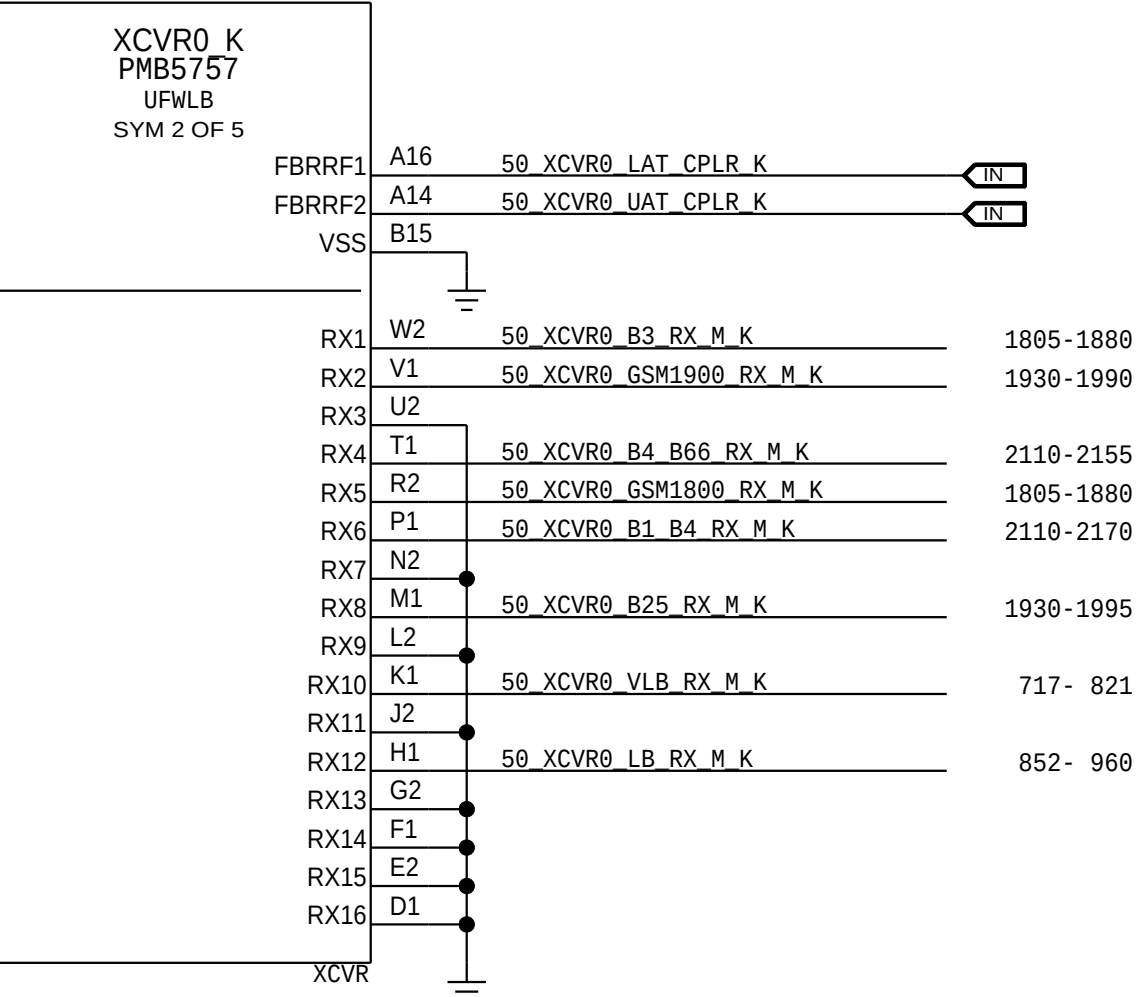
FDD TRANSMIT



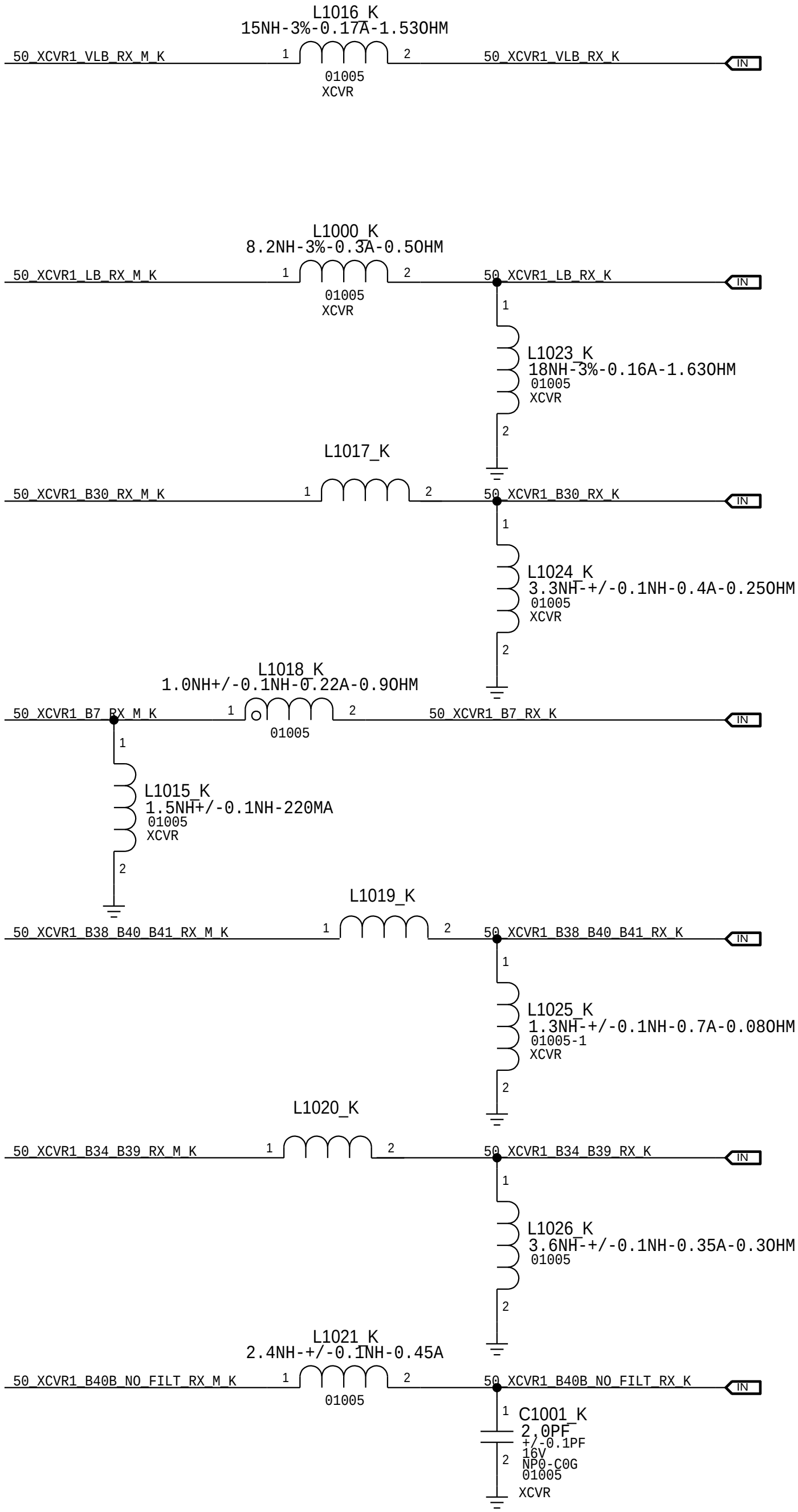
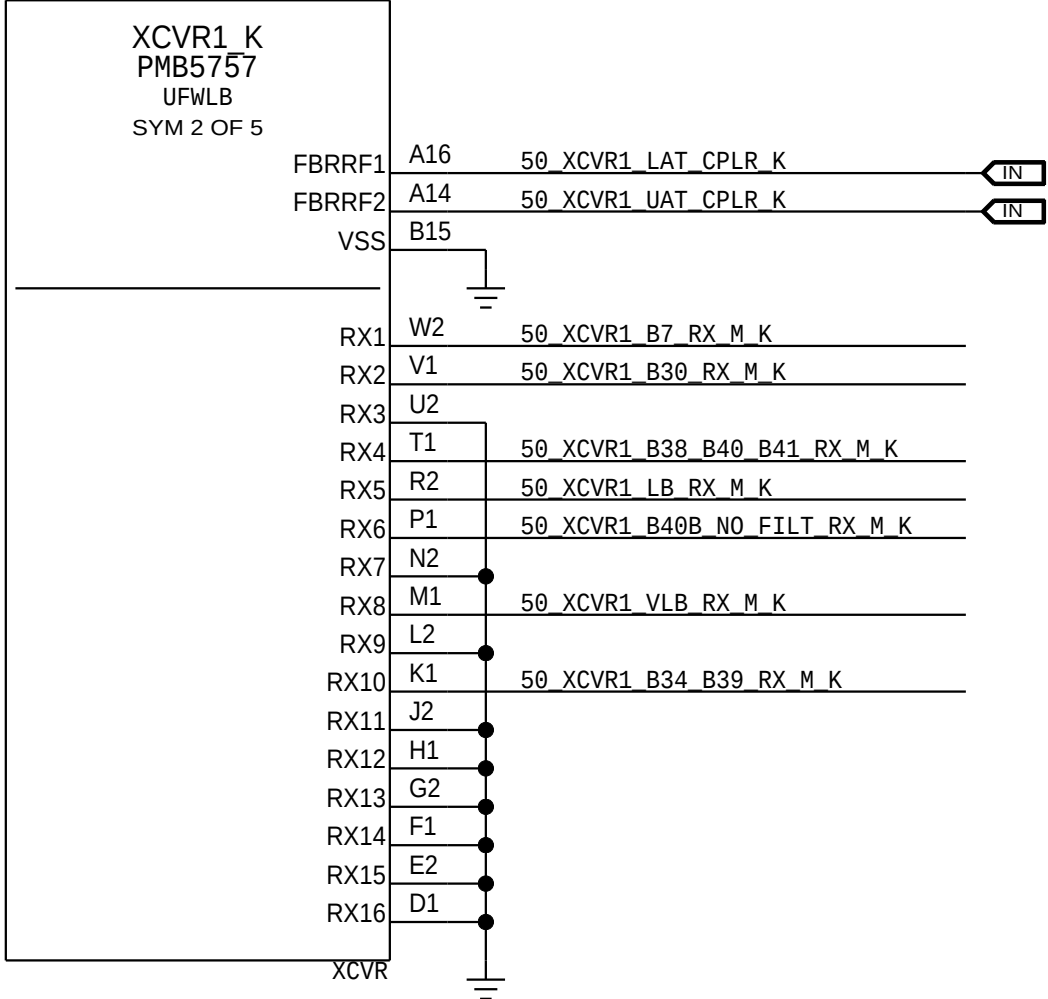
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	DRAWING NUMBER	051-02247
	REVISION	7.0.0
	BRANCH	evt-1
	PAGE	9 OF 17
	SHEET	21 OF 34

PRIMARY RECIEVE

XCVR0 PRX



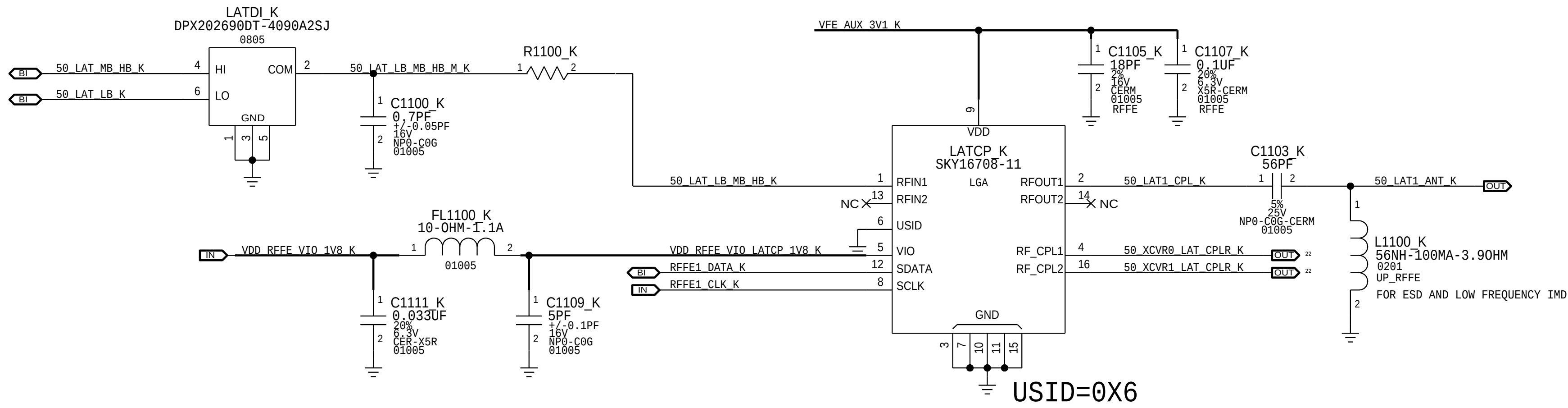
XCVR1 PRX



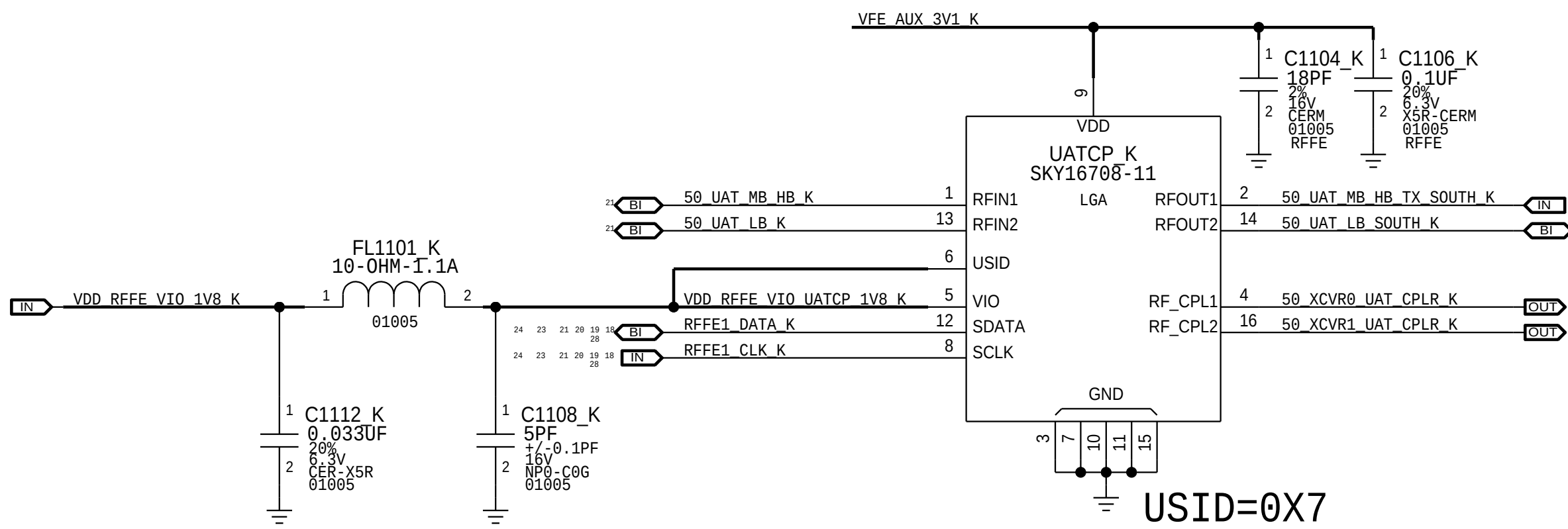
PRIMARY RECEIVE		
	DRAWING NUMBER	051-02247
	REVISION	7.0.0
	BRANCH	evt-1
	PAGE	10 OF 17
		SHEET
		22 OF 34

LOWER ANTENNA AND COUPLER

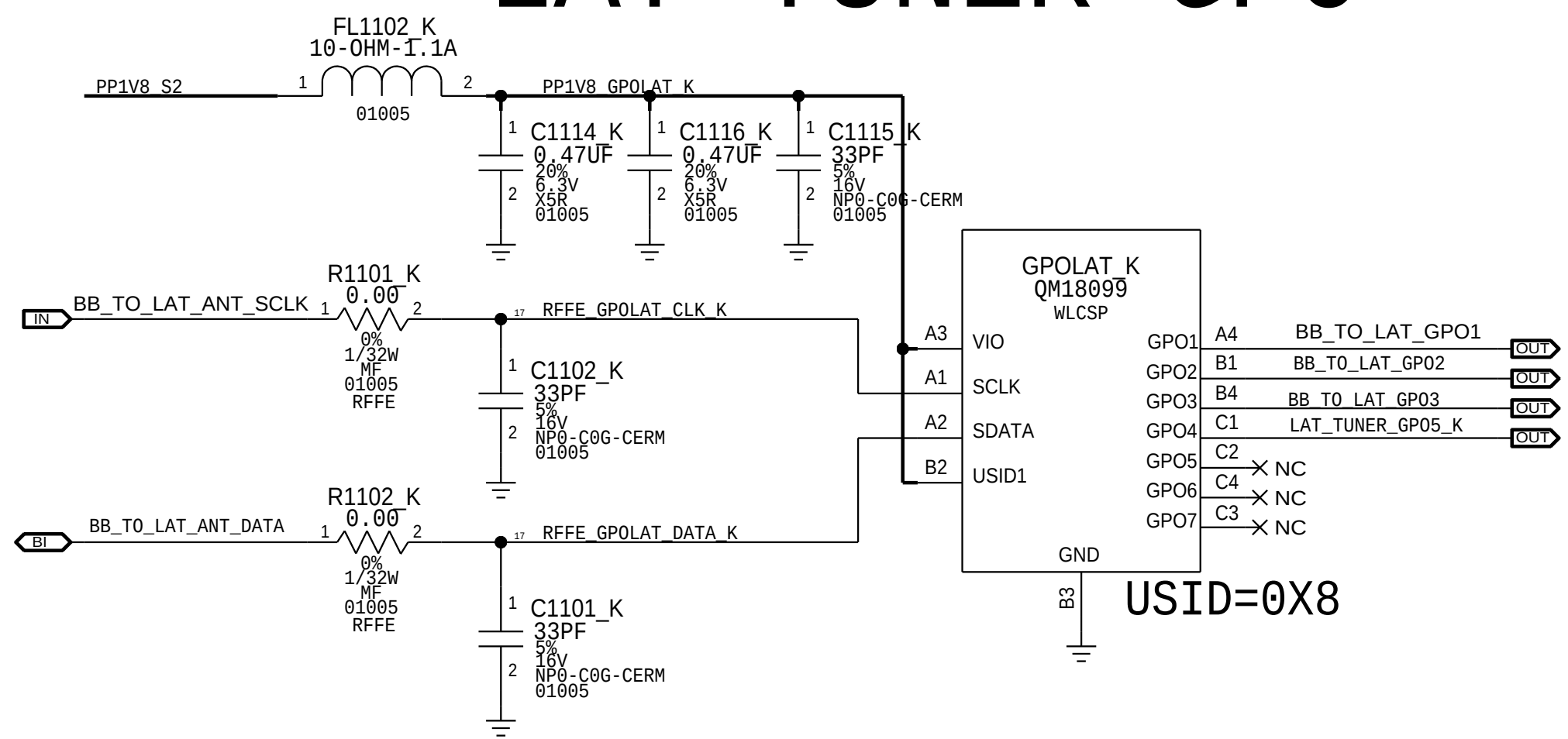
LOWER ANTENNA COUPLER



UPPER ANTENNA COUPLER



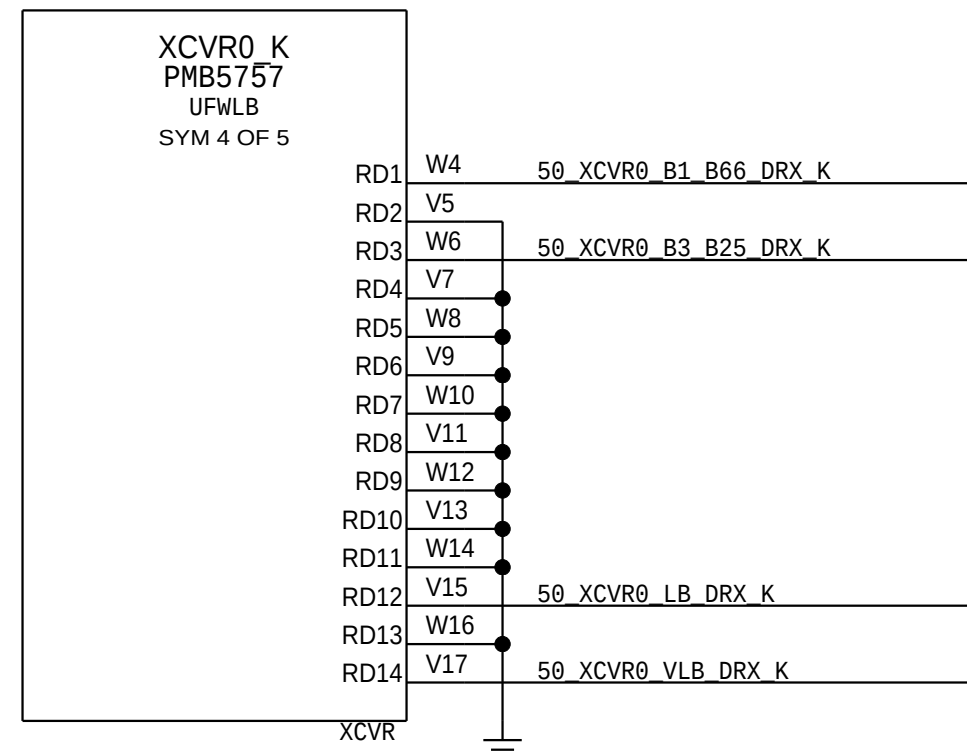
LAT TUNER GPO



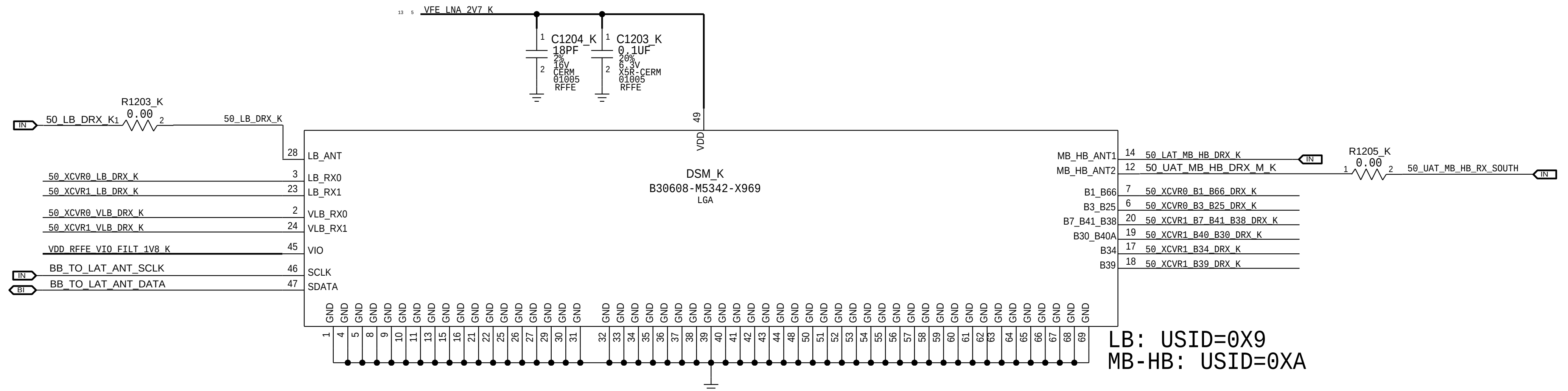
PAGE TITLE		
LOWER ANTENNA & COUPLERS		
DRAWING NUMBER		SIZE
051-02247		D
REVISION		7.0.0
BRANCH		evt-1
PAGE		11 OF 17
SHEET		23 OF 34

DIVERSITY RECEIVE ASM'S

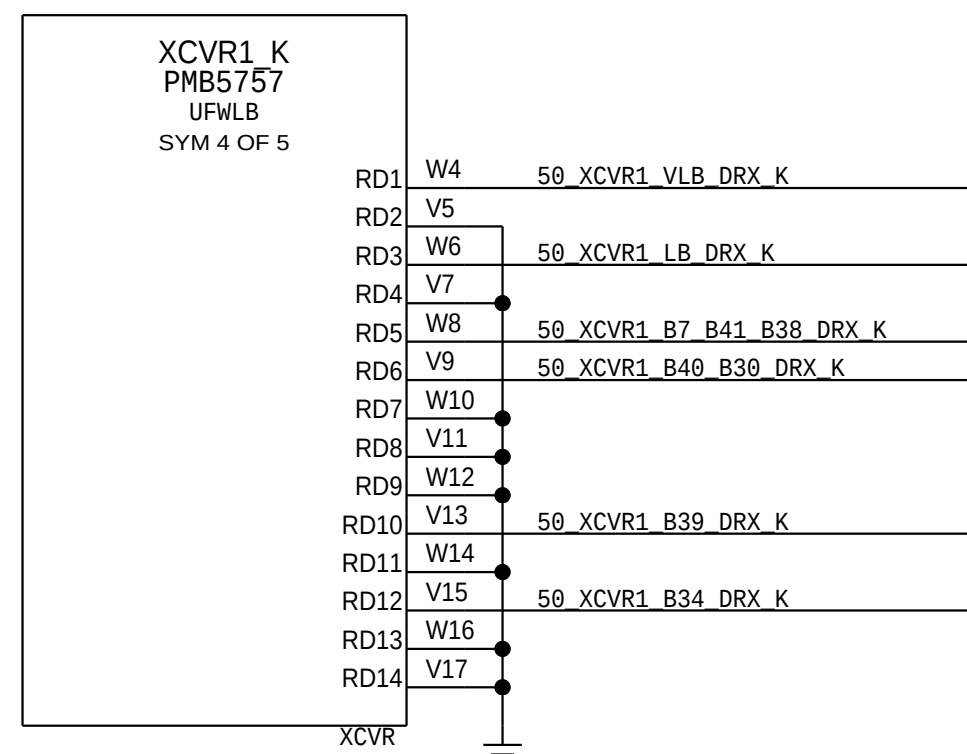
XCVR0 DRX



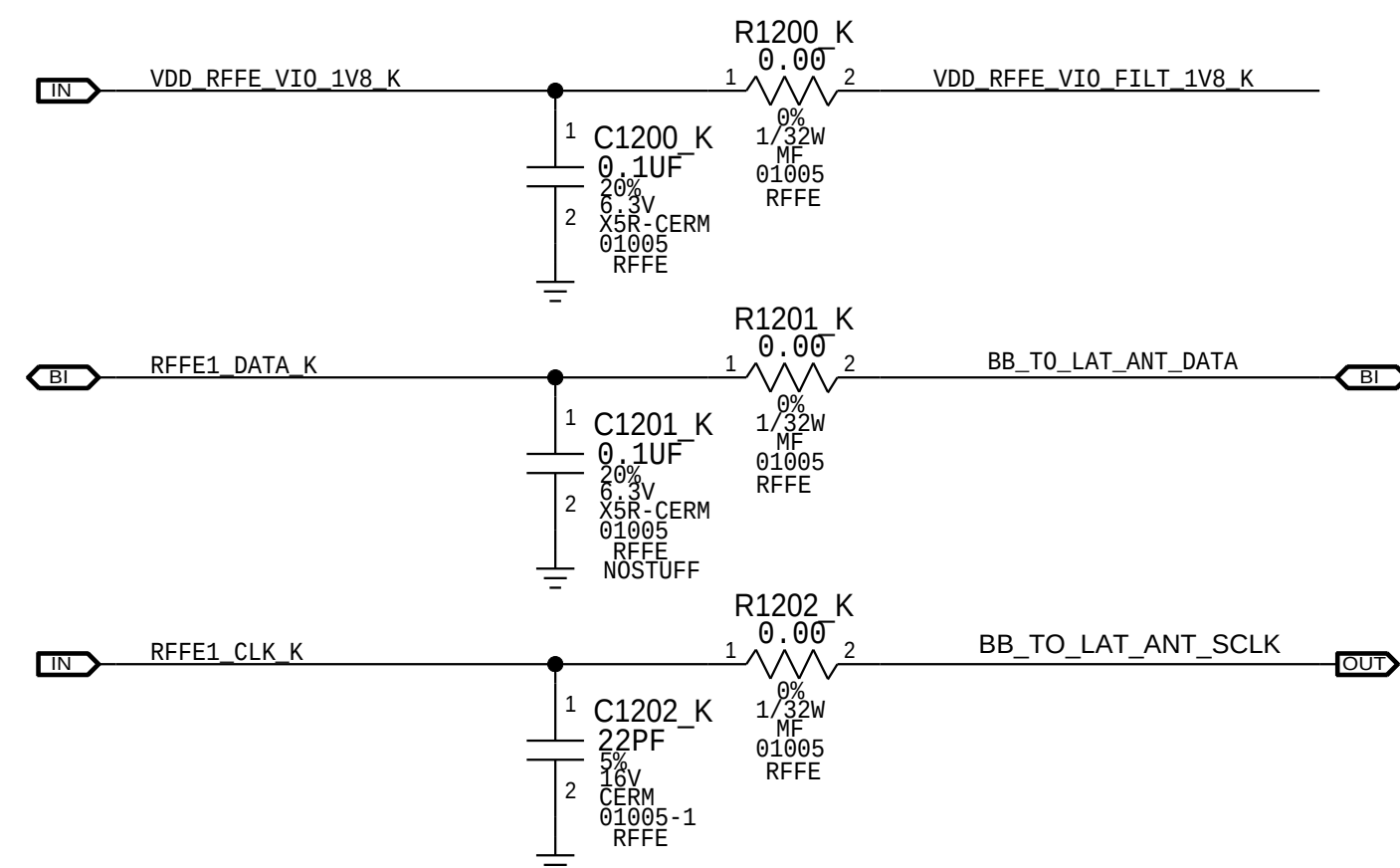
DRX DSM



XCVR1 DRX

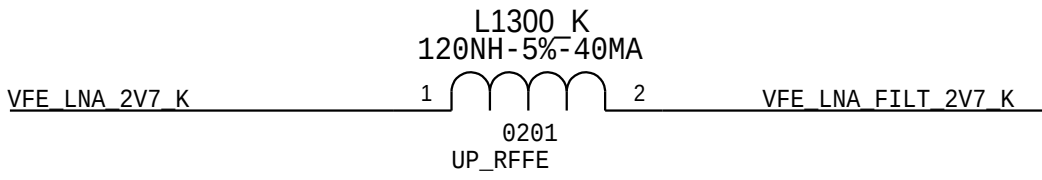


RFFE FILTERING

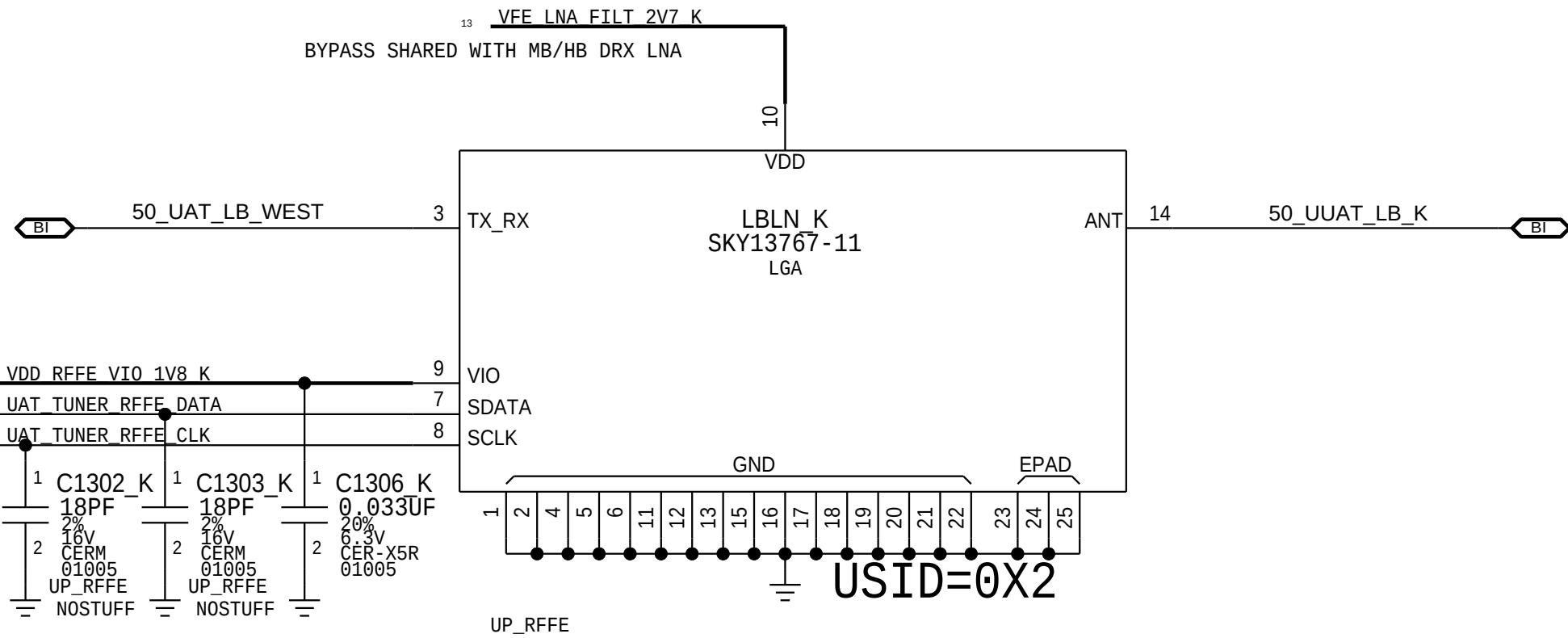


PAGE TITLE		
DIVERSITY RECEIVE ASM'S		
	DRAWING NUMBER	051-02247
	REVISION	D
	7.0.0	
	BRANCH	
	evt-1	
	PAGE	12 OF 17
	SHEET	24 OF 34

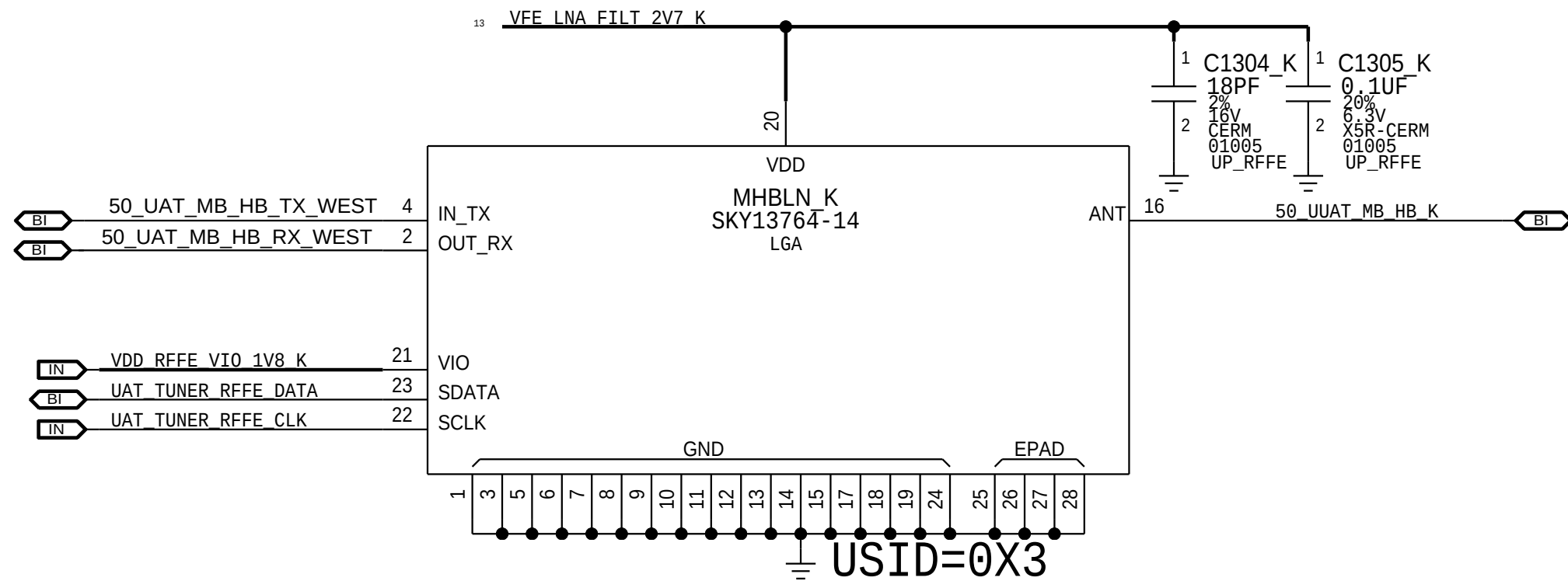
DIVERSITY RECEIVE LNAS



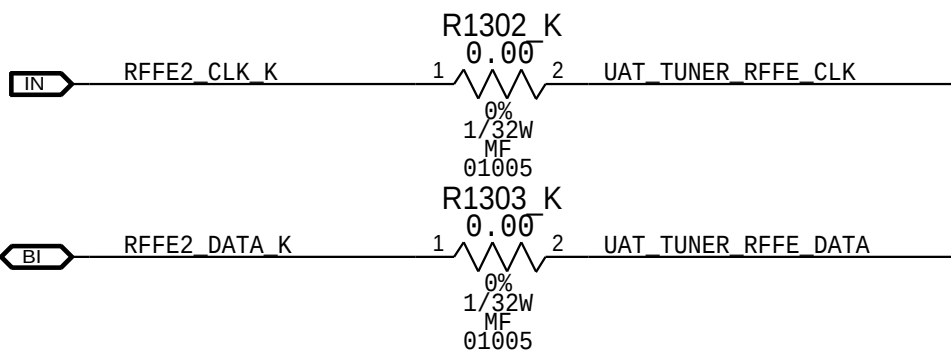
LB DRX LNA



MB/HB DRX LNA

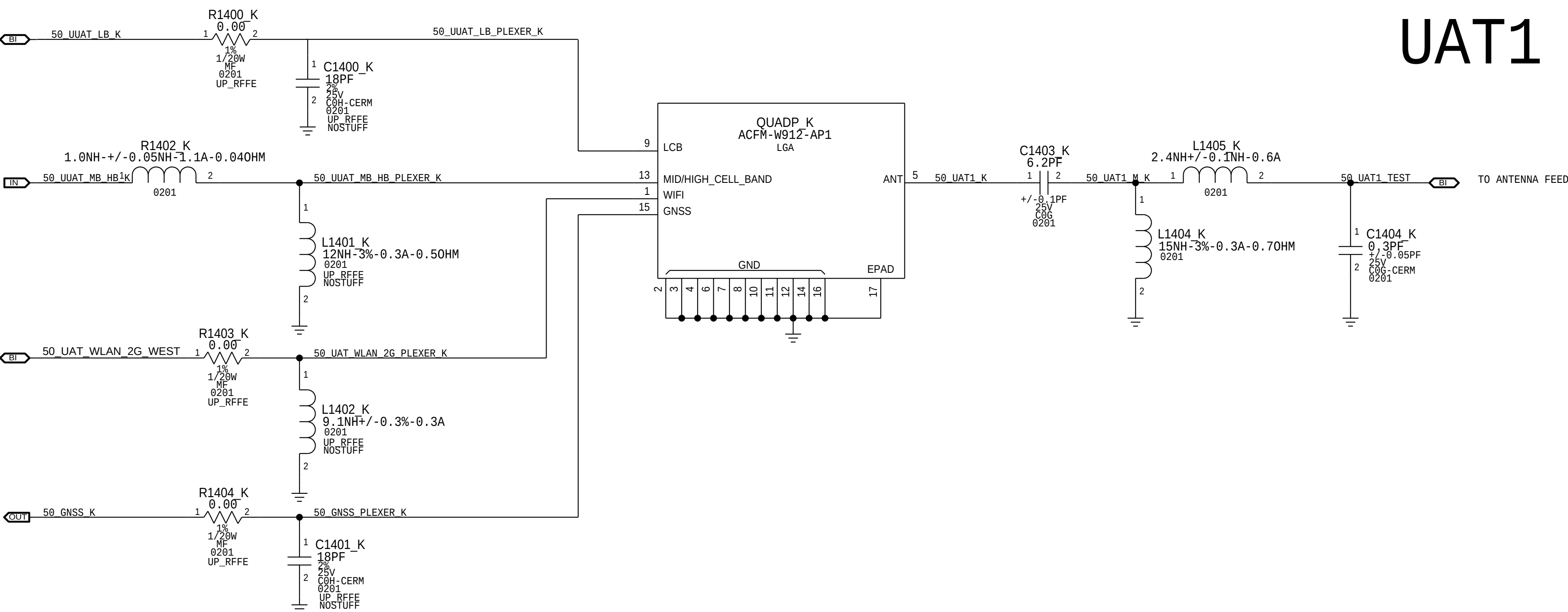


RFFE2 IMPEDANCE MATCHING

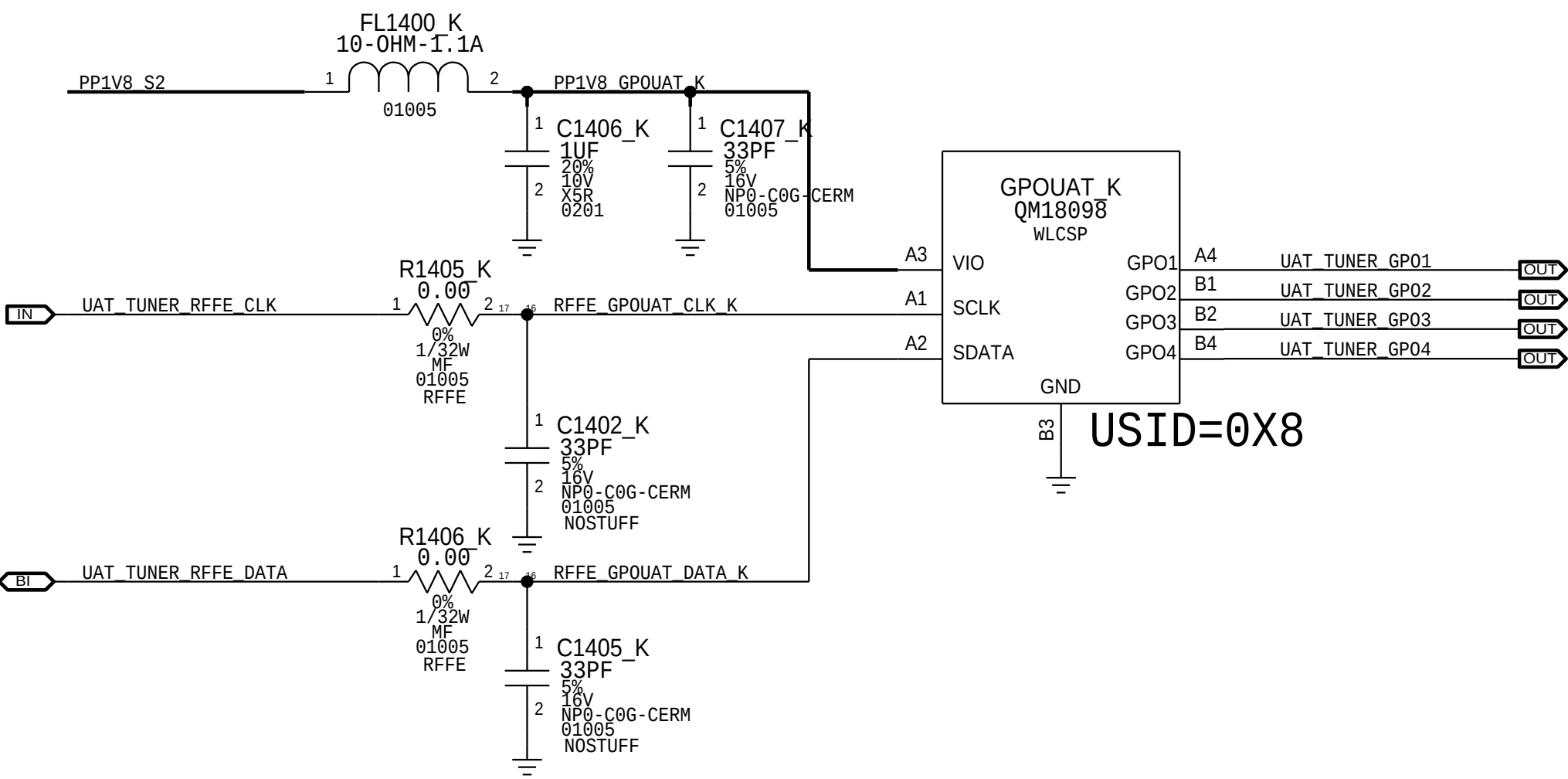


PAGE TITLE		
DIVERSITY RECEIVE LNA'S		
	DRAWING NUMBER	051-02247
	REVISION	7.0.0
	BRANCH	evt-1
	PAGE	13 OF 17
	SHEET	25 OF 34
	SIZE	D

UPPER ANTENNA FEEDS

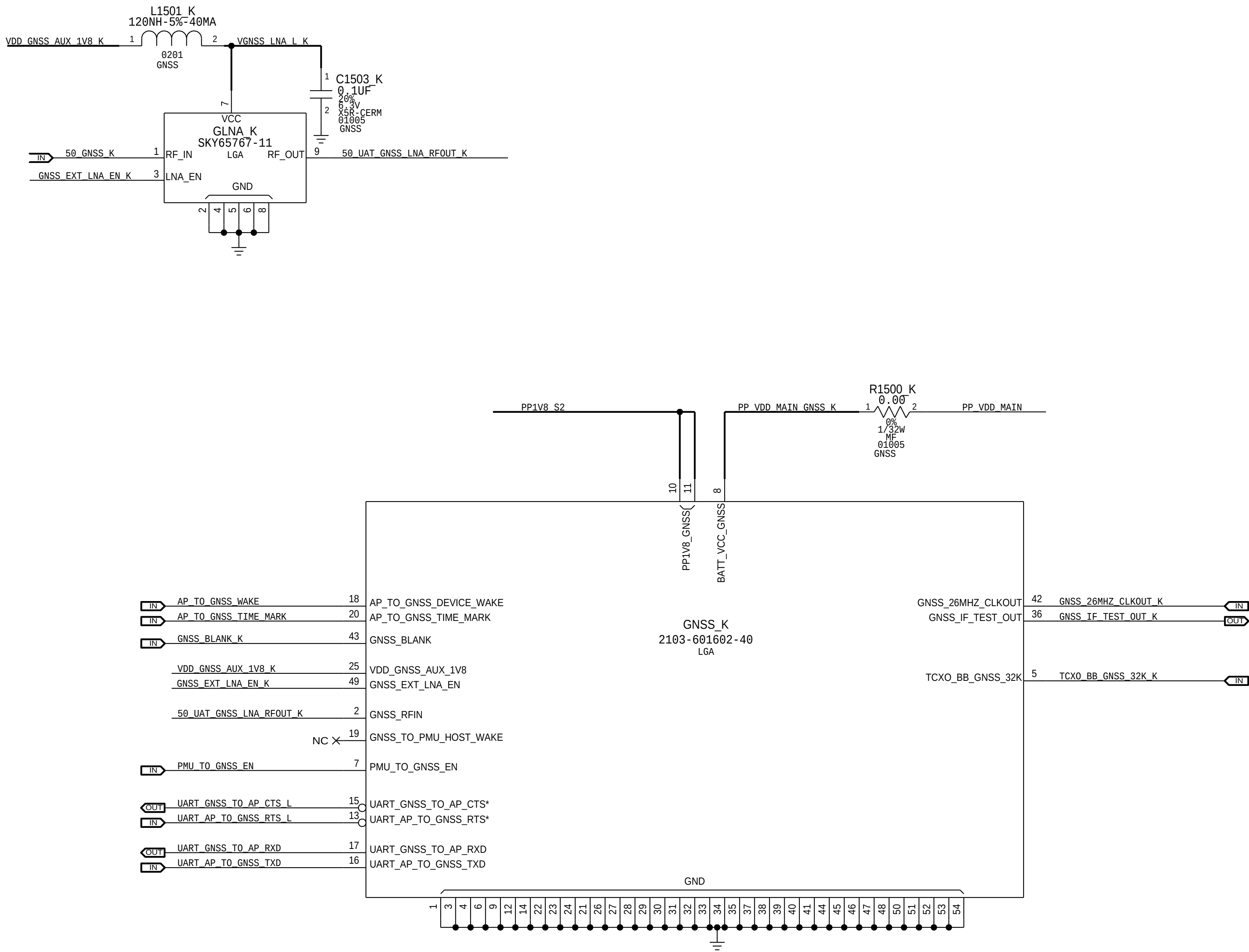


UAT TUNER GPO



PAGE TITLE		
UPPER ANTENNA FEEDS		
DRAWING NUMBER	051-02247	SIZE D
REVISION	7.0.0	
BRANCH	evt-1	
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SHEET	26 OF 34	

GNSS LNA + RECEIVER



PAGE TITLE		
GNSS		
	DRAWING NUMBER	051-02247
	REVISION	7.0.0
	BRANCH	evt-1
	PAGE	15 OF 17
	SHEET	27 OF 34
	SIZE	D

MLB TEST POINTS

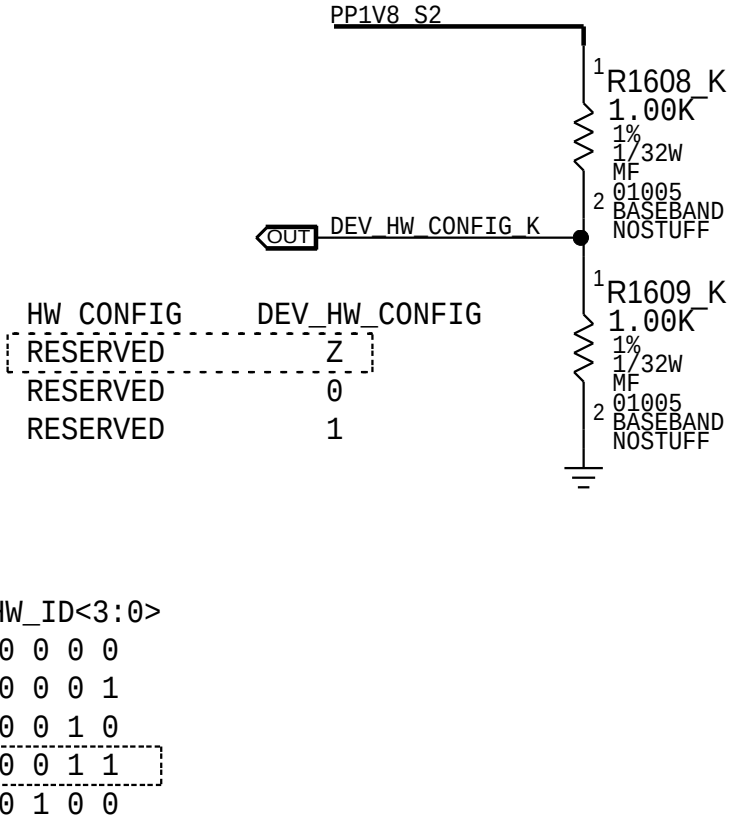
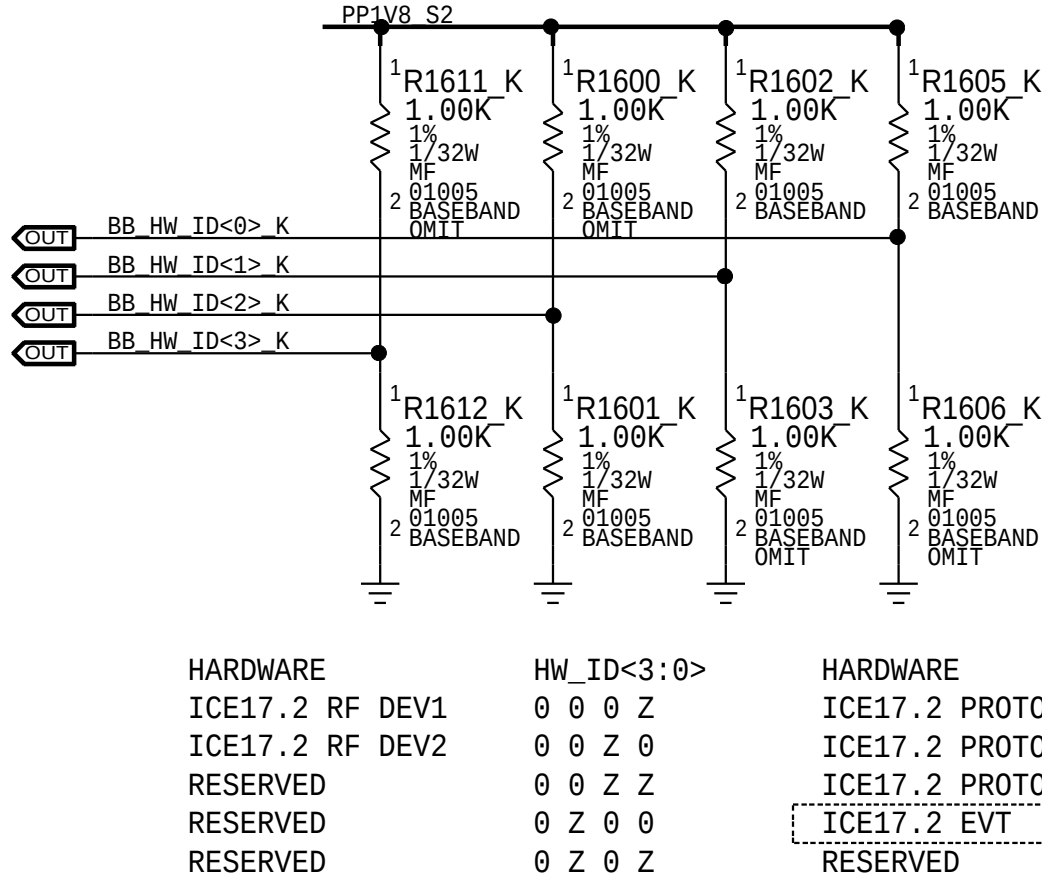
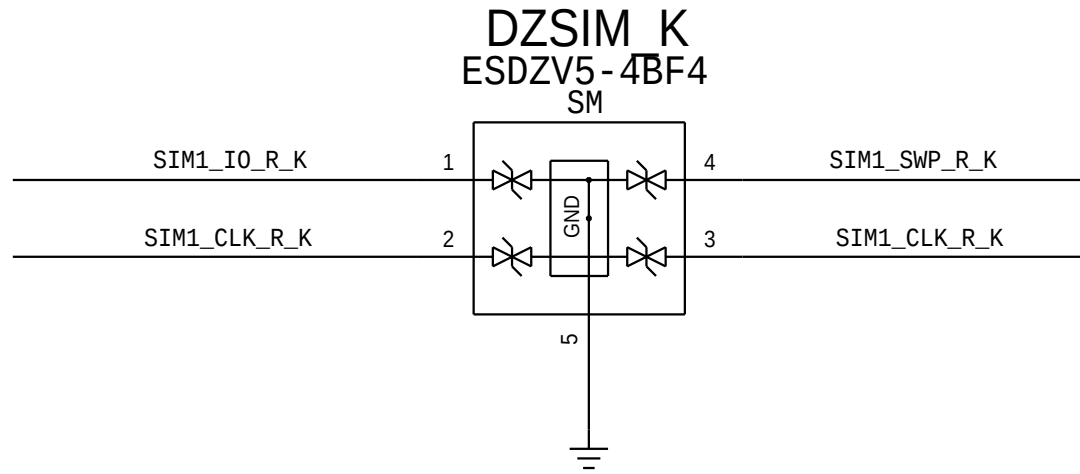
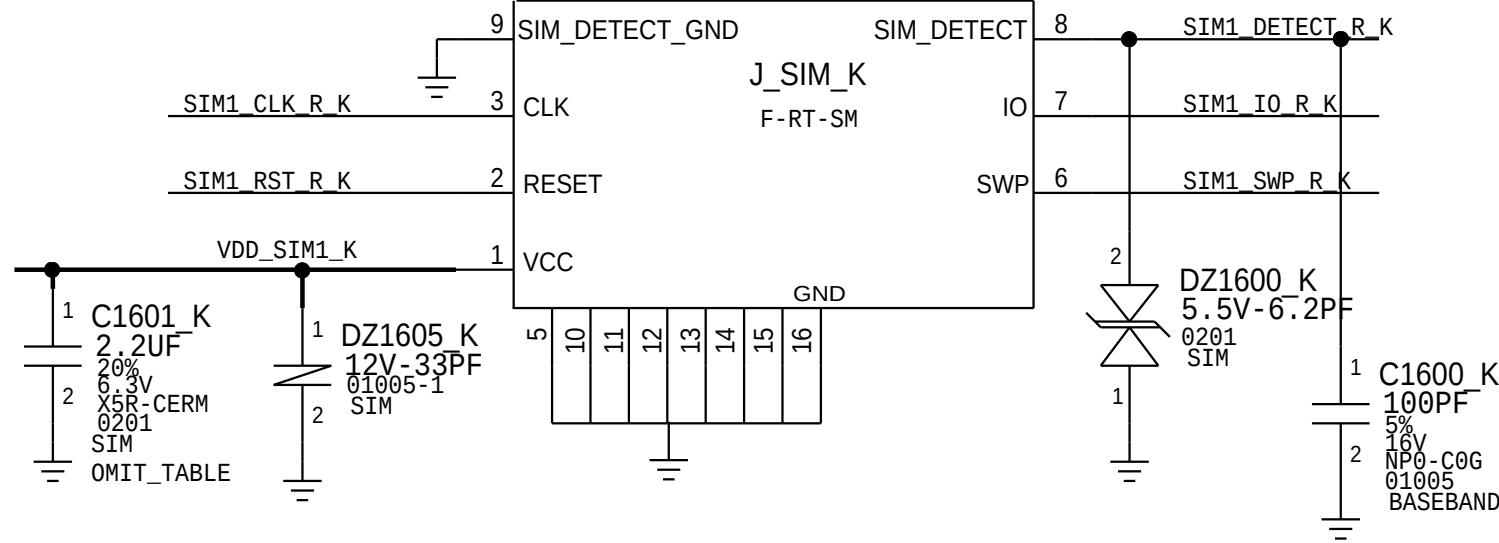
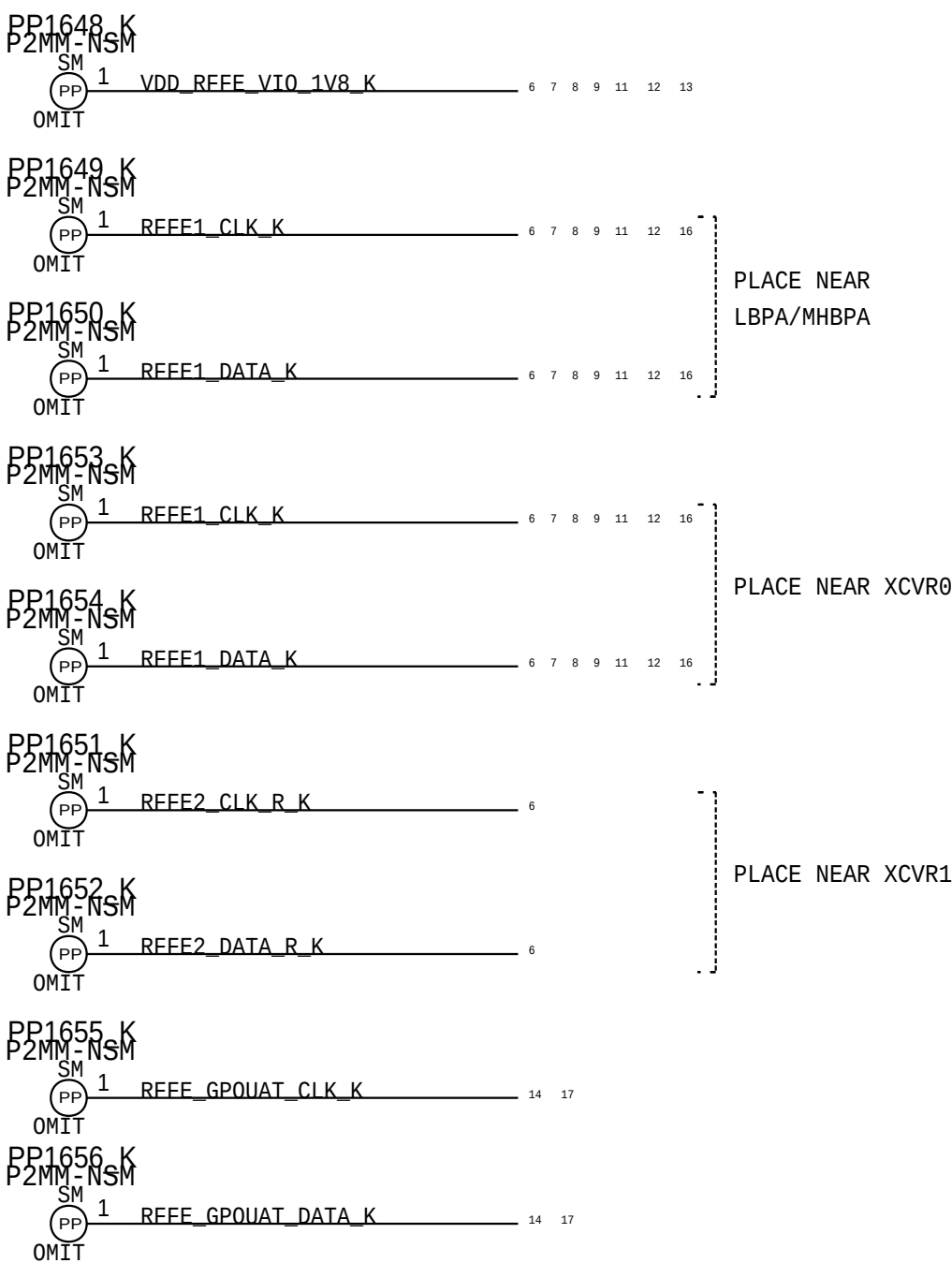
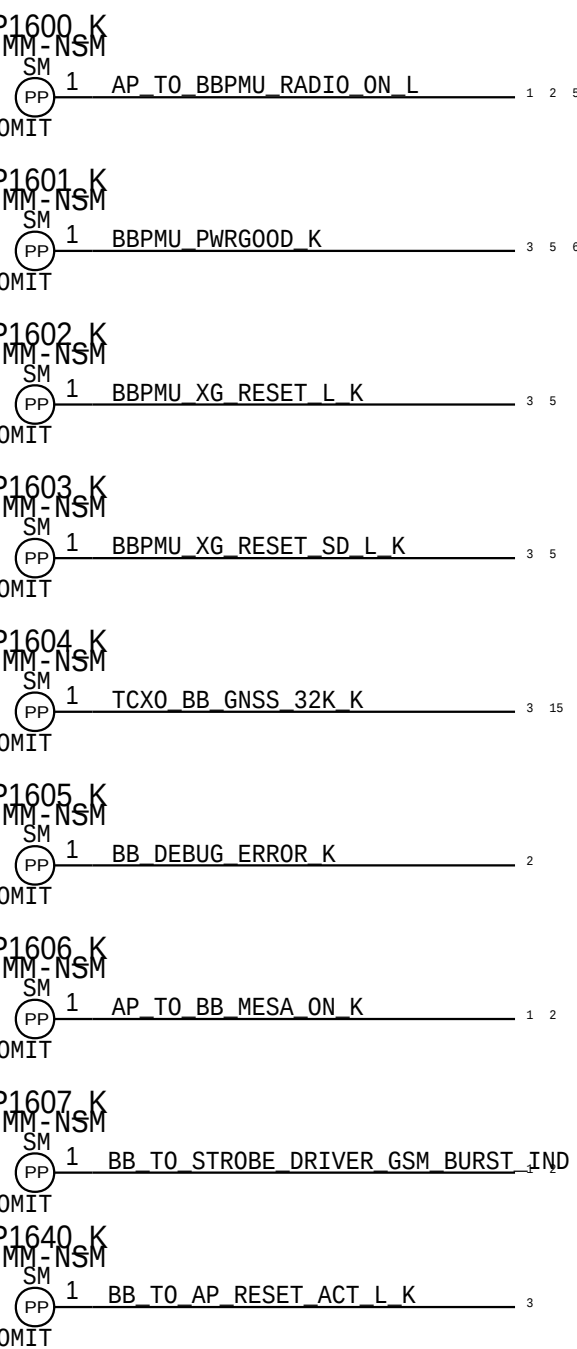
BASEBAND

RFFE

SIM CARD CONNECTOR

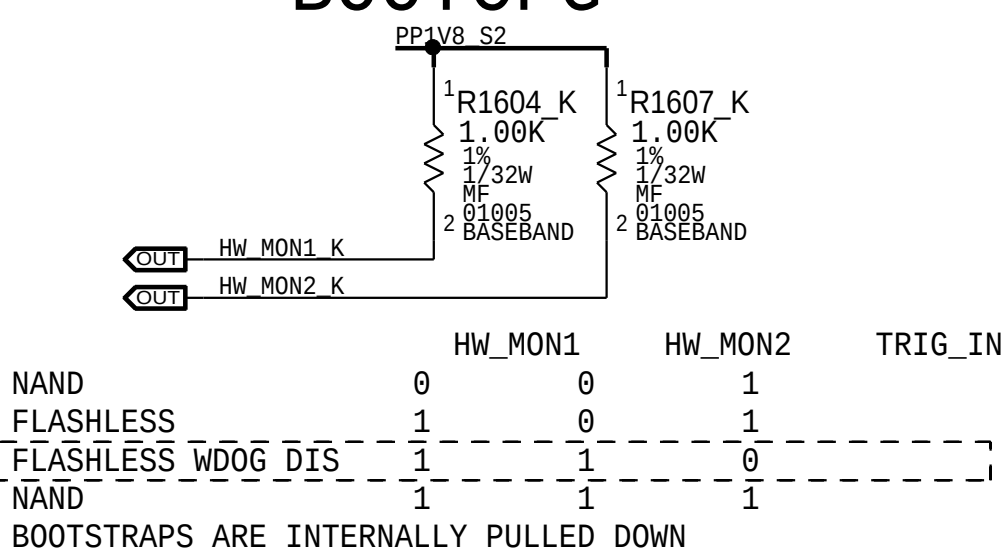
HARDWARE ID

DEV CONFIG



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RESERVED	0 Z 0 Z	RESERVED	0 1 0 0

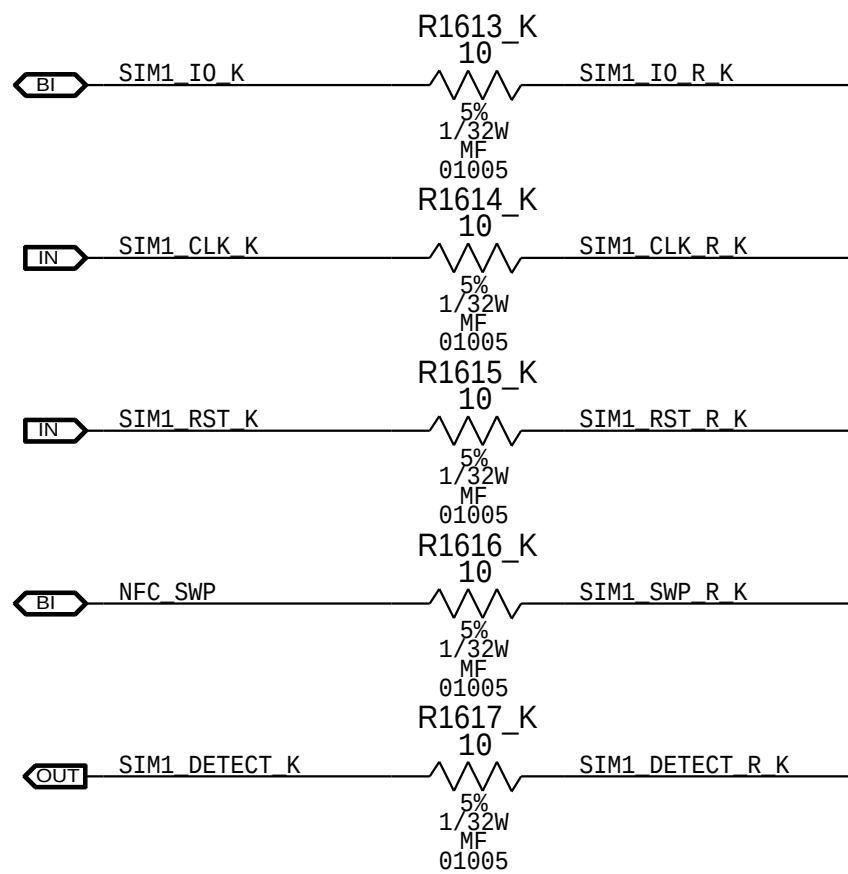
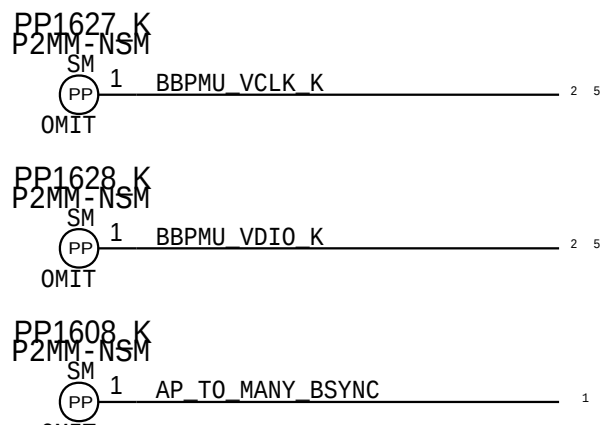
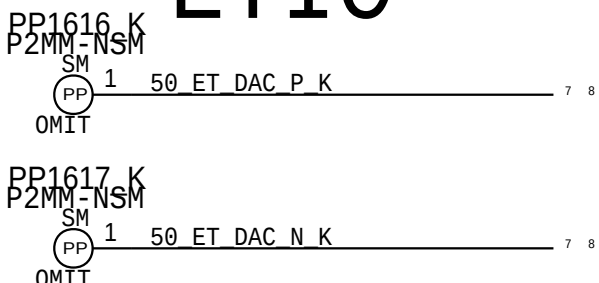
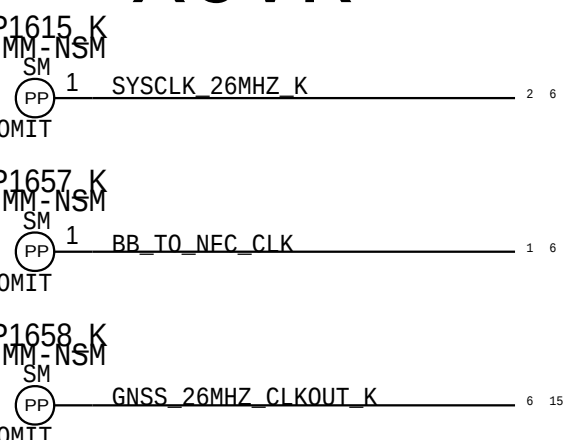
BOOTCFG



XCVR

ETIC

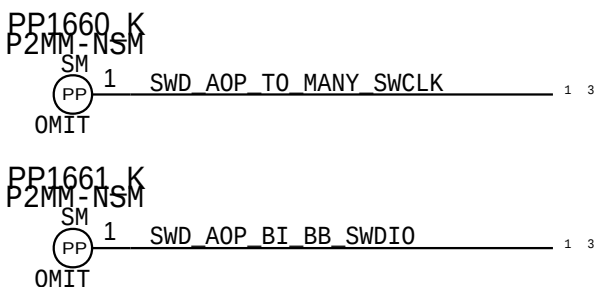
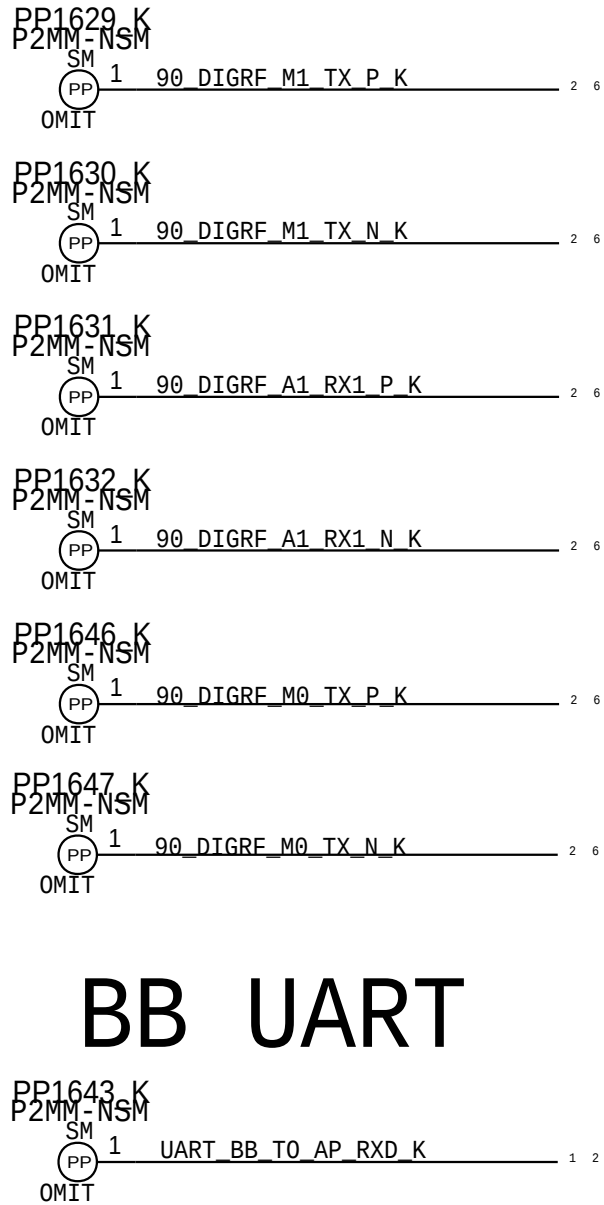
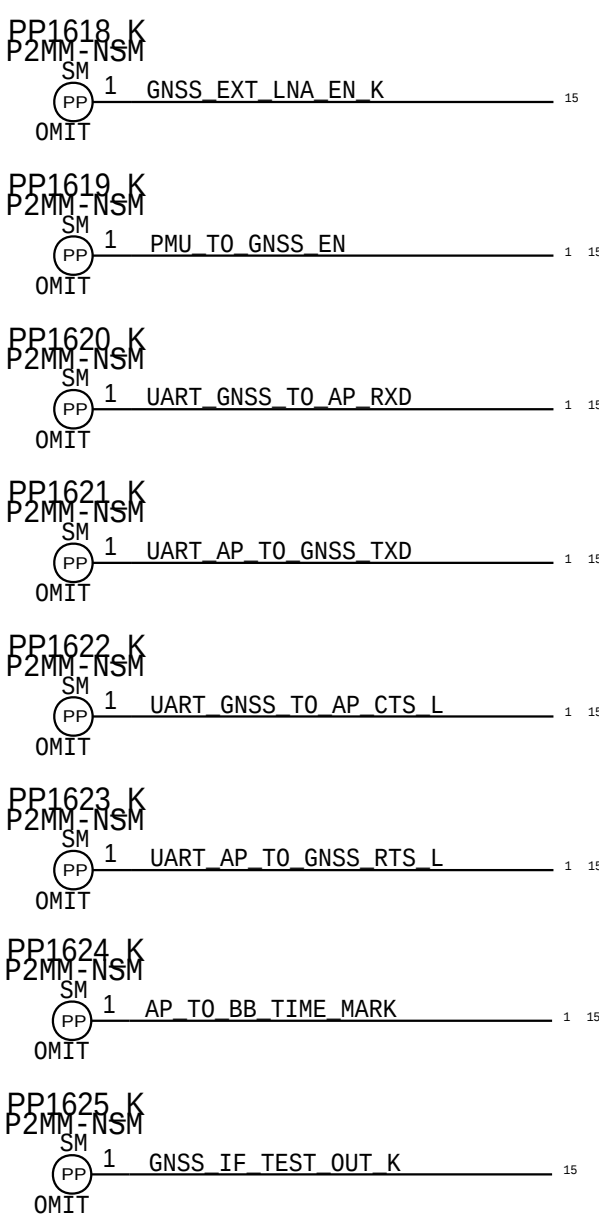
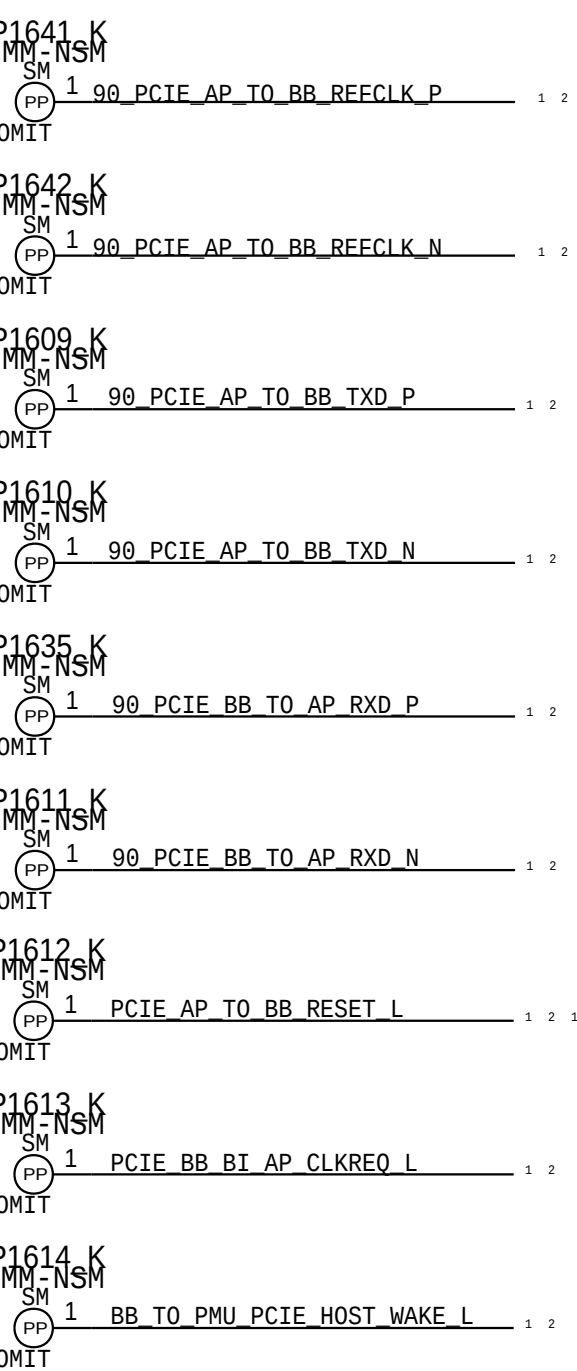
BBPMU



PCIE

GNSS

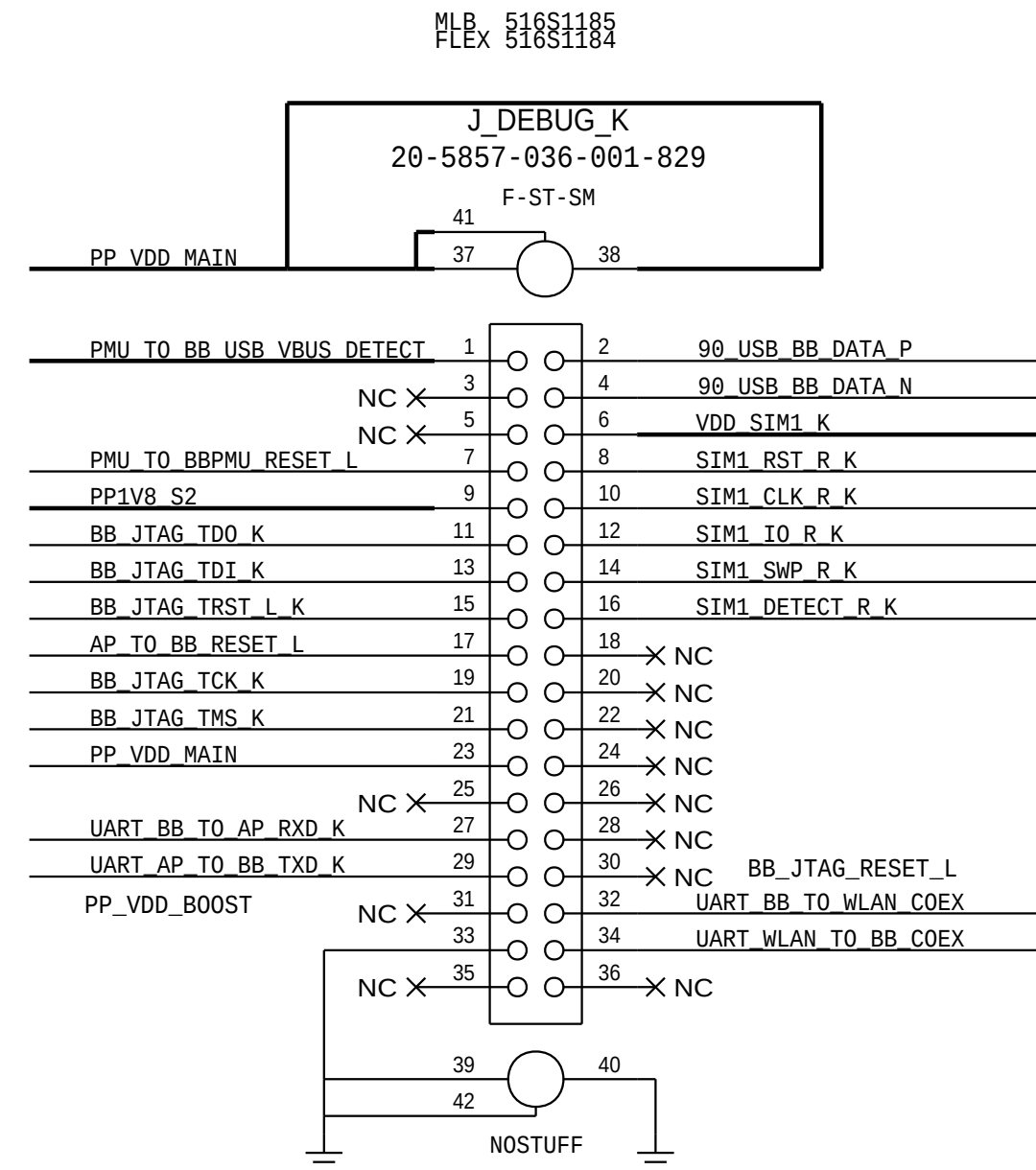
DIGRF



PCIE GND

BB UART

DEBUG CONNECTOR



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METROCIRC

D211 NORTH-SOUTH METROCIRC
339S00297

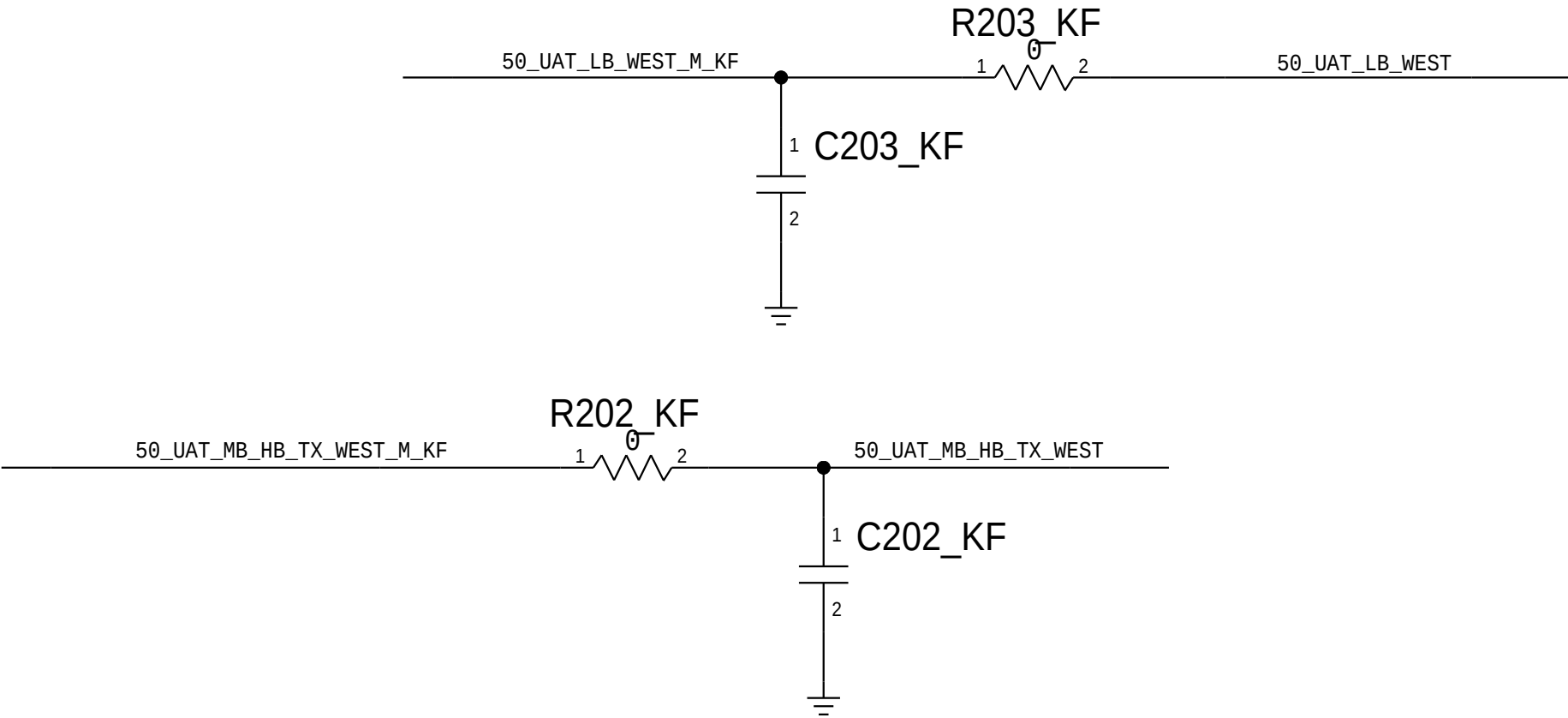
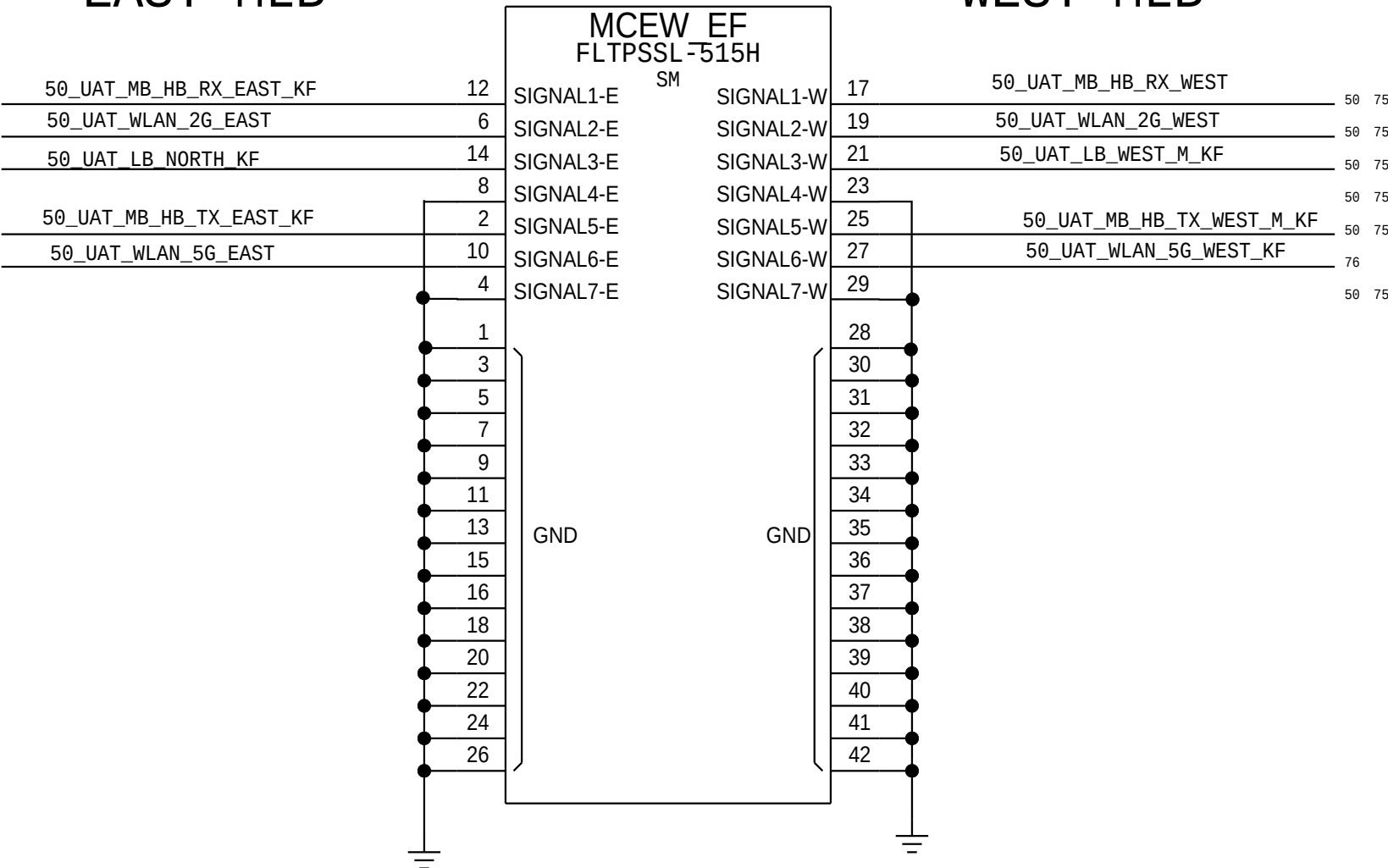
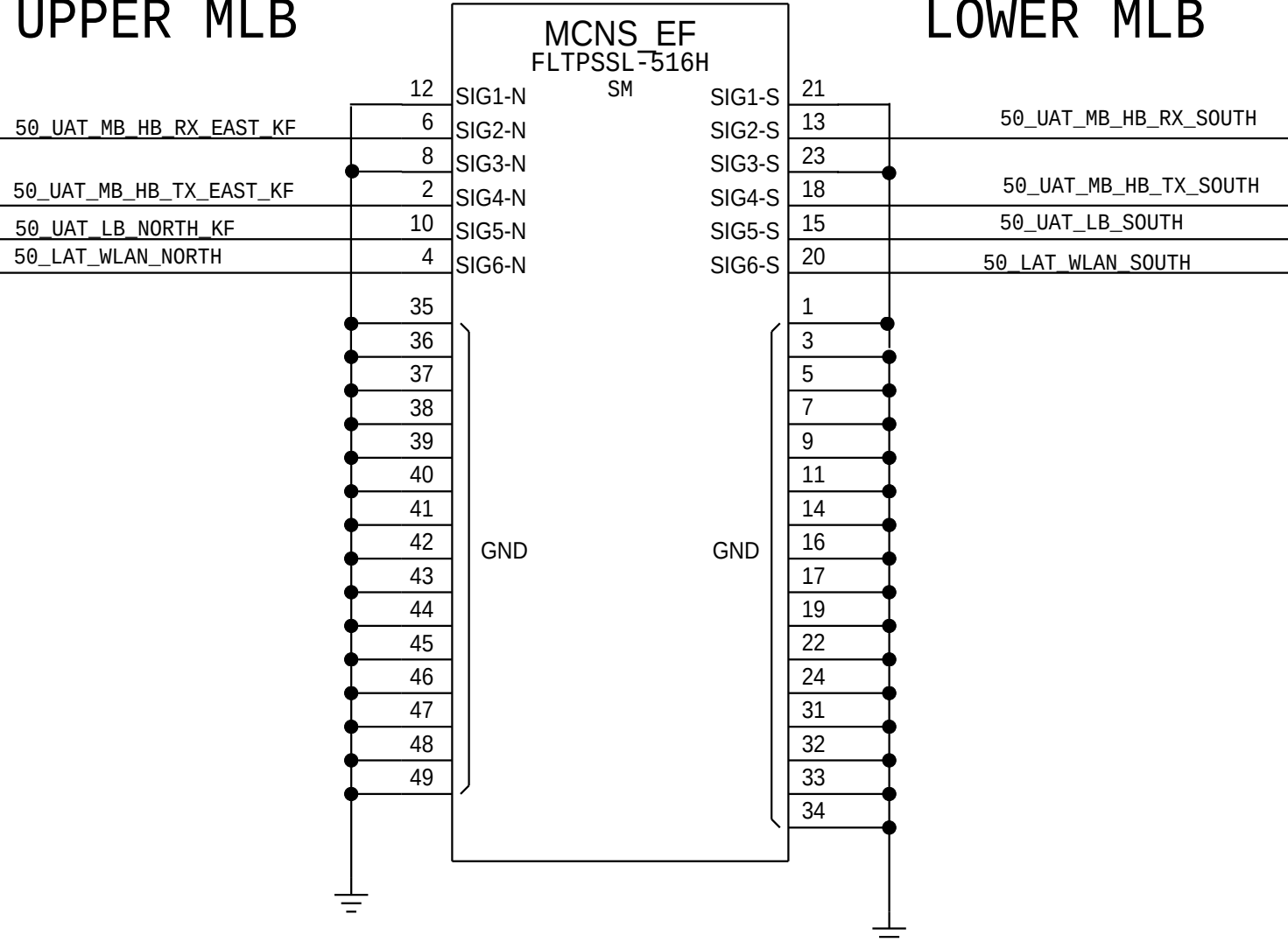
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EAST MLB

WEST MLB

UPPER MLB

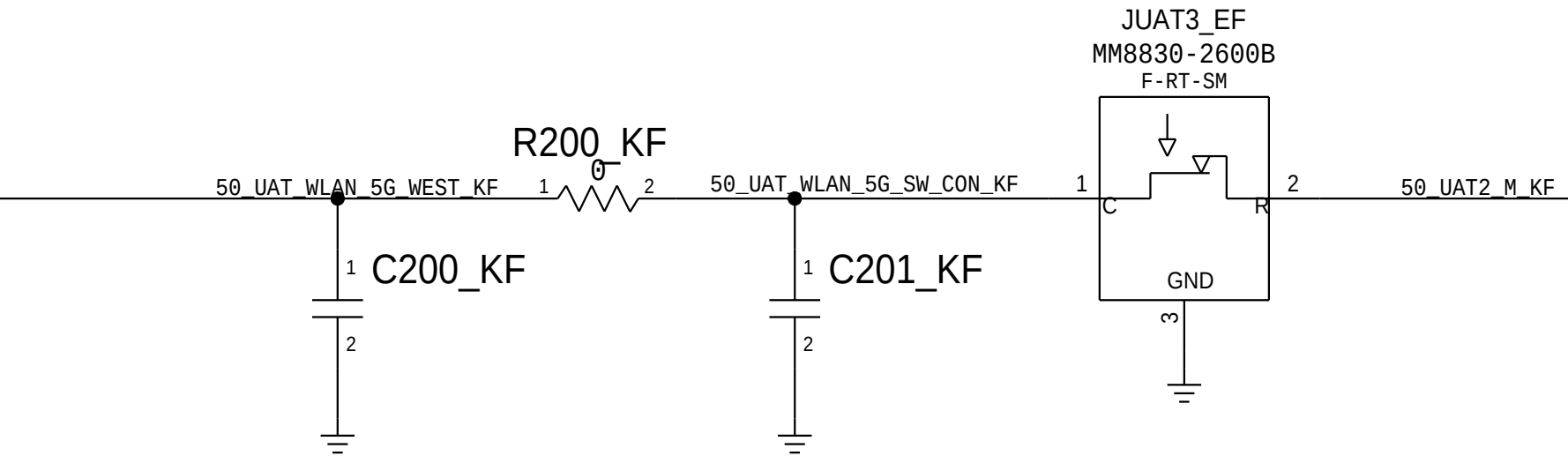
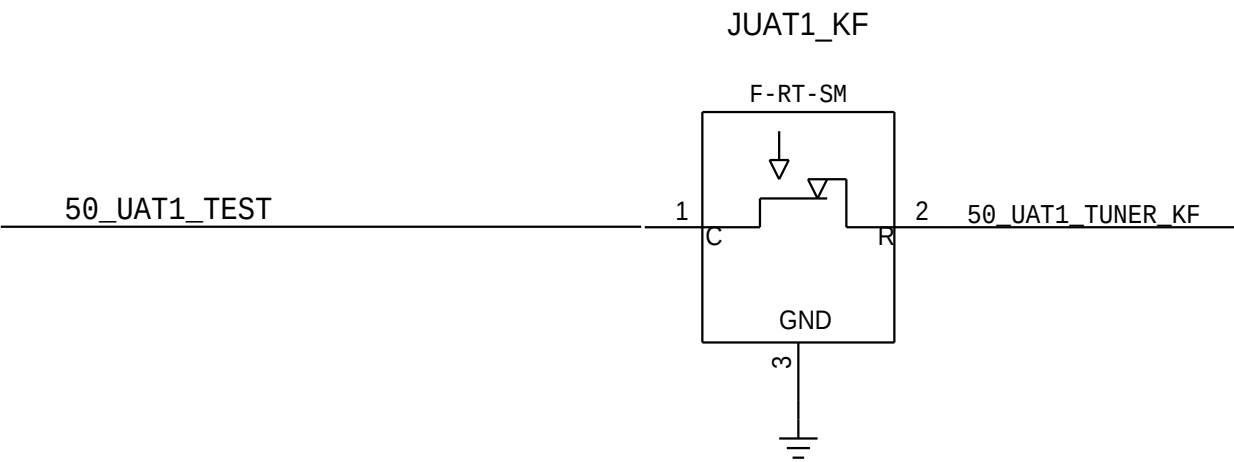
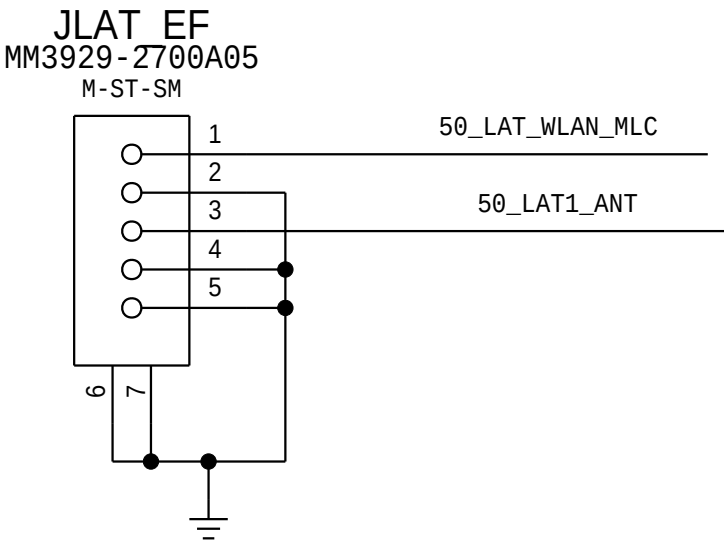
LOWER MLB



LAT METROCIRC FEED

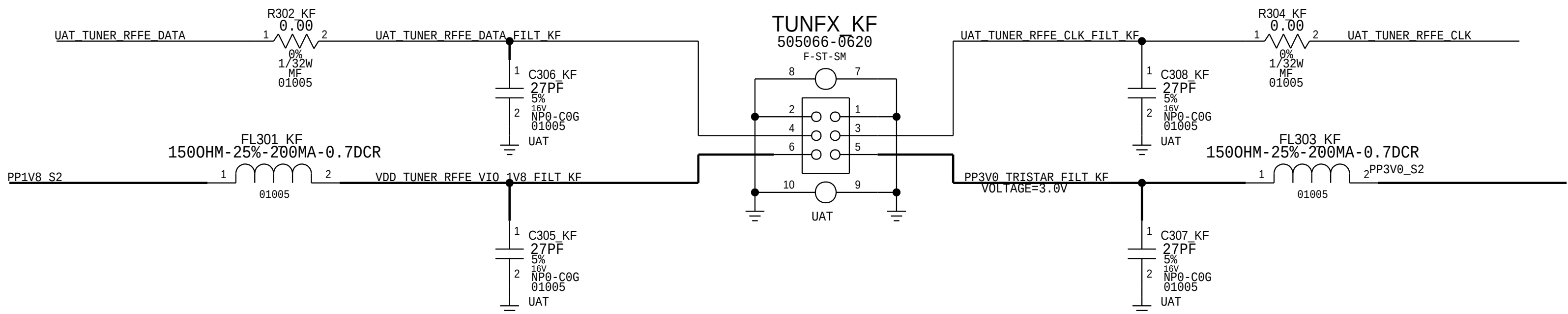
UAT1 TEST

WIFI 5GHZ TEST



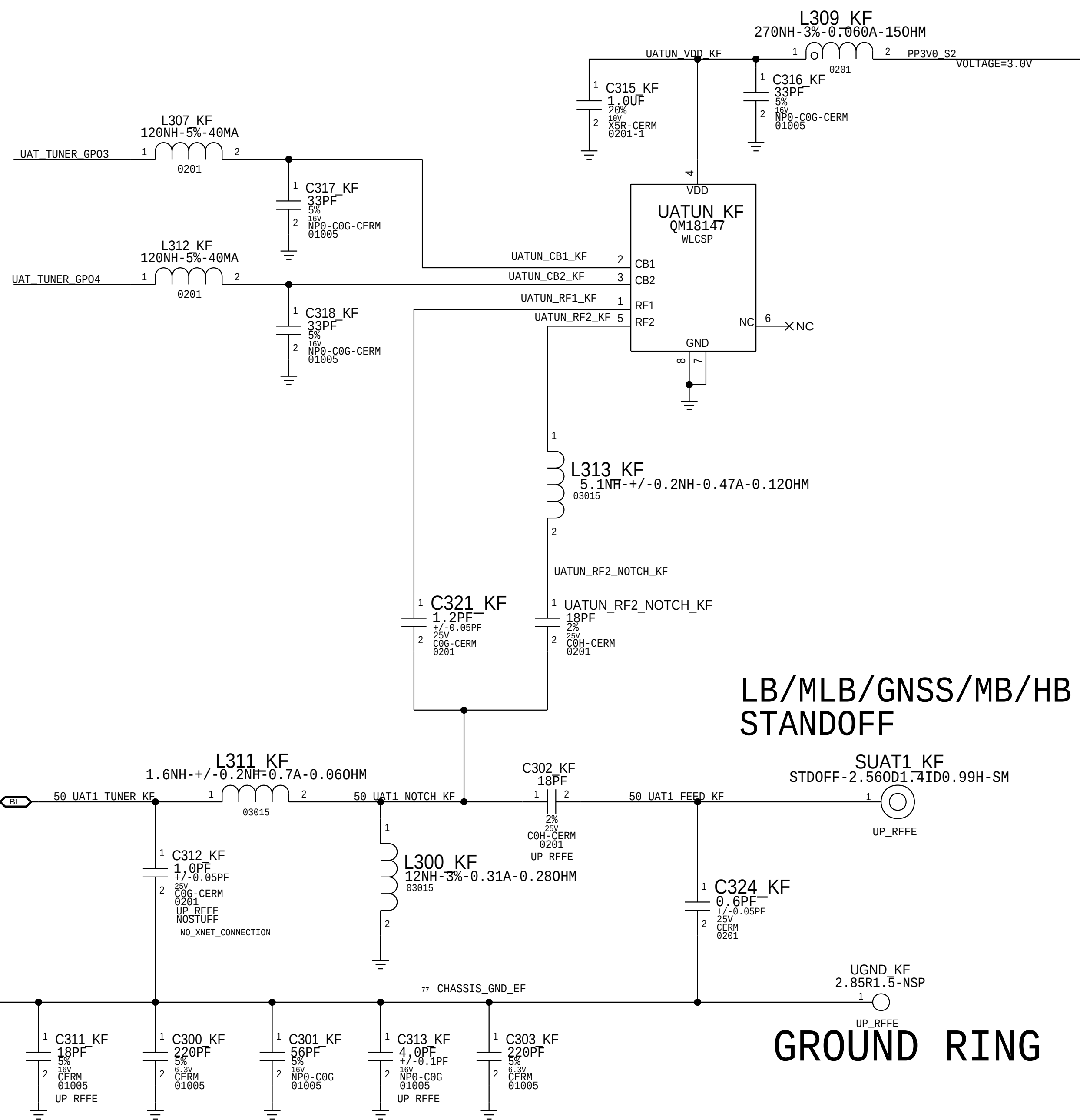
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UAT TUNER FLEX



PINOUT IS ROTATED 180
COMPARED TO D20 & D201

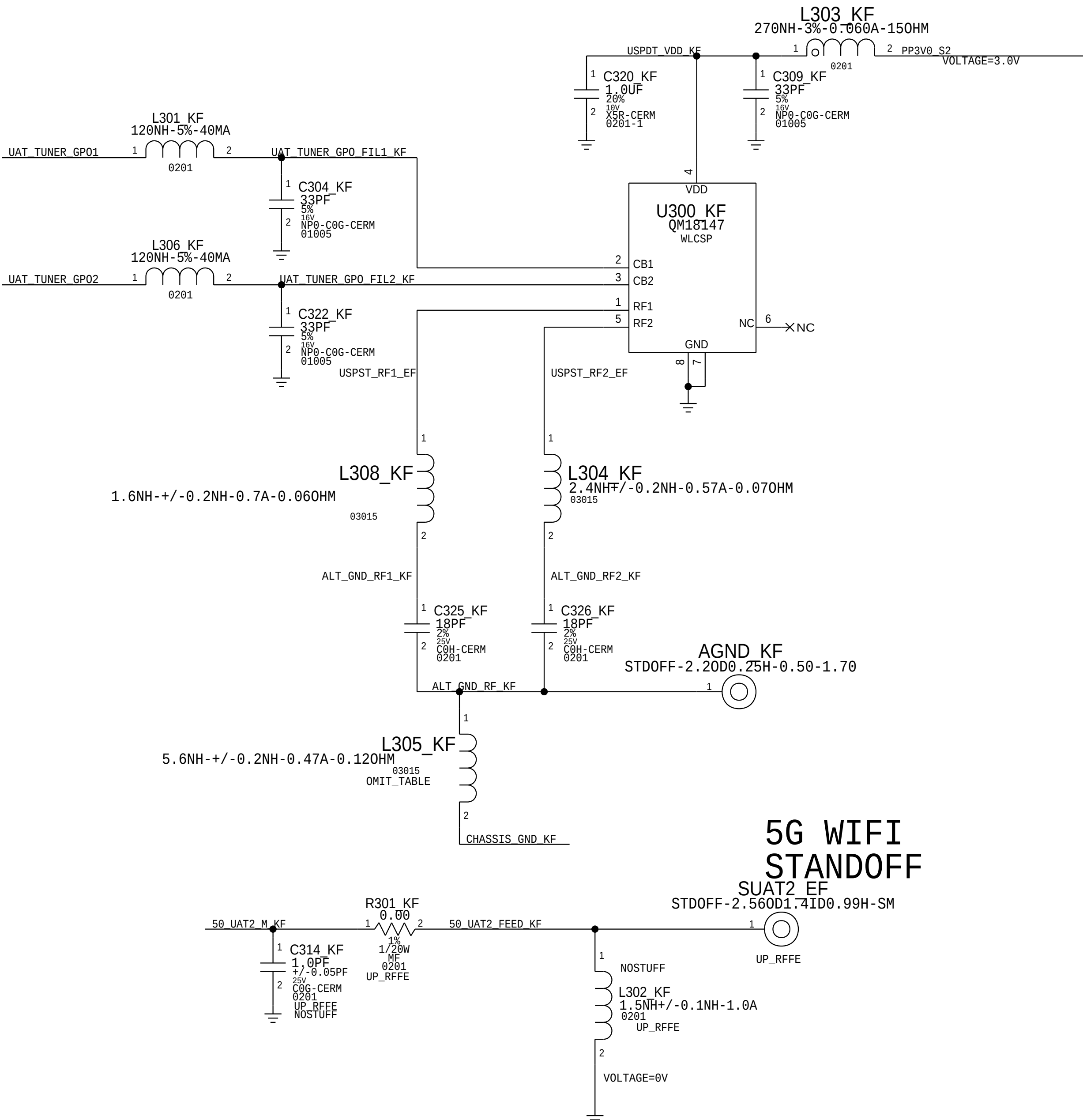
UAT TUNER



LB/MLB/GNSS/MB/HB
STANDOFF

GROUND RING

ALT UAT1 GND



5G WIFI
STANDOFF

UAT MATCH AND TUNER

DRAWING NUMBER	051-02159	SIZE	D
REVISION	10.0.0		
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	77 OF 81		